

ESTIMATING CROP EVAPOTRANSPIRATION OF VEGETABLE CROPS USING POT MICRO- LYSIMETER

A report submitted in partial fulfilment of the requirements for the degree of

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by

CHANSA RAINGAM

(D/21/AE/001)

ZINGJAR KEISHING

(D/21/AE/002)

CHANSING ZIMIK

(D/21/AE/003)

JOLLY ZINYU

(D/21/AE/005)

Under the guidance of

Prof. Aditi Bhadra



**Department of Agricultural Engineering
North Eastern Regional Institute of Science and Technology
(Deemed-To-Be University)
Nirjuli-791109, Arunachal Pradesh
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CERTIFICATE

This is to certify that the dissertation entitled, “**Estimating crop evapotranspiration of vegetable crops using pot micro-lysimeter**”, submitted in partial fulfilment of the requirements for the award of **Bachelor of Technology in Agricultural Engineering** to the **North Eastern Regional Institute of Science and Technology, Nirjuli Arunachal Pradesh**, is a record of bonified research work carried out by

Ms. Chansa Raingam	(D/21/AE/001)
Ms. Zingjar Keishing	(D/21/AE/002)
Ms. Chansing Zimik	(D/21/AE/003)
Ms. Jolly Zinyu	(D/21/AE/005)

under my/our guidance and supervision. The results embedded in this research work have not been submitted to any other University or Institute for the award of any Diploma or Degree.

Prof. Aditi Bhadra

Project Guide

Dr. Thaneswar Patel

Project Coordinator

Prof. Aditi Bhadra

Head of Department

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ABSTRACT

Irrigation plays a very crucial role in crop production and constitutes the largest portion of the water withdrawn for agriculture. Majority of the irrigation water is lost through crop evapotranspiration (ETc). Hence, there is a need for efficient water management and reliable estimation of ETc. The primary aimed of the study is to analyse the effectiveness of the technique through lysimeter to estimate crop evapotranspiration (ETc) and crop coefficient (Kc) for selected crops (i.e. Mustard, Carrot, Tomato and Green chilli). In this technique, plastic containers are used as a micro lysimeter. Thereafter the daily rainfall, reference crop evapotranspiration (Eto) and crop evaporation (ETc) is to be recorded. With the obtained Kc value, Kc curve is to be plotted against the stages of growth. Generally, FAO 56 Penman-Monteith is regarded as the standard method for estimating ETc and Kc. Comparison between the Kc curve of the crops obtained in this technique and the standard value of Kc is to be carried out. Therefore, the present technique is to confirm the significant comparability with the standard generalized Kc; indicating the approximate water requirement by the selected crops. This technique is relatively simple and requires lesser input parameters in the estimation of ETc and Kc and may be adopted as the alternatives method for determining crop evapotranspiration in regions where comprehensive meteorological information and data are limited.

CHAPTER 1

INTRODUCTION

Efficient water management is considered as a key feature in agriculture, particularly for vegetable crops that require precise irrigation to achieve optimal growth and yield. Irrigation of crops consume maximum agricultural water worldwide. This water withdrawn by agriculture is lost through crop evapotranspiration (ETc) which effect the crop water demand. Therefore, there is a need for proper water management and reliable estimation of ETc. The available methods for estimating ETc involve expensive devices and extensive meteorological information, which becomes limitation in areas with no instruments and with limited data.

1.1 Evapo-Transpiration

Evapotranspiration, which refers to the loss of water through evaporation from the soil and transpiration by plants, is an important parameter in estimating crop water requirements.

Evaporation from vegetation is generally given a more specific term -- evapotranspiration or ET for short. By definition, ET is the loss of water from a vegetated surface through the combined processes of soil evaporation and plant transpiration. The term evapotranspiration comes from combining the prefix "evapo" (for soil evaporation) with the word transpiration. Both soil evaporation and plant transpiration represent evaporative processes; the difference between the two rests in the path by which water moves from the soil to the atmosphere. Water lost by transpiration must enter the plant via the roots, then pass to the foliage where it is vaporized and lost to the atmosphere through tiny pores in the leaves known as stomata. In contrast, water lost through soil evaporation passes directly from the soil to the atmosphere. Weather parameters, crop characteristics, management and environmental aspects are factors affecting evaporation and transpiration. Evapotranspiration data are usually presented as a depth of water loss over a particular time period in a manner similar to that of precipitation. Common units for ET are inches/day or millimetres /day.

The estimation of crop evapotranspiration is an essential component of agricultural water management. Accurate estimation of evapotranspiration is crucial for sustainable irrigation water management to avoid under or over-irrigation, which can adversely affect crop yields, water use efficiency, and farmers' income. The interaction of various climatic parameters that determine crop water use and water supply from rain dictates the water requirement for crop production.

One method that has been developed for estimating crop evapotranspiration is the use of Pot Micro-Lysimeter. Lysimeters are large containers installed in the field to measure and monitor plant-water use. Pot Micro-Lysimeters are an innovative development of traditional lysimeters, which are designed for small-scale crop management. Pot Micro-Lysimeter is made up of a PVC pot filled with soil and fitted with a weighing platform and a pressure transducer to measure soil water content and plant water use. The system is also equipped with a rain gauge, a temperature gauge, and a humidity sensor to measure environmental conditions.

The Pot Micro-Lysimeter method offers high accuracy in estimating crop evapotranspiration due to its small size and the ability to measure water use at a small scale. This method also provides essential insights into the factors affecting crop evapotranspiration, including soil type, plant physiology, and environmental conditions. The use of Pot Micro-Lysimeter in estimating crop evapotranspiration is particularly useful in vegetable crop management due to the high water demand of vegetable crops and the need for accurate irrigation scheduling. Precise information about crop water needs can help farmers optimize irrigation systems, reduce water use, and achieve maximum yields.

Several methods had been developed by hydrologists for estimating ETC. The concepts of the methods were developed only for specific regional climate (Burnash, 1995). These methods can be described as empirical such as those developed by Thornthwaite (1948) and Blaney and Criddle (1950), as well as physical-based [e.g. Penman, (1948); Monteith, (1981); FAO's Penman-Monteith method by Allen et al., (1998)]. ETC can be obtained directly with the aid of lysimeters and through indirect estimation methods (Villanova et al., 2006). The above-mentioned differ in terms of accuracy and data requirement. Since many of these methods are region-specific, they only work best in the condition and region they were developed for but often fail to estimate potential ETC in other climatic conditions (Tukimat et al., 2012; Zeleke and Wade 2012).

The FAO-56 Penman-Monteith (FAO56PM) approach is considered to be the standard method for ETC estimation and KC determination in the present (Allen et al. 1998; Maina et al. 2012). Having gained acceptance from the international scientific community, the FAO56PM model is known for its precisely better results when compared to other models in various regions of the world and its good approximation to accurate lysimeter observations (Mohawesh, 2011; Maina et al., 2012). However, despite its reliability, the FAO56PM requires extensive meteorological information, data, and parameters, which pose a problem in places where data is scarce and unavailable. For most regular meteorological stations, only temperature and rainfall are measured while other meteorological parameters essential for ETC estimation are not available (Tukimat et al., 2012). The magnitude of the problem increases in remote areas with no weather stations for recording the required meteorological data and parameters. Therefore, a much simpler and parameter-less alternative that can deliver reliable results in estimating crop water requirements is desirable.

1.2 Scope of the Study and its Application

Evapotranspiration (ET) is a fundamental process in the hydrological cycle that plays a critical role in the movement of water within the environment.

One of its main contributions to the environment is climate regulation. By releasing water vapor into the atmosphere, ET influences local weather patterns and contributes to cloud formation and precipitation. Areas with high rates of evapotranspiration often experience increased rainfall, while regions with low ET may be drier.

By understanding the components and factors affecting ET, we can better manage water resources, optimize agricultural practices, and adapt to the challenges posed by climate change

and urbanization. As climate change impacts water availability and weather patterns, understanding ET becomes even more critical for agricultural resilience. Farmers can use ET data to adapt to changing conditions, ensuring that crops remain productive even in the face of droughts or excessive rainfall.

Evapotranspiration (ET) is an important component of the water cycle and agricultural water balance. Planning, managing, and regulating agricultural water resources rely on reliable estimates of agricultural water consumption. It leads to establishing a sustainable water balance, mitigates the consequences of water scarcity, and prevents the waste and overuse of scarce water resources. An accurate assessment of ET is important to improve water use efficiency and sustainable water management in agriculture because ET is a primary consumer of irrigation water and precipitation in agricultural landscapes.

Hence the understanding and the measurement of ET plays a vital role, which can also be explained by some of its application.

- **Agricultural Management:** Farmers use ET data to optimize irrigation practices, ensuring crops receive adequate water without over-irrigation, which can lead to water wastage and soil salinization.
- **Water Resource Management:** ET measurements help water managers assess water availability and plan for sustainable water use in urban and rural areas.
- **Climate Studies:** Researchers use ET data to study climate change impacts, as changes in ET patterns can indicate shifts in climate and water availability.
- **Ecosystem Health:** Monitoring ET can provide insights into ecosystem health, as changes in ET rates may indicate stress in plant communities or shifts in hydrological conditions.

Evapotranspiration is also a critical factor in determining crop water requirements. Understanding ET rates allows farmers to:

- **Optimize Irrigation:** By knowing how much water crops are losing through ET, farmers can adjust their irrigation schedules to match crop needs, improving water efficiency.
- **Select Appropriate Crops:** Different crops have varying ET rates. Farmers can choose crops that are well-suited to local climatic conditions and available water resources.
- **Implement Water Conservation Practices:** Knowledge of ET can help farmers adopt practices such as mulching, cover cropping, and conservation tillage to reduce water loss and improve soil moisture retention.

1.3 Objectives

The study was undertaken with the following objectives:

- i. To determine crop evapotranspiration of tomatoes, mustard, carrots and beans using small plastic pots as percolation micro-lysimeters and a drip irrigation set-up.
- ii. To determine crop coefficient of tomatoes, mustard, carrots and beans using FAO-56 Penman Monteith Equation and crop evapotranspiration.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Water Scarcity and Agricultural Water Use

Homer-Dixon (1993) examined the relationship between environmental scarcity and social conflict, indicating how shortages of renewable resources, such as water, arable land, and forests, can lead to social instability and violent conflict. His research was carried out by analysing historical case studies and theoretical frameworks to understand how resource scarcity contributes to conflict. He classified resource scarcity into three types: supply-induced, where resources become depleted or degraded; demand-induced, where population or consumption increases; and structural scarcity, where the distribution of resources leads to social inequalities. He then discussed how these scarcities can undermine local economies, disrupt livelihoods, and weaken government stability, thereby raising the likelihood of conflict. The study delved deeper into how the pressures of resource shortages were fueling migration and intensified ethnic or political tensions within areas receiving displaced populations. In connecting environmental stressors to social and political outcomes, Homer-Dixon's study emphasized the need for the sustainable management of resources coupled with proactive governance to forestall conflicts, especially in resource-poor regions. This approach serves as a foundation for understanding how degradation in the environment, particularly with respect to water scarcity, can fan more comprehensive social and political challenges.

Rijsberman (2004), Morison et al. (2007), and FAO (2017) conducted extensive examinations of the interplay between water scarcity and agricultural water use, providing a multidimensional understanding of the challenges and potential solutions. Rijsberman (2004) emphasized the global implications of water scarcity and noted that the problem is not only a matter of physical shortages but also of mismanagement and inequitable access to water resources. He pointed out that agriculture is the largest consumer of freshwater worldwide, accounting for 80–90% of total water resources, with figures reaching as high as 95% in less developed countries. Rijsberman contends that inefficient use of water for agricultural purposes potentiates scarcity, especially in arid regions with scarce water supply. Morison et al. (2007) discuss the compounding effects of climate change that aggravate water scarcity by changing precipitation patterns and the occurrence of extreme weather events. They emphasized the need for efficient water use in agriculture, especially advanced irrigation systems and precision agriculture, to avoid water wastage. They also called for integration of water resource management with sustainable agricultural practices for long-term food security and environmental conservation. FAO (2017) provided a comprehensive global overview of agricultural water use and the strategies needed to tackle water scarcity. The organization emphasized that improving water use efficiency in agriculture is key to addressing the growing demand for food, driven by population growth and urbanization. The FAO underscored the importance of innovative technologies, such as drip irrigation and fertigation, which optimize

water and nutrient application, reducing waste and enhancing productivity. Third, FAO emphasized that solid frameworks for water governance in agriculture have to ensure distribution equities and resource manageability among competing sectors on water use by agriculture.

2.2 Importance of Estimating Crop Evapotranspiration (ET_c)

Smith (2000) explored the relevance of crop evapotranspiration estimation by focusing on the dynamic interaction between climatic factors, water use by crops, and water availability from precipitation. He emphasized that the understanding of ET_c is essential for the determination of crop water requirements and optimization of irrigation. Smith analysed the role of comprehensive meteorological data such as temperature, humidity, wind speed, and solar radiation in accurate ET_c computation. By incorporating all these climatic factors, his work showed how accurate ET_c estimates could be very useful in developing irrigation schedules to minimize over- or under-irrigation and thus efficient utilization of water. Smith also mentioned the importance of gathering, analysing, and processing detailed meteorological datasets for decision-making in the management of water resources. His study shows that the accurate estimation of ET_c not only improves agricultural productivity but also serves to be an important factor in mitigating water scarcity problems, mainly in regions where agriculture heavily depends on irrigation.

Al-Kaisi (2000) and Tukimat et al. (2012) examined the importance of estimating crop evapotranspiration (ET_c) by exploring its role in understanding and managing the total water use of crops. Al-Kaisi (2000) emphasized that ET_c is a critical indicator encompassing two key processes: soil evaporation and plant transpiration, which together represent the total transfer of water from the Earth's surface to the atmosphere. His study demonstrated how accurately estimating ET_c informs irrigation demand, ensuring that crops receive the optimal amount of water for growth while minimizing wastage. Tukimat et al. (2012) expanded on this by overcoming the difficulties in estimating ET_c in areas with limited meteorological data. They underlined that accurate ET_c estimation is critical for irrigation scheduling, water budgeting, and long-term water resource management, especially in areas facing water scarcity. Their work pointed out the need for practical and accessible methods of computing ET_c, especially in remote areas where data collection is limited. Together, these studies highlighted the significance of ET_c as a tool for sustainable agricultural water management, providing solutions for balancing crop water needs with resource conservation in diverse climatic conditions.

Zeleeke and Wade (2012) focused on the significance of efficient agricultural water management and the role of accurate ET_c estimation in irrigation scheduling and water resource planning. According to them, accurate ET_c data is essential for establishing the right amount of water that should be applied on crops to ensure that water resources are optimally and efficiently used. Zeleeke and Wade (2012) have recommended including local climate conditions and advanced water management strategies in improving the accuracy of ET_c estimations specifically in areas of water scarcity. Both of these studies strongly recommended more accurate methods for estimating ET_c in sustaining agricultural uses of water while developing even simpler and cost-effective solutions overcoming data limitation issues in various agricultural regions.

2.3 The FAO-56 Penman-Monteith (FAO56PM) Model

Allen et al. (1998) and Maina et al. (2012) conducted significant research based on the FAO-56 Penman-Monteith (FAO56PM) model, a widely accepted method for estimating crop evapotranspiration (ET_c) and determining crop coefficients (K_c). The work by Allen et al. (1998) actually helped establish the formulation of the FAO56PM model as a widely applicable, standard method in calculating ET_c, thus emphasizing accuracy and reliability in estimations. Their work specified in good detail how one could go about using this model; the model integrates multiple meteorological factors, such as temperature, humidity, wind speed, and solar radiation, to calculate evapotranspiration. Using such a model is how these researchers showed that one can successfully calculate ET_c: an activity necessary for efficient irrigation scheduling and water resource management in agricultural production. Maina et al. (2012) developed further by validating the FAO56PM model in specific regional contexts and crop types. Field studies were carried out to evaluate the ability of FAO56PM in simulating ET_c for various crops by comparing the outcomes with measurements from lysimeters. Their results established that the FAO56PM model can accurately simulate ET_c under different climates and types of crops, and hence serves as a very important model for water management in agriculture. However, they also recognized that such a model would rely on detailed meteorological data and may face challenges in areas with fewer weather stations; therefore, alternative approaches or data collection strategies would be necessary. Both studies supported the FAO56PM model's role in enhancing agricultural water use efficiency and the widespread application of this model in global agricultural practices.

Mohawesh (2011) and Maina et al. (2012) conducted research based on the FAO-56 Penman-Monteith (FAO56PM) model, focusing on its application in estimating crop evapotranspiration (ET_c) and optimizing water use in agriculture. Mohawesh (2011) explored the effectiveness of the FAO56PM model in estimating ET_c for various crops under different climatic conditions. Mohawesh (2011) investigated the potential of the FAO56PM model in estimating ET_c for different crops under varying climatic conditions. He confirmed the accuracy of the model through a comparison of its estimations with the lysimeter study and field experiment observations. His work proves that the FAO56PM model is a reliable method to estimate ET_c, especially useful in irrigation scheduling, more so in areas where water resources are limited. Maina et al. (2012) also explored the FAO56PM model with respect to its performance in localizing regional contexts and crop type, thereby testing how effectively it can estimate ET_c under different climates, especially under those regions where meteorological data might be scarce. It therefore agreed that the FAO56PM model is generally valid to estimate ET_c for varied conditions and thus can provide correct data for irrigation practice or other forms of water use decisions. However, the group found an issue with the fact that a reliable, detailed meteorological information should exist in the area in question, and if these conditions were not met, there should be alternative methods or more simplistic models. Both studies added further credence to the FAO56PM model as an essential and reliable tool in enhancing water use efficiency in agriculture.

2.4 Limitation of FAO56PM Method

Tukimat et al. (2012) recognized that while the FAO56PM model is widely regarded as a reliable and accurate method for estimating ET_c, it requires comprehensive meteorological data, such as temperature, humidity, solar radiation, and wind speed. The FAO56PM model becomes hard to implement effectively in regions where such data is scarce, especially in

remote areas or developing countries. Tukimat et al. (2012) discussed this challenge by exploring alternative approaches to ET_c estimation that could be used in areas where meteorological stations are not available, thus highlighting the need for simpler, more accessible methods. They proposed that though the FAO56PM model is very accurate, its dependency on detailed data makes it inapplicable in some regions. Their work emphasized the need to develop and adopt alternative methods for crop water requirement estimation in areas where data availability is limited, aiming to improve water management and irrigation scheduling in such regions.

2.5 Lysimeters as a Tool for ET_c Estimation

Fisher (2004), Piccinini et al. (2009), Abdullahi et al. (2013), Abedinpour (2015), and Gebler et al. (2015) have all contributed to the advancement of lysimeters as a tool for estimating crop evapotranspiration (ET_c), with the aim of improving the accuracy and reliability of ET_c measurements in agricultural settings. Fisher (2004) described various lysimeter designs with size, operation, and shape differences, and their implications on the accuracy of ET_c measurements for crops. His findings indicated that lysimeters are very reliable for crop water use determination, but the design and configuration of the lysimeter significantly affect the quality of the data collected. Piccinini et al. (2009) and Abdullahi et al. (2013) further tested lysimeters in field conditions, comparing the direct measurements of ET_c obtained from lysimeters with data from indirect methods such as the Penman-Monteith model. They found that lysimeters gave accurate, real-time measurements that could be used as a benchmark for validating model-based estimates, especially in regions where indirect methods may have difficulties because of data limitations. These works focused attention on the specific capability of lysimeters to detect localized differences in evapotranspiration caused by unique conditions of environment and crop, thus providing insight into developing irrigation strategies specifically for a region. Based on this work, Abedinpour (2015) and Gebler et al. (2015) expanded these studies to wider crop types and environmental settings to test the suitability of lysimeters for different agriculture systems. They underlined the necessity for accurate data in water-use efficiency analysis, which involves handling the irrigation system and helping resolve the water scarcity issue by use of a lysimeter. On the other hand, they expressed concern that installation and maintenance involve expensive procedures that might limit the acceptance and wide use of these tools in resource-poor settings.

Howell (2005) focused on the use of lysimeters as a direct and reliable tool for estimating crop evapotranspiration (ET_c). In his study, he emphasized lysimeters as being one of the most precise methods for the measurement of the water lost through evapotranspiration by crops. This is because lysimeters offer real-time measurements at the field level. A lysimeter consists of a pot containing soil and crop for which water entering and leaving the system is measured to great precision, usually either through weighing or percolation methods. Howell placed an importance on the advantages offered by lysimeters in offering high-quality and reliable data of ET_c, crucial in changing irrigation practices for efficiency and better water use. He also admitted that lysimeters, although very accurate, have high installation costs and maintenance, which may not be feasible in some areas. However, Howell's study emphasized the importance of lysimeters in cross-validation of other ET_c estimation methods, such as the FAO56PM model, where direct measurements of evapotranspiration are compared with model predictions.

This makes lysimeters a crucial tool for crop water management and increasing the efficiency of water use in agriculture.

Villanova et al. (2006) conducted research on the use of lysimeters as a tool for estimating crop evapotranspiration (ET_c) by comparing lysimeter-based measurements with other indirect methods of ET_c estimation. Their study focused on the accuracy and reliability of lysimeters in providing direct, field-based measurements of water lost through evapotranspiration. They found lysimeters effective in measuring total crop evapotranspiration, especially in different environmental conditions and offering very useful data that might be applied to improve irrigation practices as well as water management. There are still some limitations of lysimeters, including high installation costs and the maintenance costs, and these are not very useful in a large-scale use or where resources are meager, according to Villanova et al. 2006. These were, however highlighted by the authors as critical to making lysimeters act as a benchmark for testing other methods, such as the FAO56 Penman-Monteith model, thus providing in situ and real-time and accurate crop water use data. Their study supported the worth of lysimeters as aids in further improving the understanding of crop water requirements and optimizing irrigation efficiency, especially where water is scarce.

López-Urrea et al. (2006) and Vaughan et al. (2007) conducted studies on lysimeter applications, focusing on the accuracy and effectiveness of lysimeters for estimating crop evapotranspiration (ET_c) in various agricultural settings. López-Urrea et al. (2006) investigated the direct measurement of ET_c using lysimeters, comparing these measurements with those taken from other indirect estimation methods, such as the FAO56 Penman-Monteith model. Their results indicated that lysimeter-based measurements of ET_c were very accurate and could be used for reliable data on irrigation scheduling and water management. They also emphasized the fact that lysimeters could provide valuable information on local crop water use, enabling more precise, region-specific irrigation strategies. Vaughan et al. (2007) also validated lysimeter measurements against model-based estimations of ET_c, confirming that lysimeters are a reliable tool for assessing crop water requirements. They concluded that lysimeters produced accurate, real-time data, which could be used to fine-tune irrigation practices, thereby enhancing water use efficiency and crop productivity. Both studies reinforced the role of lysimeters as a critical tool for crop water management, noting their ability to provide direct measurements that could be used to validate and refine other ET_c estimation methods. However, they still highlighted the problems that lysimeters entail; the installation cost is relatively high and needs extensive maintenance, which means its application is limited only in large-scale agricultural systems. But their study proved how crucial lysimeters remain, considering the present trend regarding the scarcity of water, where management practices should be improved as far as possible.

Abubakary et al. (2013), Sandhu et al. (2016), and Cal et al. (2018) conducted research on lysimeter studies and applications, contributing to the understanding of lysimeters' role in accurately estimating crop evapotranspiration (ET_c) and improving agricultural water management. Abubakary et al. (2013) focused on the design and efficiency of lysimeters in measuring ET_c for specific crop types, particularly in regions where traditional irrigation practices face challenges. Their work also made a case for lysimeter accuracy in providing up-to-date crop water-use measurements, thereby providing a sound basis for managing irrigated water use more optimally and improving irrigation-scheduling practices. Sandhu et al. (2016)

went further and addressed the relevance of lysimeters in crop water management while pointing out their relevance for cross-validation of alternative methods of ET_c computation like the FAO56 Penman-Monteith method. They have shown that lysimeter data are critical for fine-tuning model estimates to allow for more accurate irrigation decisions and higher water use efficiency. The researchers have also addressed the limitations of lysimeters, such as their high installation and operational costs, which may limit their applicability in large-scale agricultural systems, especially in resource-limited regions. Cal et al. (2018) developed and extended the previous lysimeter study by considering the potential use of lysimeters for other crop species, as well as for conditions under which they may operate. Lysimeters allowed farmers to better understand how much water their crops actually needed and the extent of irrigation requirements. Their work also established that lysimeters are necessary instruments for collecting quality, time-sensitive information, but in this regard also underlined that cost efficiency is the need of hour to make this technology useful for more significant numbers of agricultural operations. Altogether, these papers established lysimeters as significant tools that can help make water a more efficient user and allow agriculture to function sustainably, but, at the same time, recognized challenges in the more practical acceptance of lysimeter technology at large.

2.6 Challenges and Limitations

Fisher (2004), Panda et al. (2014), and Facchi et al. (2016) all addressed the challenges and limitations of using lysimeters for estimating crop evapotranspiration (ET_c) in their respective studies. Fisher (2004) studied the factors related to the practicality of using lysimeters and various factors that might influence its accuracy, including lysimeter design and size. He also acknowledged high costs in the installation and maintaining lysimeters, making it impractical for large-scale farms or in poor countries with limited resources. Fisher points out that even though they provide very accurate information from lysimeters, logistical obstacles and high costs prevent wide usage. Panda et al. (2014) and Facchi et al. (2016) also looked at these limitations of lysimeters that are resource intensive in themselves. They emphasized that the high installation costs, the need for specialized maintenance, and the technical expertise required to operate lysimeters make them impractical in many cases, especially in low-resource settings. The size and complexity of lysimeters also limit their application in large-scale agricultural systems. Despite these limitations, both studies also acknowledged the potential of lysimeters in verifying other ET_c estimation methods, especially for accurate water management and irrigation scheduling. Panda et al. (2014) and Facchi et al. (2016) recommended that alternative, more cost-effective methods or simplified lysimeter designs be explored to overcome these limitations, making accurate crop water use data more accessible to a broader range of farmers and researchers.

2.7 Integrating Climate Change and Resource Management

Morison et al. (2007) analysed the influence of climate change on the water requirements for agriculture and emphasized the importance of adopting adaptive strategies for managing water resources more efficiently. They discussed how the altered precipitation and temperature regimes in weather patterns could affect crop water use and irrigation demands. Their work stressed the integration of climate change projections into water management practices to allow agriculture to continue thriving in the face of such changes. They also called for more accurate methods of estimating crop evapotranspiration (ET_c) to effectively manage water resources under variable climatic conditions.

This further enhanced by FAO (2011) outlining strategies of improving water use efficiency in agriculture, especially considering the fact that climate change is an issue coupled with a further increase in water scarcity. The FAO asserted that climate change would make waters scarce in most areas of the world, hence integration of advanced technologies including irrigation systems that consume less water and efficient methods of crop evapotranspiration estimation were encouraged for mitigation. The work of the FAO highlighted the importance of sustainable agricultural practices that would balance water conservation with the demand for food production. Both Morison et al. (2007) and the FAO (2011) emphasized the need to integrate climate data into resource management strategies and promote sustainable water use practices to adapt to the challenges brought about by a changing climate. Overall, this work aims to suggest that incorporating climate change projections into water and resource management is essential to achieve sustainability in agricultural productivity and long-term food security.

CHAPTER 3

DATA AND METHODOLOGY

3.1 Study Site

The study was conducted at the Agricultural Experimental Farm, North Eastern Regional Institute of Science and Technology (NERIST), located at Nirjuli, Arunachal Pradesh. Geographically, the site is located at a latitude of about 27°07'54.67" N and longitude of 93°44'56.15" E and an elevation of about 149 meters above sea level. The preparation and set up of the experimental field were carried out for a period of four months.

This area falls in the northeastern region of India, which has varied agro-climatic conditions. The climate of this region is classified as humid subtropical, with high amounts of rainfall during the monsoon season, which usually occurs between June and September, and moderate to cool winters between November and February. It is characterised by distinct wet and dry seasons. The temperature ranges between 22 and 15°C (72 and 59 °F) during winter months. It receives heavy showers about 2,000 to 4,000 mm (160 in) between the months of May and September yearly. Monsoon temperature oscillates between 30 and 18 °C (86 and 64 °F). Summer temperature can soar up to 40 °C (104 °F) some days.

3.2 Experimental Area

The area for conducting the experiment was measured at 2.16 m in length, 1.55 m in width, and 0.75 m in height, so it provided a compact space and controlled environment for the experiment.

This was situated on a leveled platform under open field conditions to allow enough exposure of natural sunlight, rainfall, and variations in ambient temperatures.

Equipped with a drip irrigation system, the setup was composed of the main tank, a network of plastic pipes to supply water to the potted plants, and plastic pot containers that served as micro-lysimeters. The photo of the actual setup is shown in Figure 3.1 and a perspective plan of the experimental area is shown in Figure 3.2.



Figure 3.1. Photo of actual setup.

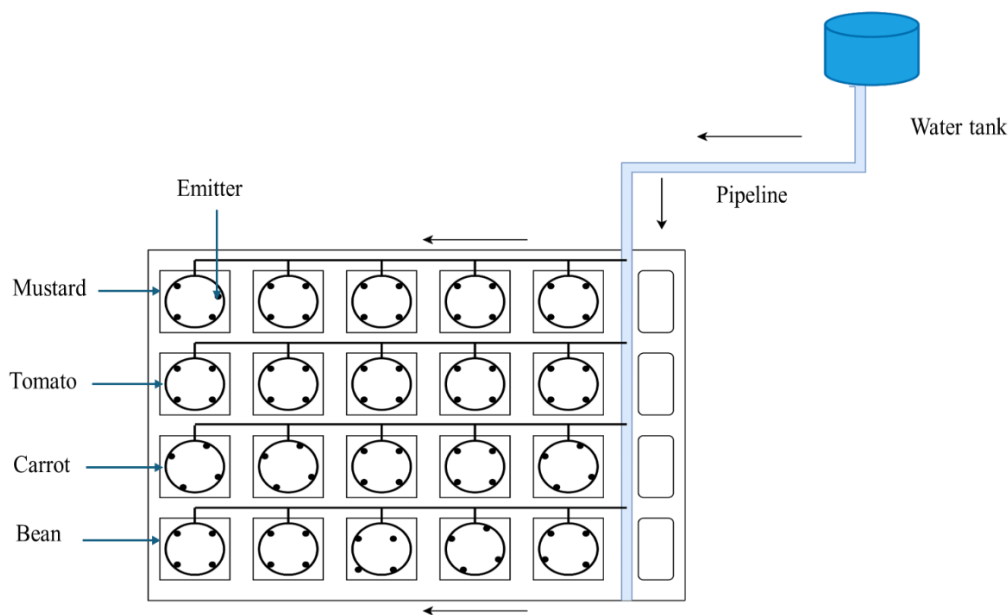


Figure 3.2. Layout of the working area.

3.3 Experimental Crops

The research included four crops comprising beans (*Phaseolus vulgaris*), carrot (*Daucus carota*), mustard (*Brassica juncea*), and tomato (*Solanum lycopersicum*). These four crops were selected based on their economic importance, differences in growth pattern and the water needs. The crops have a wide range of evapotranspiration (ET_c) characteristics and are therefore ideal for conducting tests to determine the performance of pot micro-lysimeters.

3.3.1 Beans

Bean is an edible seed or seedpod of certain leguminous plants of the family Fabaceae. Rich in protein and providing moderate amounts of iron, thiamine, and riboflavin, beans are used worldwide for cooking in either fresh or dried form.

Most varieties of bean grow either as an erect bush or as a climbing plant, but a few important kinds are of intermediate form. Many beans are summer crops that needs warm temperatures to grow; peas are an exception. Legumes are capable of nitrogen fixation and hence need less fertiliser than most plants. Maturity is typically 55–60 days from planting to harvest. As the pods mature, they turn yellow and dry up, and the beans inside change from green to their mature colour. Many beans are vines needing external support, such as "bean cages" or poles.

HISTORY

Beans in an early cultivated form were grown in Thailand from the early seventh millennium BCE, predating ceramics. Most of the kinds of beans commonly eaten today are part of the genus *Phaseolus*, which originated in the Americas.

CLIMATIC PARAMETERS

- i. Temperature: Beans are warm-season crops, preferring temperatures between 65°F (18°C) and 85°F (30°C). They cannot tolerate extreme temperatures, including:

- a. Frost: Beans are sensitive to frost, and temperatures below 50°F (10°C) can damage or kill the plants.
 - b. High temperatures: Prolonged exposure to temperatures above 90°F (32°C) can lead to reduced yields, increased disease susceptibility, and reduced quality.
- ii. Soil: Beans prefer well-draining, fertile soils with a pH range of 6.0 to 7.5. The ideal soil conditions for beans include:
 - a. Good aeration: Beans require adequate soil aeration to promote root growth and water uptake.
 - b. Moderate organic matter: Beans benefit from moderate levels of organic matter, which helps retain soil moisture and suppress pests.
 - c. Adequate nutrients: Beans require sufficient nitrogen, phosphorus, and potassium to support optimal growth.
- iii. Water: Beans require adequate moisture, especially during fruiting and pod development. However, excessive water can lead to root rot and disease susceptibility.
- iv. Sunlight: Beans require full sun to produce optimally. They need:
 - a. Direct sunlight: Beans require direct sunlight to undergo photosynthesis and produce energy.
 - b. Limited shade: Beans can tolerate some shade, but extended periods of shade can lead to reduced yields and altered plant architecture.

GROWING STAGES:

Beans can be started indoors 2-3 weeks before the last frost date.

- i. Warmth: Keep the soil temperature at 70°F to 80°F (21°C to 27°C) to promote germination and seedling growth.
- ii. High humidity: Cover the seedlings with clear plastic or a cloche to maintain high humidity and promote healthy growth.

FIELD CONDITIONS:

Various conditions in the field while planting beans contribute to its healthy growth. Factors like careful row spacing (12-18 inches), seed depth (1-2 inches) and Mulching (2-3 layers). The ideal climate and growing conditions for Beans are shown in Table 3.1.

Table 3.1. Ideal Climate and Growing Conditions for Beans

Factor	Ideal Range
Temperature	65°F (18°C) – 85°F (30°C)
Soil pH	6.0 – 7.5
Soil Aeration	Good
Soil Organic Matter	Moderate
Soil Nutrients	Adequate
Water	Adequate moisture

3.3.2 Carrot

Carrot are versatile root vegetables that survive a broad range of climatic conditions but prefer cooler temperature of 15°C to 18 °C. Carrots are typically grown as annuals, harvested at the end of the first growing season before they can flower. Carrot is a major vegetable crop in India. Haryana, Andhra Pradesh, Karnataka, Punjab and Uttar Pradesh are major carrot growing states.

The carrot seeds usually start to sprout in 14 – 21 days. As the carrot grow, keeping the soil right is a key. Carrot demands well- drained soils free from stones and heavy clumps to allow smooth elongation of the taproot. Carrots need about an inch of water per week when young, but as the roots mature, increase water to 2 inches per week.

CLIMATIC PARAMETERS:

- i. Temperature: Sowing (18-23°C)
Growing (7-23°C)
- ii. Sunlight: Carrots grow best in full sunlight, receiving at least 6 hours of direct sunlight per day. They can also perform well in partial shade, where sunlight is available for at least 4 hours.
- iii. Humidity: 60-80% humidity optimal for carrot root development and health.
- iv. Rainfall: The ideal rainfall for carrot crops is 75-100cm

GROWING STAGES:

- i. Germination: 1-3 weeks
- ii. Seedling: 3-5 weeks
- iii. Vegetative growth: 5-10 weeks
- iv. Maturation: 10weeks to harvest

SOIL INFORMATION:

- i. Soil type: Carrot best grows in Sandy loam or loamy soil.
- ii. Soil pH: It requires soil pH ranging from 6.0 to 8.0
- iii. Sowing dept: the seeds should be sown at a dept of 0.6 to 1 cm.
- iv. Soil moisture: -40 to -50 cbars.

3.3.3 Mustard

Mustard is a quick-growing crop that can be harvested within 60-70 days. It has a rosette of large light or dark green leaves and is considered a hardy leaf vegetable. It is a versatile crop that is grown for its seeds, leaves, and oil. It is a cool-season crop that grows well in areas with a mean temperature of around 10-25 °C. The ideal sowing time for mustard is from October to November in India.

CLIMATIC PARAMETERS

- i. Temperature: The temperature needed for the mustard crop is from 10 °C to 25 °C.
- ii. Sunlight: It needs a clear sky and no frost.
- iii. Humidity: High humidity, cold, and rainfall during flowering are hazardous to mustard crops.
- iv. Rainfall: It prefers moderate rainfall of around 750-1000 mm annually.

GROWING STAGES

1. Initial stage: This stage starts from seed sowing and lasts for 0-5 days. It involves seed germination, and the ideal temperature for germination is 7-30 degrees Celsius.
2. Seedling stage: This stage starts after the first true leaf appears, and it lasts for 5-15 days. Optimum temperature for this stage is 15-20 degrees Celsius.
3. Vegetative stage: This stage starts from the fifth day onwards and lasts for 20-25 days. During this stage, the plant develops more leaves and branches. The ideal temperature for this stage is 20-25 degrees Celsius.
4. Reproductive stage: This stage starts from the 30th day onwards and lasts for 30-35 days. During this stage, flowering and pod formation occur. The ideal temperature for this stage is 20-25 degrees Celsius.
5. Maturity stage: This stage starts around the 60th day and lasts for 10-12 days. The pods mature and dry out and can be harvested when they turn brown in colour.

SOIL INFORMATION

- i. Soil type: Mustard grows well in loamy soil.
- ii. Soil pH: It requires soil pH ranging from 6.0 to 7.5.
- iii. Sowing depth: Seeds are sown at a maximum depth of 6 cm for improved germination.
- iv. Soil moisture: Mustard crops grow best in well-drained soil.

3.3.4 Tomato

Tomato is a rapidly growing crop with a growing period of 90 to 150 days. It is a daylength neutral plant. The seed is generally sown in nursery plots and emergence is within 10 days. Seedlings are transplanted in the field after 25 to 35 days. Water requirements related to reference evapotranspiration (ET_o) in mm/period are given by the crop factor (K_c) for different crop development stages.

CLIMATIC PARAMETERS

- i. Temperature: Tomatoes grow well at temperatures between 70°F to 85°F (21°C to 29°C) and are sensitive to frost. Temperature below 50°F (10°C) can cause damage to the plant. Sow when soil and air temperature rise steadily.
- ii. Sunlight: It requires 6 to 8 hours of direct sunlight per day. For best growth, choose a location that receives full sun.
- iii. Humidity: Tomatoes prefer a low humidity environment because excessive levels tend to increase the risk of diseases such as fungal infections.
- iv. Rainfall: Tomatoes require low to medium rainfall.

GROWING STAGES

- i. Initial stage: 0-4 weeks after sowing, Seed germination occurs within 6-12 days
- ii. Development stage: 4-8 weeks after sowing (3-4 weeks after transplanting)
- iii. Mid stage: 8-12 weeks after sowing
- iv. Maturity stage: 12-20 weeks after sowing, depending on the variety and growing conditions

Water requirements related to reference evapotranspiration (ET_o) in mm/period are given by the crop factor (K_c) for different crop development stages, or during:

- i. the initial stage 0.4-0.5 (10 to 15 days),

- ii. the development stage 0.7-0.8 (20 to 30 days),
- iii. the mid- season stage 1.05-1.25 (30 to 40 days),
- iv. the late-season stage 0.8-0.9 (30 to 40 days), and
- v. at harvest 0.6-0.65.

SOIL INFORMATIONS

- i. Soil type: well-draining loamy soils rich in organic matter give tomatoes ideal conditions. Any addition of compost or manure contributes to better fertility and moisture of the soil.
- ii. Soil pH: Tomatoes prefer well-draining soil with a pH of 6.0 to 7.0.
- iii. Sowing dept: The ideal depth to plant tomato seeds is between 0.32cm- 0.63 cm into the soil.
- iv. Soil moisture: To ensure tomato plants are receiving the right amount of moisture, aim for a moisture meter reading between 30% and 70% on most moisture meters. Total water requirements after transplanting, of a tomato crop grown in the field for 90 to 120 days, are 400 to 600 mm, depending on the climate.

The crops selected for this study is shown in Figure 3.5.



Figure 3.3. Seeds of (A)Beans, (B)Mustard, (C)Tomato & (D)Carrot.

3.4 Experimental Pots

The experimental study of ETc crop carried out with the 20 pots served as micro-lysimeters, with 5 pots allotted for each crop, tomato, mustard, carrot and green chilli, with precision under controlled conditions. The pots were resistant, non-porous plastic, so that no leakage of water occurred, and each pot had an upper surface area of $30 \times 30 \text{ cm}^2$, a lower surface area of $25 \times 25 \text{ cm}^2$, and a height of 30 cm. The pots were arranged with an inter-pot spacing of 8-10 cm to minimize interference in water uptake and to ensure that each plant had adequate room for growth.

A small drainage hole was provided at the bottom of each pot to manage excess water. A funnel, securely glued to the base of the pot, was connected to a pipe that directed excess water into 10 litre capacity containers placed below each pot.

The pots were filled with silty clay loam soil which has been sieved to remove debris and stones. The soil in the pots was uniformly packed and filled up to a height of approximately 21 cm.

The use of pots ensured an accurate control over soil, water, and environmental factors; this significantly diminished variability and ensured a greater precision of the outcome. Furthermore, the portability of the pot enabled easy observation and adjustments, making them suitable for evapotranspiration studies on the crops.

3.5 Determination of soil type using Hydrometer

The hydrometer method is one commonly used method to accurately determine particle size distribution in a soil sample. A hydrometer is an instrument used to measure the specific gravity of the soil. Hydrometer Test is based on Stokes Law. Hydrometer test is the process by which fine-grained, clay, and Silts are graded. The hydrometer method is useful only for measuring particles with a grain diameter of 2 mm or less. Based on Stoke's Law, it is known that sand size particles (0.05 mm to 2 mm) fall from suspension rapidly. Smaller silt sized particles (0.002 mm to 0.05 mm) remain in suspension longer, but eventually fall from suspension. Clay sized particles (less than 0.002 mm) are small enough to remain in suspension indefinitely. This test is performed if the grain sizes are too small for Sieve analysis.

When a soil specimen is dispersed in water, the particles settle at different velocities, depending on their shape, size, weight, and viscosity of water. Once percent sand, silt, and clay are known for a sample, the soil can be classified by textural class using the textural triangle.

This test was conducted on 17 December, 2024 at Soil and Water Conservation Lab, NERIST.

Apparatus required in the test includes soil samples, weighing balance, magnetic stirrer, hydrometer, pipette and funnel, distilled water, 50 g oven dry soil, Sodium hexametaphosphate, thermometer, watch.

The hydrometer method for determining soil type begins with preparing a sample by sieving it through a 2 mm sieve and weighing approximately 50 g of the fine soil fraction. The soil is then mixed with a dispersing agent (e.g., sodium hexametaphosphate) and distilled water, and stirred or shaken for 10-15 minutes to ensure the particles are fully dispersed. The soil slurry is transferred into a 1000mL sedimentation cylinder, and distilled water is added to bring the total volume to exactly 1000 mL. The cylinder is shaken vigorously to ensure an even suspension. Hydrometer readings are taken at specific time intervals (5 minutes and 45 minutes) while noting the temperature of the suspension for necessary corrections. A blank test with only water and dispersing agent is also performed to correct any possible interference.

These steps allow for the calculation of the soil's particle size distribution, which is crucial for determining the soil type and classification based on its texture.

The results obtained are shown in Table 3.2.

Table 3.2. Soil Texture Analysis of Experimental Samples

Sample	% Sand	% Silt	% Clay	Types of soil
S1	91.28	6	2.72	Sandy Loam
S2	88.28	6	5.72	Sandy Loam
Average	89.78	6	4.22	Sandy Loam

3.5.1 Calculation

1. Stoke's Law:

$$V = \frac{d^2 g (\rho_p - \rho_s)}{18\eta} \quad 3.1$$

where,

V = Settling velocity, cm/s

g = Gravitational acceleration, cm/s²

d = Diameter of the particles

ρ_p = Density of the particle, g/cc

ρ_s = Density of the water, g/cc

2. Percent Finer:

The percentage of soil particles finer than each measured particle size are calculated using the formula

$$\text{Percent fine} = \frac{R \times 1000}{W} \quad 3.2$$

where R is the corrected hydrometer reading (g/L) and W is the weight of the dry soil sample (g).

This method is effective for determining the fine particle fraction of soils and classifying soil texture, particularly for soils rich in clay and silt.

3.6 Determination of soil field capacity moisture content

Soil moisture measurements are important to estimate the evapotranspiration and depth of irrigation to be applied at different stages of the crop as well as optimum irrigation scheduling.

Depending on the soil texture, soil moisture is held in soil pores by soil moisture suction or tension. The amount of moisture a soil holds at various tensions indicates the ease or difficulty with which moisture can be removed from the soil. In light soils, soil moisture can be taken up by plants by exerting less suction whereas clay soils have to apply more tension to extract the same amount of water. The clay soil can store higher amount of soil moisture as compared to sandy soil. However, release of water from a sandy soil is much easier than from the clay soil.

The principal method for soil moisture determination in common use is where the water is removed by oven dry method. Accuracy and reproducibility of water content measurements, assuming that weighing precision is consistent with desired precision of water content measurement, depend upon the drying technique and the care with which it is used.

Soil field capacity moisture content refers to the amount of soil moisture or water held in the soil after it has been thoroughly saturated and allowed to drain freely under the influence of gravity for a period of time. At this point, the soil holds the maximum amount of water that is available to plants, primarily in the micro pores, while gravitational water in the micro pores has drained away. At field capacity, the large pores are filled with air and micro pores are filled with water. It represents a critical threshold in soil-water relationships and is widely used in irrigation planning, water balance studies, and crop water requirement assessments. The soil moisture tension at field capacity varies from soil to soil and generally ranges from 1/10 to 1/3 atmosphere or bar which is equivalent to 100 to 333 cm (1 atmosphere 1 bar = 10 m) head of water. Sandy soils will hold less amount of water than the clay soils.

To determine the field capacity moisture content of the experimental soil under natural conditions, a test plot was first thoroughly saturated with water to ensure complete wetting of the soil profile. After saturation, the plot was covered with a plastic sheet to minimize water loss through evaporation, ensuring that changes in moisture content were due solely to gravitational drainage. The plot was then left to drain naturally under gravity for 72 hours. After the drainage period, five soil samples were collected from different depths within the root zone to account for vertical variation in moisture retention.

Each soil sample was immediately placed into a pre-weighed moisture box, and the combined weight of the moist soil and box was recorded. The samples were then oven-dried at 105°C for 24 hours, after which the weight of the dry soil and box was measured. The weight of the empty moisture box was subtracted from both the wet and dry measurements to obtain the true wet and dry weights of the soil alone. The moisture content on a wet basis was calculated for each sample using the formula given in Equation 3.3.

$$\text{Moisture Content (\%)} = \left(\frac{\text{Wet weight} - \text{Dry weight}}{\text{Wet weight}} \right) \times 100 \quad 3.3$$

Finally, the average of the five moisture content values was taken to represent the field capacity of the soil profile. This average was used as a benchmark for determining irrigation need and for interpreting water movement, including percolation losses, throughout the experiment.

3.7 Measurement of data

3.7.1 Climatic Data

To estimate ETo using the FAO-56 Penman-Monteith method, the following climatic and geographical data shown in Table 3.3 are required.

Table 3.3. Meteorological Parameters and Their Sources

Symbols	Parameters	Units	Source
Tmax, Tmin	Daily maximum and minimum air temperature	°C	AWS
RHmax, RHmin	Relative humidity (max and min)	%	AWS
N	Sunshine hours	MJ/m ² /day	AWS
U _z	Wind Speed	m/s	AWS
φ	latitude 27.132N	Degrees	AWS
z	Height of wind measurement = 9	Meter	AWS
Z	Elevation above sea level	Meter	GPS
R	Rainfall	mm	AWS

3.7.2 Percolation

Water was supplied to the plants in the pot using surface emitters that allowed for controlled and uniform irrigation. After irrigation, any water that was not absorbed by the soil or taken up by the plants percolated downward and was drained out of the pots through an integrated drainage system. The drained water was collected in containers placed beneath each pot. The collected water was regularly monitored and transferred to a measuring cylinder for measurement. The volume of water supplied to the plants in the pot and the water drainage from the pot were measured every time to estimate the actual water consumption.

The measurement was carried out over a continuous period starting from 20th December 2024 to 24th April 2025. During this period, excess water draining from the pots after each irrigation event was collected and recorded in millilitres (ml). It was then converted into depth by dividing the collected volume by the surface area of the pots as shown in Equation 3.4.

$$Depth(mm) = \frac{Volume (ml)}{762.5} \times 10 \quad 3.4$$

The regular monitoring of drained water helped in identifying how much water moved beyond the root zone for each crop type. If little or no water was collected after irrigation, it was assumed that the supplied water was fully retained in the soil or utilized by the plant. On the other hand, greater drainage volumes indicated a lower water retention capacity of the soil or limited uptake by the plant. Over time, these measurements offered insights into the water use efficiency and soil-water dynamics associated with each crop.

3.7.3 Cross-Sectional Area of pot

The experimental pots which served as micro-lysimeter has a height of 30 cm and the upper opening of the pot measured $30 \times 30 \text{ cm}^2$, while the bottom measured $25 \times 25 \text{ cm}^2$. To account for the varying width from top to bottom, the average cross-sectional surface area of the pot was calculated using the mean of the top and bottom surface areas, resulting in an average area of 762.5 cm^2 . This average area was used in converting the volume of collected water into depth (mm).

3.7.4 Emitter discharge

The discharge rate of the emitters was measured to ensure uniform water application and to quantify the volume of water delivered to each pot. To determine the discharge rate of the emitters, three emitters were randomly selected from the system for testing. Water was allowed to flow through each emitter, and the discharged volume was collected in a measuring cylinder over a fixed duration of 5 minutes. After the collection period, the volume of water discharged from each emitter was measured in millilitres. The volumes recorded for the three emitters were 139 ml, 133 ml, and 133 ml, respectively. The average volume discharged over 5 minutes was therefore 135 ml. This average was then used to calculate the emitter discharge rate in litres per hour (L/h) by the following calculation.

$$\text{Discharge rate} = \left(\frac{135 \text{ ml}}{5 \text{ min}} \right) \times 60 = 1620 \text{ ml/hr} = 1.62 \text{ L/hr}$$

Thus, the average emitter discharge rate was found to be 1.62 L/h, which was used as the standard for estimating water supply to each pot throughout the experiment. This measurement ensured that the emitters were functioning consistently and provided a basis for estimating the volume of water supplied to each pot during irrigation.

3.7.5 Opening and Closing time of Valves

In this experiment, the opening and closing of valves were automatically regulated using a sensor-based irrigation system. The system was designed to activate the valves based on real-time soil moisture readings. When the soil moisture content dropped below a predetermined threshold, the sensor would trigger the valve to open, allowing water to flow through the emitters. Once the moisture level reached the desired range, the sensor would automatically close the valve, thereby stopping the irrigation process. This automation enabled precise and efficient water application, eliminating the need for manual operation and reducing water wastage.

To quantify the average irrigation duration, the time for which the valve remained open during each irrigation event was recorded across multiple cycles. To monitor and assess water usage, the valve opening duration was recorded daily for each crop. These timings varied slightly depending on the crop's water demand and soil moisture retention. The recorded durations for each irrigation event were then used to calculate the average valve opening time over the experimental period. This time log was used to calculate the volume of water delivered during each irrigation event. By combining the average valve opening time with the known emitter discharge rate, the total volume of water applied per pot was determined. This volume was then converted into depth of irrigation by using the cross-sectional area of the pots. This approach

allowed the actual irrigation depth to be quantified for each event, ensuring accurate estimation of water input.

3.8 Determination of ETo

3.8.1 Data requirement and sources

The input data were obtained from an Automatic Weather Station (AWS) installed at the study location. The AWS provides continuous and reliable meteorological measurements, which were used to calculate the reference evapotranspiration (ETo) on a daily basis.

3.8.2 The FAO-56 Penman-Monteith Equation

The FAO Penman-Monteith method is recommended as the sole standard method for the definition and computation of the reference evapotranspiration. This method requires radiation, air temperature, air humidity and wind speed data which can be collected from meteorological data provided.

The FAO-56 Penman Monteith Method used in this study is written in the Equation 3.5 below.

$$ET_0 = \frac{0.408 \Delta (R_n - G) + \gamma \frac{900}{(T_{mean} + 273)} u_2 (e_z^0 - e_z)}{\Delta + \gamma (1 + 0.34 u_2)} \quad 3.5$$

where,

ET_0 = Reference evapotranspiration (mm day⁻¹)

R_n = Net radiation at the crop surface (MJ m⁻² day⁻¹)

G = Soil heat flux density (MJ m⁻² day⁻¹) assumed to be zero for 24hr calculation

T_{mean} = Mean daily air temperature at 2 m height (°C)

u_2 = Wind speed at 2 m height (m s⁻¹)

e_z^0 = Mean saturation vapour pressure (kPa)

e_z = Actual vapour pressure (kPa)

$e_z^0 - e_z$ = Saturation vapour pressure deficit (kPa)

Δ = Slope vapour pressure curve (kPa °C⁻¹)

γ = Psychrometric constant (kPa °C⁻¹)

3.8.3 Estimation of Net Radiation (R_n)

Net radiation is the difference between incoming net shortwave radiation and outgoing net longwave radiation.

$$R_n = (R_{ns} - R_{nl}) \quad 3.6$$

where,

R_{ns} = Incoming net shortwave radiation

R_{nl} = Outgoing net longwave radiation

$$R_{ns} = (1 - \alpha)R_s \quad 3.7$$

where,

R_s = short wave or solar radiation receives at earth surface on a horizontal plane that is not reflected from the surface

α = shortwave reflectance or albedo, the fraction of solar wave reflected by the surface, and its value is assumed to be 0.23 for green grass reference crop.

Solar radiation (R_s) is calculated using

$$R_s = \left(a + b \times \frac{n}{N}\right) R_a \quad 3.8$$

where,

a & b = Angstrom values ($a=0.23$, $b=0.49$)

n = Sunshine hour

N = Daylight hour

R_a = Extra-terrestrial radiation for daily periods

The daylight hours, N , is given by:

$$N = \frac{24}{\pi} \times \omega_s \quad 3.9$$

The extra-terrestrial radiation for each day of the year and for different latitudes is estimated from the solar constant, the solar declination and the time of the year by

$$R_a = \frac{24 \times (60)}{\pi} \times G_{sc} \times d_r [\omega_s \times \sin(\varphi) \sin(\delta) + \cos(\varphi) \cos(\delta) \sin(\omega_s)] \quad 3.10$$

where,

R_a = Extra-terrestrial radiation ($\text{MJ m}^{-2} \text{ day}^{-1}$)

G_{sc} = Solar constant = $0.0820 \text{ (MJ m}^{-2} \text{ min}^{-1}\text{)}$

d_r = Inverse relative distance Earth-Sun(m)

ω_s = Sunset hour angle (rad)

φ = Latitude (0.473 rad)

δ = Solar declination (rad)

Sunshine hour angle (ω_s) is calculated by using the expression

$$\omega_s = \cos^{-1}(-\tan \varphi * \tan \delta) \quad 3.11$$

The inverse relative distance Earth-Sun, dr , and the solar declination, δ , are given by:

$$dr = 1 + 0.033 \cos\left(\frac{2\pi}{365} \times D\right) \quad 3.12$$

$$\delta = 0.4039 \times \sin\left(\frac{2\pi}{365}\right) \times (284 + D) \quad 3.13$$

where,

D = The number of the day in the year

The calculation of the clear-sky radiation, R_{so} , when $n = N$, is required for computing net longwave radiation.

$$R_{so} = (a_s + b_s)R_a \quad 3.14$$

where,

R_{so} = Clear-sky solar radiation (MJ m⁻² day⁻¹)

$(a_s + b_s)$ = fraction of extra-terrestrial radiation reaching the earth on clear-sky days ($n = N$)

$a_s = 0.23$ & $b_s = 0.49$

Net longwave radiation (R_{nl}) is the rate of longwave energy emission proportional to the absolute temperature of the surface raised to the fourth power. This relation is expressed quantitatively by the Stefan-Boltzmann law. The net energy flux leaving the earth's surface is, however, less than that emitted and given by the Stefan-Boltzmann law due to the absorption and downward radiation from the sky. Water vapour, clouds, carbon dioxide and dust are absorbers and emitters of longwave radiation.

The outgoing net longwave is given by

$$R_{nl} = \left(1.35 \times \frac{R_s}{R_{so}} - 0.35\right) (0.34 - 0.14\sqrt{e_z}) \sigma \left(\frac{T_{kmin}^4 + T_{kmax}^4}{2}\right) \quad 3.15$$

Where,

R_{nl} = Net outgoing longwave radiation (MJ m⁻² day⁻¹)

σ = Stefan-Boltzmann constant (4.903 10⁻⁹ MJ K⁻⁴ m⁻² day⁻¹)

T_{kmax} = Maximum absolute temperature during the 24-hour period (K = °C + 273.16)

T_{kmin} = Minimum absolute temperature during the 24-hour period (K = °C + 273.16)

e_a = Actual vapour pressure (kPa)

R_s/R_{so} = Relative shortwave radiation (limited to ≤ 1.0)

R_s = Calculated solar radiation (MJ m⁻² day⁻¹)

R_{so} = Calculated clear-sky radiation (MJ m⁻² day⁻¹)

3.8.4 Mean Saturation Vapour Pressure (e^o_z)

Mean saturation vapour pressure refers to the average pressure exerted by water vapor in the air when the air is saturated over a given period of time. It is related to air temperature. The relationship is expressed by:

$$e_z^o = 0.6108 \times \exp\left(\frac{17.27 \times T_{mean}}{T_{mean} + 237.3}\right) \quad 3.16$$

where,

e_z^o = Mean saturation vapour pressure at the air temperature T [kPa]

T_{mean} = Mean air temperature [°C]

T_{mean} can be calculated as

$$T_{mean} = \frac{T_{min} + T_{max}}{2} \quad 3.17$$

where,

T_{min} - Mean daily min temperature (°C)

T_{max} - Mean daily maximum temperature (°C)

3.8.5 Slope of Saturation Vapour Pressure Curve (Δ)

For the calculation of evapotranspiration, the slope of the relationship between saturation vapour pressure and temperature, Δ is required. It is given by the expression

$$\Delta = \frac{4098 \times e_z^o}{(T_{mean} + 237.3)^2} \quad 3.18$$

3.8.6 Actual Vapour Pressure (e_z)

The actual vapour pressure is the vapour pressure exerted by the water in the air. When the air is not saturated, the actual vapour pressure will be lower than the saturation vapour pressure. The difference between the saturation and actual vapour pressure is called the vapour pressure deficit or saturation deficit and is an accurate indicator of the actual evaporative capacity of the air. The actual vapour pressure can also be calculated from the relative humidity. Dew point temperature when available (T_d , °C) provide the best estimate of e_z as

$$e_z = 0.6108 \exp\left(\frac{17.27 \times T_d}{T_d + 237.3}\right) \quad 3.19$$

where,

T_d = dewpoint temperature (°C)

The dewpoint temperature (T_d) is the temperature to which the air needs to be cooled to make the air saturated. The actual vapour pressure of the air is the saturation vapour pressure at the dewpoint temperature. The drier the air, the larger the difference between the air temperature and dewpoint temperature. It is given as:

$$T_d = \frac{4098.794 \times T_{mean} + 237.3(T_{mean} + 237.3) \times \ln\left(\frac{RH_{mean}}{100}\right)}{4098.794 - (T_{mean} + 237.3) \times \ln\left(\frac{RH_{mean}}{100}\right)} \quad 3.20$$

where,

RH_{mean} = Daily mean relative humidity (%)

3.8.7 Psychrometric constant (γ)

The psychrometric constant, γ , is given by:

$$\gamma = \frac{C_p \times P}{\varepsilon \times \lambda} \quad 3.21$$

Where,

γ = Psychrometric constant (kPa °C⁻¹)

P = Atmospheric pressure (kPa)

λ = Latent heat of vaporization, 2.45 (MJ kg⁻¹)

C_p = Specific heat at constant pressure, 1.013×10^{-3} (MJ kg⁻¹ °C⁻¹)

ε = Ratio molecular weight of water vapour/dry air = 0.622.

The atmospheric pressure, P , is the pressure exerted by the weight of the earth's atmosphere. Evaporation at high altitudes is promoted due to low atmospheric pressure as expressed in the psychrometric constant. The effect is, however, small and in the calculation procedures, the average value for a location is sufficient. A simplification of the ideal gas law, assuming 20°C for a standard atmosphere, can be employed to calculate P :

$$P = P_o \times \left(\frac{293 - 0.0065 \times Z}{293} \right)^{5.26} \quad 3.22$$

where,

P = Atmospheric pressure (Kpa)

P_o = Atmospheric pressure at mean sea level ($z=0$), 101.3 kPa

Z = Elevation above sea level (m), 233m for our case

The latent heat of vaporization (λ) represents the amount of energy required to convert a unit mass of water from liquid to water vapour at constant pressure and temperature. This energy drives the evaporation and transpiration processes that constitute evapotranspiration. The value of the latent heat varies as a function of temperature. At a high temperature, less energy will be required than at lower temperatures. It is obtained from the expression

$$\lambda = 2.501 - 0.002361 \times T_{mean} \quad 3.23$$

3.8.8 Wind Speed (u_2)

In the calculation of ET_o wind speed measured at a height of 2 meters above the ground is required. Wind speed data may be recorded at different heights depending on the placement of meteorological instruments. Therefore, to standardize wind speed measurements to the required 2-meter height, a logarithmic wind profile equation is applied.

The adjusted wind speed at 2 meters is calculated from a measured wind speed (u_z) at height 900 cm using the following formula

$$u_2 = u_z \frac{4.87}{\ln(67.8 \times z - 5.42)} \quad 3.24$$

where,

u_z = Measured wind speed at z m above ground surface (m s^{-1})

u_2 = Wind speed at 2 m above ground surface (m s^{-1})

z = Height of measurement above ground surface (m)

3.9 Determination of irrigation (Q)

The depth of irrigation is measured to supply sufficient water to the crops and keep the soil content to field capacity in the root zone.

The time log is calculated from the opening and closing time of the valves. The average value of the emitter discharge of various crops is considered. With the parameters mentioned, the depth of irrigation is determined using the equation

$$\text{Depth} = \frac{Q \times t}{\text{CSA of the tub}} \times \text{no. of emitter} \quad 3.25$$

where,

Q = Discharge of the emitter, L/hr

t = Time required

CSA = Cross sectional area of the tub, m^2

3.10 Determination of ETc

The ETc was based on the calculations of the water balance of the system. This was estimated daily by using the plastic pots as micro-lysimeters. In this way, water provided by irrigation (Q), stored by the substrate (W), and drained (D) was calculated daily

$$ETc = Q + R - D \quad 3.26$$

where,

Q is the discharge (water applied to plants in the experimental pot)

R is rainfall

D is the drainage

During the experiment, the soil was maintained at field capacity since the vegetable crops were continuously supplied with water using drip irrigation. Hence, the effect of stored water (W) in the soil was no longer included in Equation

3.11 Determination of Kc

Crop coefficient (K_c) is a crucial parameter in agriculture used to predict evapotranspiration (ET), which is the amount of water that crops need.

The crop coefficient concept was first introduced by Jensen (Jensen, 1968) to relate the ET observed for a specific crop to that of a well-calibrated reference crop under similar condition. The estimation of crop coefficients (K_c) was extended for variations under different climatic conditions and growth stages. Daily measurements of evaporation (Epan) and rainfall were recorded from the evaporation pan and rain gauge installed in the experimental area,

respectively. Taking into account the aforementioned parameters, Kc value is estimated using the equation

$$K_C = \frac{ETc}{ETo} \quad 3.27$$

where,

K_C = Crop coefficient

ETc = Crop Evapotranspiration

ETo = Reference evapotranspiration

Kc values vary depending on the crop type and its developmental stages with multiple values possible for a single crop throughout its growth.

The KC values of vegetable crops (Mustard, tomato, carrot and beans) were obtained and the KC curve for each crop was generated by plotting the KC with respect to the stages of growth. Comparison between KC curves of crops are obtained using the present technique with the generalized KC curve by the FAO56 Penman-Monteith (FAO56PM), the standard method for estimating ETC and KC.

4.1 Determination of Soil Textural Composition

To determine the textural class of soil, we need to know the percentage of, sand, silt and clay in the soil sample. These percentages are typically obtained through methods like the hydrometer method or sieve analysis. Once you have these percentages, the textural class is determined using a soil textural triangle shown in the Figure 4.1 below.

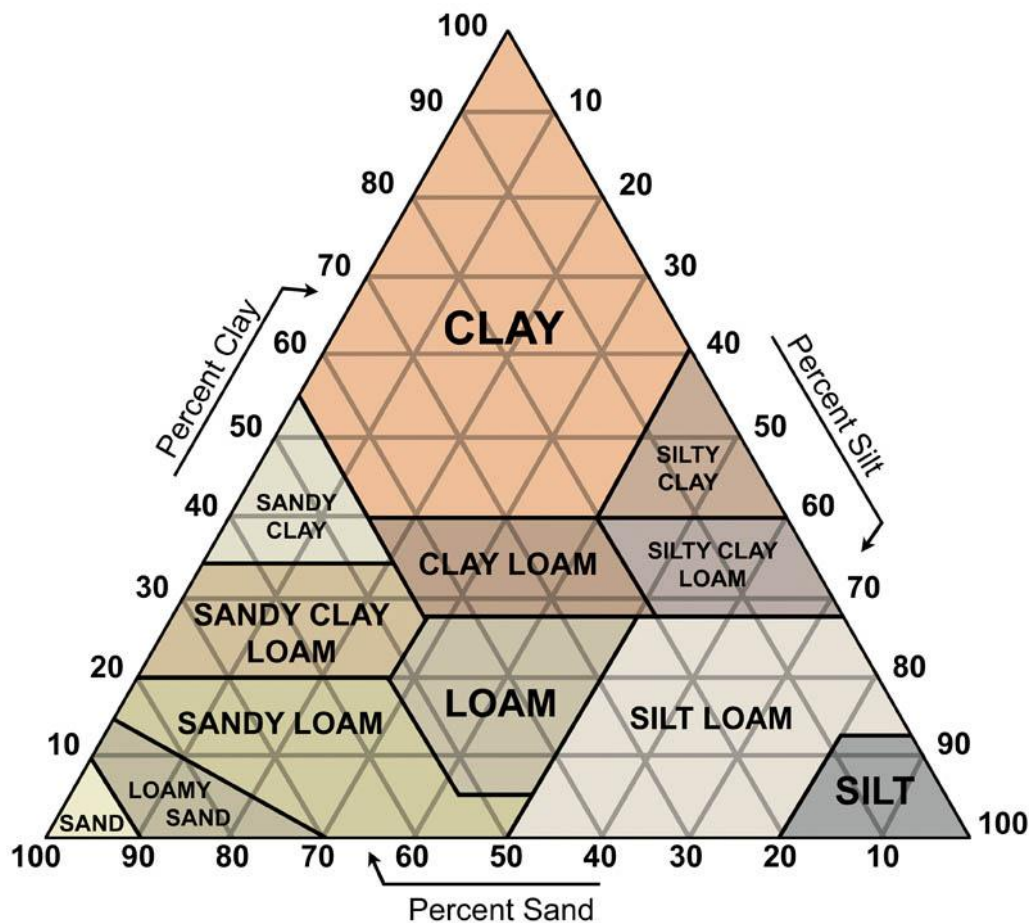


Figure 4.1. Soil Textural Triangle.

The triangle is divided into regions where each side represents one of the soil particles: sand, silt, or clay. You plot the percentage of each particle (sand, silt, and clay) on the corresponding axis of the triangle. The point where the three percentages meet will indicate the soil's textural class. For example, a soil with 70% sand, 20% silt, and 10% clay would fall into the Sandy Loam category. The textural class helps assess the soil's properties, including its water retention, drainage, and aeration characteristics, which are important for agricultural and construction purposes.

The results obtained is shown in Table 4.1.

Table 4.1. Soil Textural Analysis

Sample	% Sand	% Silt	% Clay	Soil textural class
S1	86.28	7	6.72	Loamy Sand
S2	86.28	7.7	6	Loamy Sand
Average	86.28	7.35	6.36	Loamy Sand

The soil sample analysed was found to be predominantly loamy sand in texture based on the particle size distribution obtained through the hydrometer method.

4.2 Determination of Soil Field Capacity Moisture Content

The field capacity of the experimental soil was determined through a gravimetric method using samples collected after a three-day drainage period following saturation of the test plot. A total of five soil samples were collected from different depths of the root zone to account for vertical variation in moisture retention. The soil samples were weighted before and after oven-drying at 105°C for 24 hours, and the moisture content was calculated on a wet basis.

The calculated values are shown in Table 4.2

Table 4.2 Moisture Content Analysis of Samples

Sample No.	Wet Weight (g)	Dry Weight (g)	Moisture Content (%)
1	99.81	74.55	25.32%
2	132.33	96.72	26.89%
3	118.92	86.50	27.27%
4	141.38	105.07	25.68%
5	187.58	141.26	24.68%

Average field capacity moisture content:

$$\text{Average} = \frac{25.32 + 26.89 + 27.27 + 25.68 + 24.68}{5} = 25.97\%$$

Therefore, the field capacity of the soil, calculated on a wet basis, was found to be approximately 25.97%. This value reflects the soil's ability to retain moisture after gravitational water has drained and represents the upper limit of water availability for plants.

4.3 Variation in Climatic Parameters

The climatic conditions during the experimental period play a crucial role in influencing plant growth, evapotranspiration, irrigation needs, and percolation. In this study, the simulation period extended from 20th December 2024 to 24th April 2025, during which daily and monthly climatic data were recorded and analyzed.

The variation in key climatic parameters, including rainfall, temperature (maximum, minimum, and average), relative humidity, wind speed, and sunshine hours, was assessed on both monthly and daily basis. These parameters are summarized in the following tables and figures.

A comprehensive daily climatic dataset was maintained throughout the simulation period. This is presented in Table 4.3, which records the daily variation in temperature, rainfall, wind speed, sunshine hours, and relative humidity.

Table 4.3. Day-wise record of weather parameters during the entire simulation period

Date	T _{min} (°C)	T _{max} (°C)	T _{mean} (°C)	Rainfall (mm)	Windspeed (km/hr)	Sunshine hrs (hr)	RH (%)
20-12-2024	26.5	10.5	18.5	0	1.1	4.833	77.3
21-12-2024	25.5	10.7	18.1	0	1.0	4.608	79.1
22-12-2024	25.9	11.6	18.75	0	1.1	4.096	82.4
23-12-2024	24.8	12	18.4	0	1.0	4.808	79.5
24-12-2024	24.8	10.9	17.85	0	1.1	4.860	77.5
25-12-2024	24.9	10	17.45	0	1.0	4.842	76.5
26-12-2024	24.8	9.9	17.35	0	1.0	4.831	76.0
27-12-2024	25.7	10.6	18.15	0	1.0	4.832	77.6
28-12-2024	26.3	10.7	18.5	0	1.0	4.798	76.6
29-12-2024	26.3	10.2	18.25	0	0.9	1.102	78.7
30-12-2024	26.3	13.2	19.75	0	1.1	4.309	78.7
31-12-2024	26.2	11.1	18.65	0	1.0	4.673	80.5
01-01-2025	25.5	12.2	18.85	0	1.0	4.658	79.7
02-01-2025	25.1	11.9	18.5	0	1.1	4.786	77.9
03-01-2025	25.6	11.3	18.45	0	1.1	4.826	77.6
04-01-2025	26	11.1	18.55	0	1.1	4.858	75.6
05-01-2025	26.4	11	18.7	0	1.0	4.813	77.3
06-01-2025	27.2	11.2	19.2	0	1.1	4.054	78.6
07-01-2025	25.8	12.1	18.95	0	0.9	2.624	78.8
08-01-2025	25.3	16.1	20.7	0	1.2	0.353	91.0
09-01-2025	18.8	14	16.4	0	1.0	4.286	85.9
10-01-2025	22.1	13.3	17.7	0	1.0	4.604	85.1
11-01-2025	23.4	11.9	17.65	0	1.0	3.676	79.9
12-01-2025	0	0	0	0	1.1	4.416	77.8
13-01-2025	25.4	11.1	18.25	0	1.0	2.714	78.9
14-01-2025	24.2	13.4	18.8	0	1.0	0.765	78.7
15-01-2025	0	0	0	0	1.1	4.373	78.1
16-01-2025	25.3	11.3	18.3	0	1.1	4.750	75.7
17-01-2025	25.9	11.2	18.55	0	1.0	4.249	76.9
18-01-2025	25.3	11.4	18.35	0	1.0	3.943	75.5
19-01-2025	0	0	0	0	1.0	4.263	78.9
20-01-2025	25.8	12.5	19.15	0	1.0	4.512	80.0
21-01-2025	26.1	12.7	19.4	0	1.0	4.211	78.1

22-01-2025	26.6	13.1	19.85	0	1.0	4.001	79.1
23-01-2025	26.9	13.7	20.3	0	1.0	3.427	81.2
24-01-2025	25.9	14.3	20.1	0	1.1	1.580	85.1
25-01-2025	24.2	16	20.1	0	1.0	1.789	85.9
26-01-2025	23.2	16.3	19.75	0	1.0	3.271	83.2
27-01-2025	24.6	13.9	19.25	0	1.0	3.703	79.9
28-01-2025	26.2	13	19.6	0	0.8	0.559	84.9
29-01-2025	24.1	14.4	19.25	0	0.7	0.025	86.3
30-01-2025	23	14.2	18.6	0	1.0	0.000	91.2
31-01-2025	19.6	15.9	17.75	0	1.3	0.363	89.0
01-02-2025	20.5	15.1	17.8	0	0.6	0.030	88.2
02-02-2025	21.1	15.3	18.2	0	0.9	1.039	85.2
03-02-2025	23.3	14.6	18.95	0	1.0	2.429	81.4
04-02-2025	24.8	14.7	19.75	0	1.0	2.694	81.8
05-02-2025	25.3	13	19.15	0	0.9	2.545	83.3
06-02-2025	25.4	13.7	19.55	0	0.8	0.170	88.9
07-02-2025	22.3	17.6	19.95	0	1.0	0.004	89.2
08-02-2025	22.7	15.3	19	0	1.0	2.228	84.4
09-02-2025	24.8	13.6	19.2	0	1.0	3.069	78.8
10-02-2025	26.9	12.6	19.75	0	1.0	2.518	79.2
11-02-2025	24.8	13.5	19.15	0	0.9	0.158	95.5
12-02-2025	17.1	14.9	16	0	1.2	0.313	86.1
13-02-2025	20.8	14.9	17.85	0.5	1.4	0.369	83.6
14-02-2025	19.6	14.9	17.25	23	1.3	1.892	80.4
15-02-2025	23	14.3	18.65	0	0.8	0.977	78.6
16-02-2025	23.8	15.3	19.55	0	1.0	0.491	76.6
17-02-2025	23.8	14.5	19.15	0	1.0	0.037	79.3
18-02-2025	22.9	13.4	18.15	0	1.1	0.955	78.9
19-02-2025	23.8	13.2	18.5	0	1.3	2.320	78.2
20-02-2025	24.9	13.1	19	0	1.1	2.063	73.6
21-02-2025	26.1	14.8	20.45	0	1.2	3.047	72.4
22-02-2025	26.9	13	19.95	0	1.1	0.205	86.8
23-02-2025	20.8	15.4	18.1	0	1.1	0.248	84.2
24-02-2025	0	0	0	0	1.7	0.939	58.6
25-02-2025	0	0	0	0	1.2	0.000	69.6
26-02-2025	0	0	0	0	1.2	3.920	68.2
27-02-2025	26.5	13.9	20.2	0	1.1	3.155	73.2
28-02-2025	28	13.2	20.6	0	1.0	0.846	72.0
01-03-2025	28.3	16.9	22.6	0.5	0.8	0.049	88.8
02-03-2025	19.2	16.7	17.95	0	1.3	1.528	84.3
03-03-2025	25.3	16.5	20.9	0.611	1.4	0.375	85.9
04-03-2025	23.8	17.4	20.6	0	1.1	0.020	76.3
05-03-2025	25.8	17	21.4	0	1.1	0.000	79.9
06-03-2025	23.4	17.6	20.5	0	1.5	0.000	75.6
07-03-2025	25.3	15.7	20.5	0	1.2	0.000	73.7
08-03-2025	27.4	13.9	20.65	0	1.0	0.000	70.2
09-03-2025	25.9	17.8	21.85	0	1.2	0.000	71.3
10-03-2025	26.9	16.3	21.6	0	1.1	0.000	73.8
11-03-2025	29	14.5	21.75	0	1.0	0.000	75.3
12-03-2025	28.2	15.9	22.05	0	1.0	0.000	86.3

13-03-2025	20.3	18	19.15	0	1.0	0.000	95.3
14-03-2025	18.9	16.7	17.8	0	1.2	0.000	90.8
15-03-2025	20.9	16.6	18.75	0	1.0	0.000	89.5
16-03-2025	25.1	17.6	21.35	0.5	0.8	0.000	97.0
17-03-2025	0	0	0	0.7	1.0	0.983	88.4
18-03-2025	27.5	18.2	22.85	0.5	2.3	2.754	68.0
19-03-2025	31.6	16.1	23.85	0.5	1.3	2.496	72.2
20-03-2025	28.9	14.9	21.9	0.5	0.8	0.187	82.4
21-03-2025	24	16.9	20.45	0.5	1.0	0.119	79.9
22-03-2025	24.7	17.9	21.3	0.5	0.8	0.240	81.9
23-03-2025	24.6	18.5	21.55	0.5	1.4	1.562	77.9
24-03-2025	27.9	18.1	23	0.5	1.0	1.246	75.9
25-03-2025	29.8	17.2	23.5	2.5	1.3	3.113	69.9
26-03-2025	31	15.4	23.2	2	1.3	2.894	71.6
27-03-2025	30.8	16.3	23.55	2	1.3	2.322	74.3
28-03-2025	30.9	17.8	24.35	1.5	1.1	0.148	75.1
29-03-2025	28.9	21.1	25	1	0.8	0.000	91.2
30-03-2025	24.4	20.5	22.45	1	1.1	0.236	81.9
31-03-2025	28.9	19.4	24.15	0.5	1.2	2.079	74.5
01-04-2025	30.3	17.6	23.95	0.5	1.0	0.238	76.1
02-04-2025	0	0	0	0.5	1.5	1.078	72.9
03-04-2025	29.6	21.1	25.35	0	1.4	0.340	75.3
04-04-2025	30	20.9	25.45	0	1.2	0.709	74.3
05-04-2025	29.6	20.1	24.85	368.5	1.3	0.644	72.0
06-04-2025	31.4	17.7	24.55	3.5	1.4	1.482	78.9
07-04-2025	28.4	19.1	23.75	1.5	1.1	0.556	77.4
08-04-2025	30.3	19.2	24.75	1	1.2	1.855	73.8
09-04-2025	32.5	19.4	25.95	1.5	1.1	0.121	76.7
10-04-2025	31	21.3	26.15	1	1.1	0.619	76.9
11-04-2025	31.8	21.5	26.65	1	0.9	0.103	83.3
12-04-2025	27.4	21.7	24.55	0.5	1.3	0.927	76.2
13-04-2025	30.6	20.4	25.5	0.5	1.3	1.506	79.1
14-04-2025	31.2	20.7	25.95	0	1.1	1.770	74.4
15-04-2025	32.1	19	25.55	0	1.0	0.293	75.0
16-04-2025	32.1	20.9	26.5	0.5	1.7	0.792	74.0
17-04-2025	30.8	20.7	25.75	9	1.2	0.380	84.0
18-04-2025	29.4	20.8	25.1	2	1.1	1.682	79.4
19-04-2025	30.6	20	25.3	1	1.1	1.211	79.0
20-04-2025	30.8	21.6	26.2	1.5	1.4	0.083	86.7
21-04-2025	27.6	21.7	24.65	1	0.8	0.010	82.2
22-04-2025	31	21.4	26.2	1.5	1.2	0.001	85.8
23-04-2025	28.8	23	25.9	2	1.3	0.000	95.2
24-04-2025	28.9	22.2	25.55	0.5	1.3	0.000	97.1

Figure 4.2 shows the daily rainfall distribution during the simulation period. No rainfall occurred in December and January, while light and scattered rainfall events were recorded in February and March. A sharp increase in rainfall was observed in April, with a few heavy showers. These variations influenced soil moisture levels and reduced the need for irrigation in the later stages of the experiment.

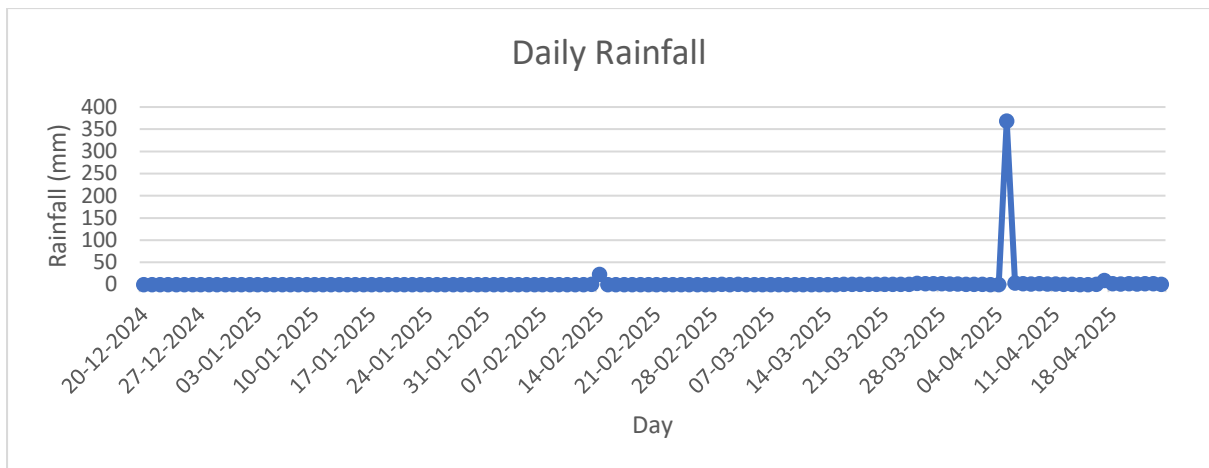


Figure 4.2 Daily Variation in Rainfall (mm) during the Simulation Period.

Table 4.4 shows the monthly variation in rainfall (mm), highlighting very low rainfall during December to March, with a sharp increase in April, while Figure 4.3 illustrates the monthly rainfall trend.

Table 4.4. Variation in Rainfall (mm) for the simulation period

Monthly Sum	
Monthly sum	Rainfall
December (2024)	0
January (2025)	0
February (2025)	23.5
March (2025)	5.8
April (2025)	399

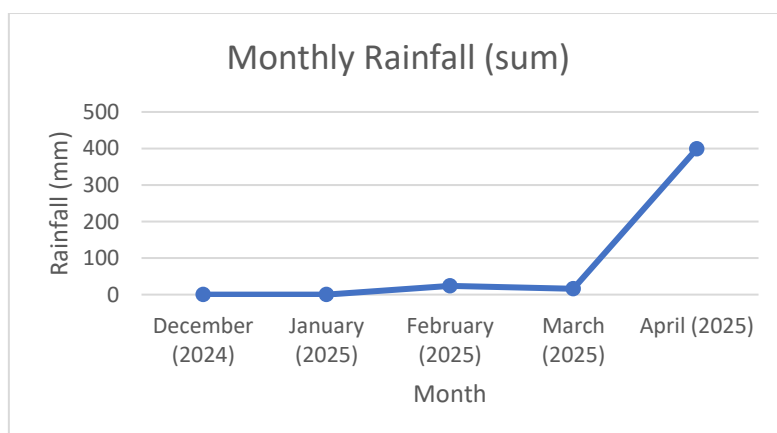


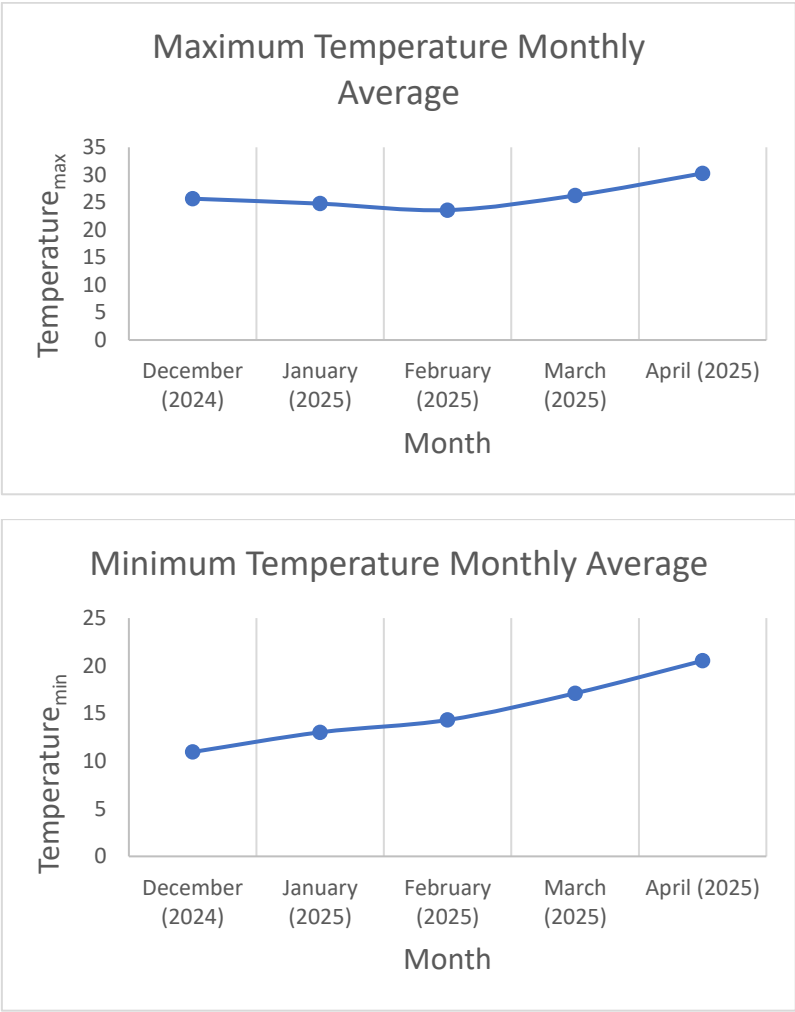
Figure 4.3 Monthly Variation in Rainfall (mm) during the Simulation Period.

Table 4.5 presents the monthly variation in temperature, including maximum, minimum, and average values, reflecting seasonal warming from December to April.

Table 4.5. Variation in Temperature (°C) for the simulation period

Monthly Average			
Month	T _{max}	T _{min}	T _{average}
December (2024)	25.66	10.95	18.31
January (2025)	24.76	13.02	18.89
February (2025)	23.59	14.31	18.95
March (2025)	25.34	16.77	21.06
April (2025)	30.27	20.52	25.39

Figure 4.4 shows the monthly trends in maximum, minimum, and average temperatures.



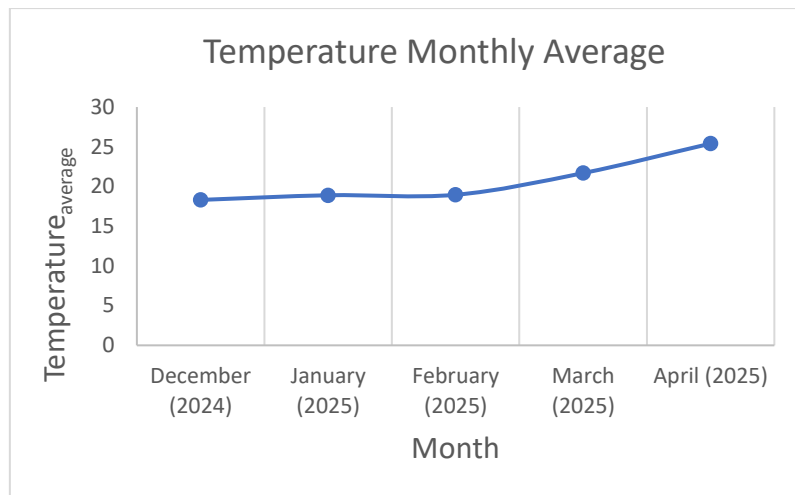
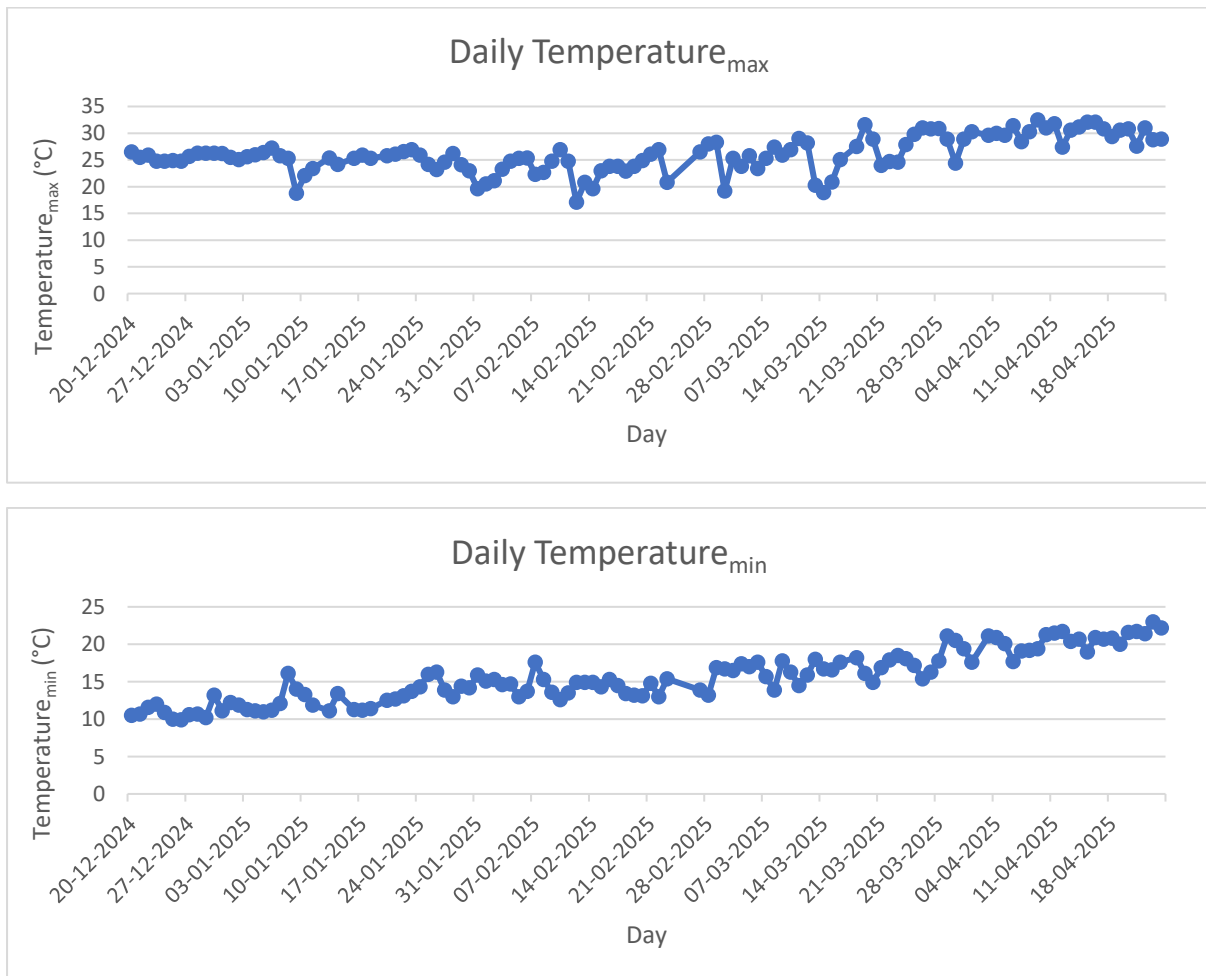


Figure 4.4. Monthly Variation in Maximum, Minimum, and Average Temperature (°C).

Figure 4.5 illustrates the day-to-day fluctuations in temperature, with a gradual increase from December 2024 to April 2025.



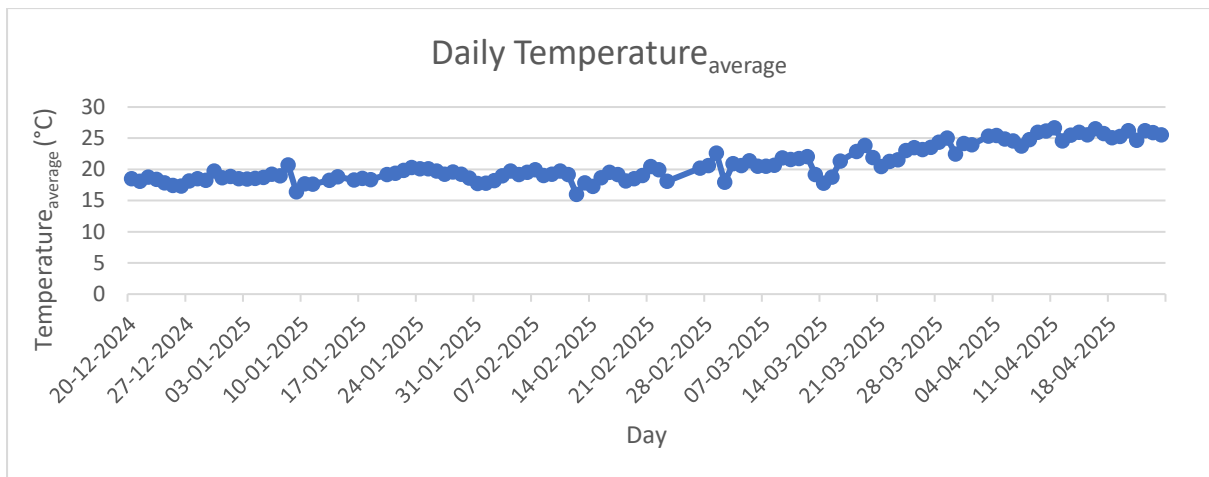


Figure 4.5. Daily Variation in Maximum, Minimum, and Average Temperature (°C).

Table 4.6 displays the monthly average relative humidity (%), showing relatively high humidity throughout the period while, Figure 4.6 shows the monthly average relative humidity during the simulation period and Figure 4.7 illustrates the daily variation in relative humidity.

Table 4.6. Variation in Relative Humidity (%) for the simulation period

Monthly Average	
Month	Relative Humidity
December (2024)	78.4
January (2025)	81.0
February (2025)	79.9
March (2025)	80.9
April (2025)	79.41

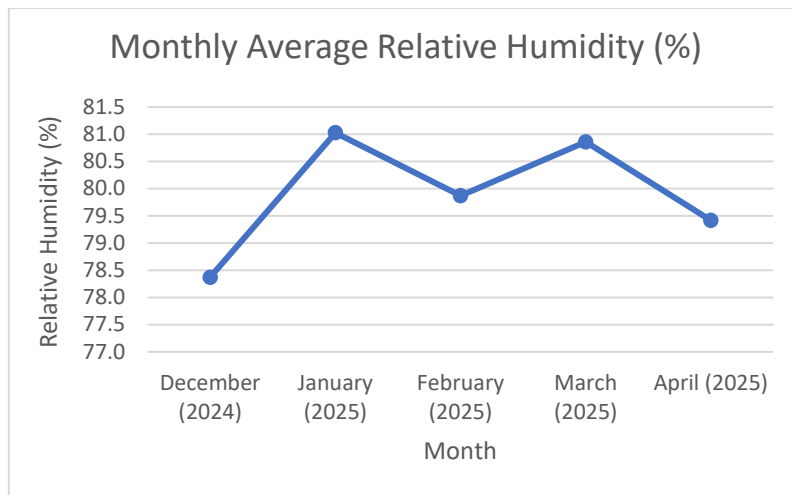


Figure 4.6. Monthly Average Relative Humidity (%) during the Simulation Period.

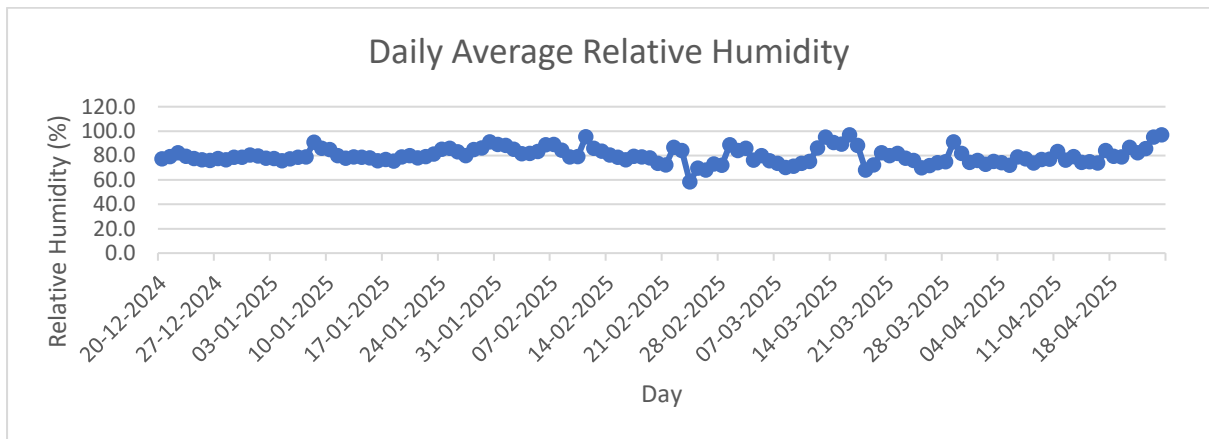


Figure 4.7. Daily Average Relative Humidity (%) during the Simulation Period.

Table 4.7 provides the monthly variation in wind speed (km/hr), indicating relatively stable and low wind conditions. The monthly and daily average variation in wind speed is shown in Figure 4.8 and Figure 4.9.

Table 4.7. Variation in Wind speed (km/hr) for the simulation period

Monthly Average	
Month	Wind Speed
December	1.0
January	1.0
February	1.1
March	1.1

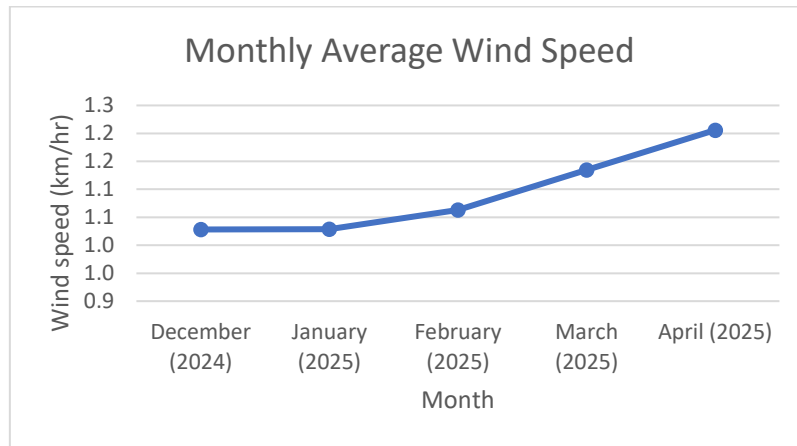


Figure 4.8. Monthly average variation in wind speed.

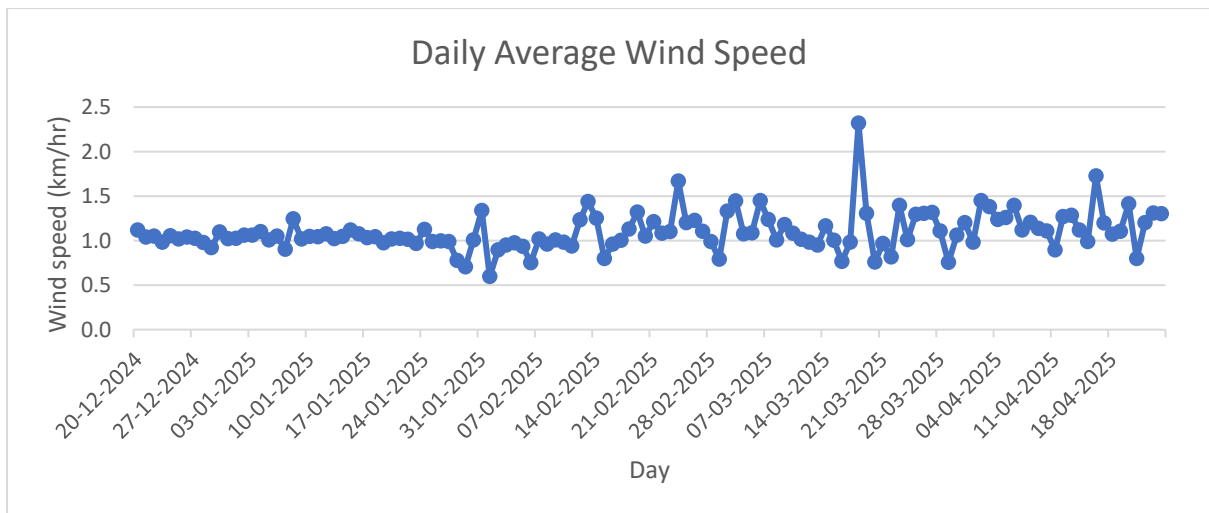


Figure 4.9. Daily average variation in wind speed.

The monthly average sunshine hours recorded during the simulation period is shown in Table 4.8. It is also represented graphically in Figure 4.10 whereas it's daily variation in sunshine hours is illustrated in Figure 4.11.

Table 4.8. Variation in Sunshine hours (hr) for the simulation period

Monthly Average	
Month	Sunshine hours
December	4.4

January	3.2
February	1.4
March	1.1
April (2025)	0.7

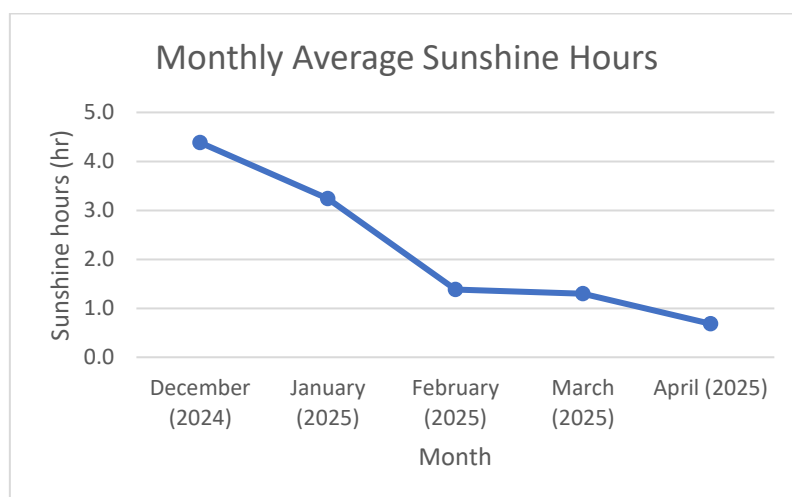


Figure 4.10. Monthly Average Sunshine Hours during the Simulation Period.

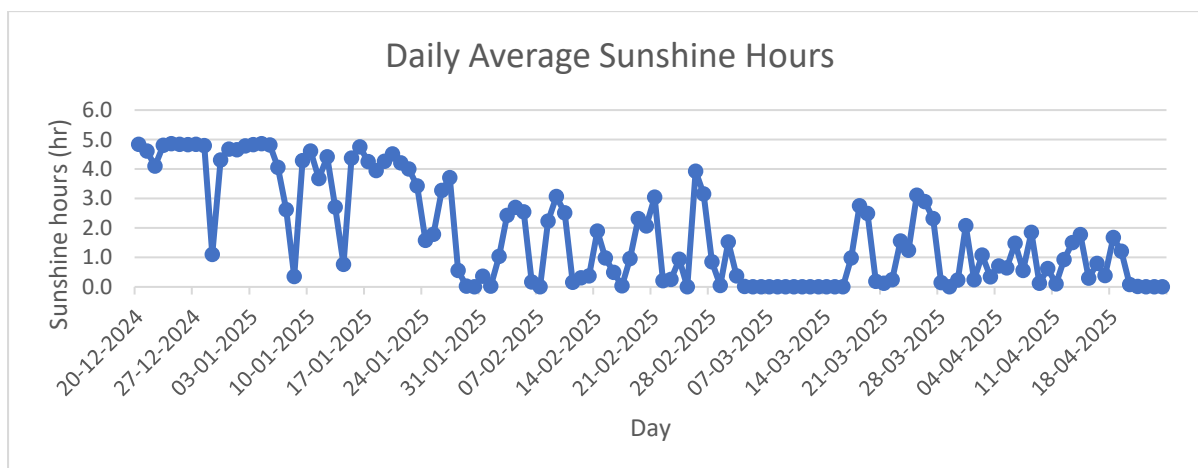


Figure 4.11. Daily Average Sunshine Hours during the Simulation Period.

4.4 Determination of ETo

The report is to analyze the variation of ETo based on observed data to understand its seasonal patterns for period from November 2 ,2024 to march 24, 2025. The data for this analysis was gathered from AWS, and daily ETo values are calculated for a period of five months. The values were averaged for each month to create a monthly dataset. Using Microsoft Excel, the monthly and daily variation of ETo was analyzed.

The monthly variation of ETo is presented in Table 4.9 and illustrated in Figure 4.12, which plots ETo values against the corresponding monthly period.

Table 4.9. Monthly ETo

Months	ETo
Nov	1.77
Dec	1.43
Jan	1.43
Feb	1.52
Mar	1.69

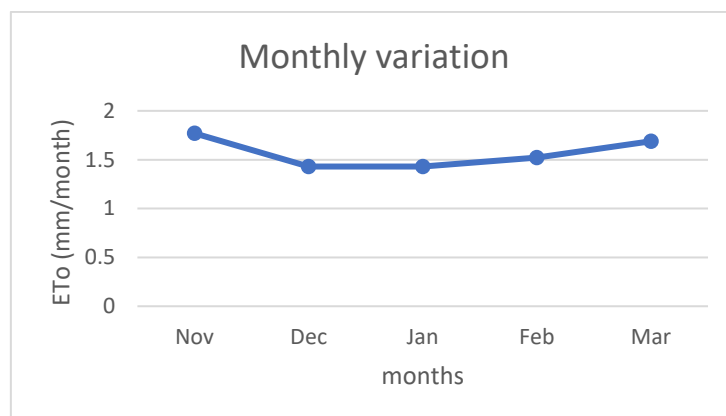


Figure 4.12. Monthly variation of ETo from Nov2024 to March2025.

Daily variation of ETo is presented in Figure 4.13 showing fluctuations in evapotranspiration rates throughout the observation period.

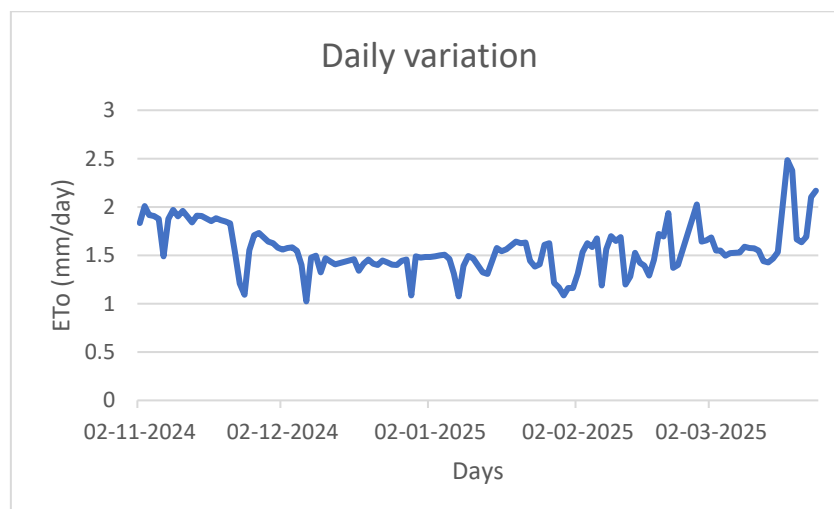


Figure 4.13. Daily variation of ETo from 2ndNov, 2024 to 24thMarch, 2025.

The required parameters derived from the input data have been calculated and are presented in the Table 4.10 below.

Table 4.10. Estimation of Eto

Date	Tmean	Omega	D	Delta	N	dr	Ra	Rs	Rns	Rso	Td
20-12-2024	18.50	1.35	355.00	-0.41	10.29	1.03	21.43	11.50	8.86	15.43	14.41
21-12-2024	18.10	1.35	356.00	-0.41	10.29	1.03	21.43	11.20	8.62	15.43	14.41
22-12-2024	18.75	1.35	357.00	-0.41	10.29	1.03	21.44	10.51	8.09	15.44	15.71
23-12-2024	18.40	1.35	358.00	-0.41	10.29	1.03	21.45	11.48	8.84	15.44	14.78
24-12-2024	17.85	1.35	359.00	-0.41	10.29	1.03	21.47	11.56	8.90	15.46	13.86
25-12-2024	17.45	1.35	360.00	-0.41	10.30	1.03	21.49	11.54	8.89	15.47	13.27
26-12-2024	17.35	1.35	361.00	-0.41	10.30	1.03	21.51	11.54	8.89	15.49	13.04
27-12-2024	18.15	1.35	362.00	-0.41	10.30	1.03	21.54	11.55	8.90	15.51	14.16
28-12-2024	18.50	1.35	363.00	-0.41	10.31	1.03	21.57	11.52	8.87	15.53	14.34
29-12-2024	18.25	1.35	364.00	-0.40	10.31	1.03	21.61	6.48	4.99	15.56	14.24
30-12-2024	19.75	1.35	365.00	-0.40	10.32	1.03	21.65	10.89	8.38	15.59	15.89
31-12-2024	18.65	1.35	366.00	-0.40	10.32	1.03	21.69	11.40	8.78	15.62	15.24
01-01-2025	18.85	1.35	1.00	-0.40	10.33	1.03	21.70	11.39	8.77	15.62	15.24
02-01-2025	18.50	1.35	2.00	-0.40	10.33	1.03	21.75	11.58	8.92	15.66	14.55
03-01-2025	18.45	1.35	3.00	-0.40	10.34	1.03	21.80	11.66	8.98	15.70	14.50
04-01-2025	18.55	1.35	4.00	-0.40	10.35	1.03	21.86	11.73	9.03	15.74	14.17
05-01-2025	18.70	1.36	5.00	-0.40	10.36	1.03	21.92	11.70	9.01	15.78	14.61
06-01-2025	19.20	1.36	6.00	-0.39	10.36	1.03	21.98	10.67	8.22	15.83	15.39
07-01-2025	18.95	1.36	7.00	-0.39	10.37	1.03	22.05	8.93	6.88	15.88	15.16
08-01-2025	20.70	1.36	8.00	-0.39	10.38	1.03	22.12	5.64	4.34	15.93	19.17
09-01-2025	16.40	1.36	9.00	-0.39	10.39	1.03	22.20	11.09	8.54	15.98	14.03
10-01-2025	17.70	1.36	10.00	-0.39	10.40	1.03	22.28	11.57	8.91	16.04	15.17
11-01-2025	17.65	1.36	11.00	-0.38	10.42	1.03	22.36	11.33	8.72	16.10	14.13
13-01-2025	18.25	1.37	13.00	-0.38	10.44	1.03	22.54	9.01	6.94	16.23	14.50
14-01-2025	18.80	1.37	14.00	-0.38	10.45	1.03	22.63	8.44	6.50	16.30	15.08
16-01-2025	18.30	1.37	16.00	-0.37	10.48	1.03	22.83	12.01	9.25	16.44	13.99
17-01-2025	18.55	1.37	17.00	-0.37	10.49	1.03	22.94	11.34	8.73	16.52	14.47
18-01-2025	18.35	1.38	18.00	-0.36	10.51	1.03	23.05	11.66	8.98	16.59	14.11
20-01-2025	19.15	1.38	20.00	-0.36	10.54	1.03	23.27	11.86	9.13	16.76	15.63
21-01-2025	19.40	1.38	21.00	-0.35	10.56	1.03	23.39	11.48	8.84	16.84	15.48
22-01-2025	19.85	1.38	22.00	-0.35	10.57	1.03	23.51	11.22	8.64	16.93	16.12
23-01-2025	20.30	1.39	23.00	-0.35	10.59	1.03	23.64	8.93	6.88	17.02	18.31
24-01-2025	20.10	1.39	24.00	-0.34	10.61	1.03	23.76	8.24	6.35	17.11	17.52
25-01-2025	20.20	1.39	25.00	-0.34	10.62	1.03	23.89	8.45	6.51	17.20	17.78
26-01-2025	19.75	1.39	26.00	-0.33	10.64	1.03	24.03	10.64	8.19	17.30	16.80
27-01-2025	19.25	1.40	27.00	-0.33	10.66	1.03	24.16	11.04	8.50	17.40	15.65
28-01-2025	19.60	1.40	28.00	-0.32	10.68	1.03	24.30	6.66	5.13	17.50	16.96
29-01-2025	19.25	1.40	29.00	-0.32	10.70	1.03	24.44	6.28	4.84	17.60	16.88
30-01-2025	18.60	1.40	30.00	-0.32	10.72	1.03	24.59	5.66	4.36	17.70	17.17
31-01-2025	17.75	1.41	31.00	-0.31	10.74	1.03	24.74	6.45	4.96	17.81	15.89

01-02-2025	17.80	1.41	32.00	-0.31	10.76	1.03	24.89	6.55	5.04	17.92	15.81
02-02-2025	18.20	1.41	33.00	-0.30	10.78	1.03	25.04	7.79	6.00	18.03	15.66
03-02-2025	18.95	1.41	34.00	-0.30	10.80	1.03	25.19	9.71	7.48	18.14	15.69
04-02-2025	19.75	1.42	35.00	-0.29	10.82	1.03	25.35	10.20	7.85	18.25	16.59
05-02-2025	19.15	1.42	36.00	-0.29	10.84	1.03	25.51	10.01	7.71	18.37	16.27
06-02-2025	19.55	1.42	37.00	-0.28	10.86	1.03	25.67	10.64	8.19	18.48	17.67
07-02-2025	19.95	1.43	38.00	-0.28	10.89	1.03	25.83	5.95	4.58	18.60	18.13
08-02-2025	19.00	1.43	39.00	-0.27	10.91	1.03	26.00	9.65	7.43	18.72	16.28
09-02-2025	19.20	1.43	40.00	-0.27	10.93	1.03	26.17	10.82	8.33	18.84	15.46
10-02-2025	19.75	1.43	41.00	-0.26	10.95	1.03	26.33	10.01	7.71	18.96	16.06
11-02-2025	19.15	1.44	42.00	-0.26	10.98	1.03	26.51	10.60	8.16	19.08	18.42
12-02-2025	16.00	1.44	43.00	-0.25	11.00	1.02	26.68	6.73	5.18	19.21	13.66
13-02-2025	17.85	1.44	44.00	-0.24	11.02	1.02	26.85	6.80	5.23	19.33	15.09
14-02-2025	17.25	1.45	45.00	-0.24	11.05	1.02	27.03	9.42	7.25	19.46	13.82
15-02-2025	18.65	1.45	46.00	-0.23	11.07	1.02	27.20	8.02	6.18	19.59	14.85
16-02-2025	19.55	1.45	47.00	-0.23	11.09	1.02	27.38	7.25	5.58	19.72	15.31
17-02-2025	19.15	1.46	48.00	-0.22	11.12	1.02	27.56	6.42	4.94	19.85	15.52
18-02-2025	18.15	1.46	49.00	-0.22	11.14	1.02	27.74	8.13	6.26	19.98	14.42
19-02-2025	18.50	1.46	50.00	-0.21	11.17	1.02	27.93	10.44	8.04	20.11	14.62
20-02-2025	19.00	1.47	51.00	-0.20	11.19	1.02	28.11	10.05	7.74	20.24	14.20
21-02-2025	20.45	1.47	52.00	-0.20	11.22	1.02	28.29	11.53	8.88	20.37	15.27
22-02-2025	19.95	1.47	53.00	-0.19	11.24	1.02	28.48	6.93	5.34	20.50	17.70
23-02-2025	18.10	1.48	54.00	-0.19	11.27	1.02	28.66	7.21	5.55	20.64	13.21
27-02-2025	20.20	1.49	58.00	-0.16	11.37	1.02	29.41	12.09	9.31	21.17	15.23
28-02-2025	20.60	1.49	59.00	-0.15	11.40	1.02	29.60	8.53	6.57	21.31	15.32
01-03-2025	22.60	1.50	60.00	-0.15	11.43	1.02	29.78	8.35	6.43	21.44	20.72
02-03-2025	17.95	1.50	61.00	-0.14	11.45	1.02	29.97	9.83	7.57	21.58	15.29
03-03-2025	20.90	1.50	62.00	-0.13	11.48	1.02	30.16	7.71	5.94	21.71	18.42
04-03-2025	20.60	1.51	63.00	-0.13	11.51	1.02	30.35	7.60	5.86	21.85	16.44
05-03-2025	21.40	1.51	64.00	-0.12	11.53	1.02	30.53	7.02	5.41	21.98	17.77
06-03-2025	20.50	1.51	65.00	-0.11	11.56	1.01	30.72	7.07	5.44	22.12	16.06
07-03-2025	20.50	1.52	66.00	-0.11	11.59	1.01	30.91	7.11	5.47	22.25	15.55
08-03-2025	20.65	1.52	67.00	-0.10	11.61	1.01	31.09	7.15	5.51	22.39	15.03
09-03-2025	21.85	1.52	68.00	-0.09	11.64	1.01	31.28	7.19	5.54	22.52	16.46
10-03-2025	21.60	1.53	69.00	-0.09	11.67	1.01	31.47	7.24	5.57	22.66	16.72
11-03-2025	21.75	1.53	70.00	-0.08	11.69	1.01	31.65	7.28	5.61	22.79	17.17
12-03-2025	22.05	1.53	71.00	-0.07	11.72	1.01	31.83	7.32	5.64	22.92	19.68
13-03-2025	19.15	1.54	72.00	-0.06	11.75	1.01	32.02	7.36	5.67	23.05	18.41
14-03-2025	17.80	1.54	73.00	-0.06	11.78	1.01	32.20	7.41	5.70	23.18	16.23
15-03-2025	18.75	1.55	74.00	-0.05	11.80	1.01	32.38	7.45	5.74	23.31	17.01
16-03-2025	21.35	1.55	75.00	-0.04	11.83	1.01	32.56	7.49	5.77	23.44	20.85
18-03-2025	22.85	1.56	77.00	-0.03	11.89	1.01	32.92	12.55	9.67	23.70	16.55
19-03-2025	23.85	1.56	78.00	-0.02	11.91	1.01	33.10	12.14	9.35	23.83	18.57
20-03-2025	21.90	1.56	79.00	-0.02	11.94	1.01	33.27	7.99	6.15	23.96	18.82
21-03-2025	20.45	1.57	80.00	-0.01	11.97	1.01	33.45	7.94	6.11	24.08	16.83
22-03-2025	21.30	1.57	81.00	0.00	12.00	1.01	33.62	8.23	6.34	24.21	18.12
23-03-2025	21.55	1.57	82.00	0.01	12.02	1.01	33.79	10.81	8.32	24.33	17.28
24-03-2025	23.00	1.58	83.00	0.01	12.05	1.01	33.96	10.97	8.44	24.45	18.52
25-03-2025	23.5	1.58	84	0.02	12.08	1	34.13	13.29	10.24	24.57	17.71
26-03-2025	23.2	1.58	85	0.03	12.1	1	34.29	13.24	10.2	24.69	17.81
27-03-2025	23.55	1.59	86	0.03	12.13	1	34.46	12.23	9.42	24.81	18.71
28-03-2025	24.35	1.59	87	0.04	12.16	1	34.62	8.29	6.39	24.93	19.65

29-03-2025	25	1.6	88	0.05	12.19	1	34.78	8.01	6.17	25.04	23.46
30-03-2025	22.45	1.6	89	0.05	12.21	1	34.94	8.57	6.6	25.16	19.2
31-03-2025	24.15	1.6	90	0.06	12.24	1	35.1	11.97	9.21	25.27	19.33
01-04-2025	23.95	1.61	91	0.07	12.27	1	35.25	8.91	6.86	25.38	19.49
03-04-2025	25.35	1.61	93	0.08	12.32	1	35.56	8.86	6.82	25.6	20.66
04-04-2025	25.45	1.62	94	0.09	12.35	1	35.71	9.63	7.42	25.71	20.53
05-04-2025	24.85	1.62	95	0.1	12.38	1	35.86	9.54	7.34	25.82	19.46
06-04-2025	24.55	1.62	96	0.1	12.41	1	36	11.26	8.67	25.92	20.65
07-04-2025	23.75	1.63	97	0.11	12.43	1	36.14	9.58	7.38	26.02	19.57
08-04-2025	24.75	1.63	98	0.12	12.46	1	36.29	11.88	9.14	26.13	19.77
09-04-2025	25.95	1.63	99	0.12	12.49	1	36.42	12.53	9.65	26.23	21.54
10-04-2025	26.15	1.64	100	0.13	12.51	1	36.56	9.83	7.57	26.32	21.72
11-04-2025	26.65	1.64	101	0.14	12.54	0.99	36.69	8.68	6.68	26.42	23.66
12-04-2025	24.55	1.64	102	0.14	12.57	0.99	36.83	10.25	7.89	26.52	20.09
13-04-2025	25.5	1.65	103	0.15	12.59	0.99	36.96	11.39	8.77	26.61	21.6
14-04-2025	25.95	1.65	104	0.16	12.62	0.99	37.08	12.13	9.34	26.7	21.04
15-04-2025	25.55	1.66	105	0.16	12.64	0.99	37.21	9.12	7.02	26.79	20.8
16-04-2025	26.5	1.66	106	0.17	12.67	0.99	37.33	10.2	7.85	26.88	21.49
17-04-2025	25.75	1.66	107	0.18	12.7	0.99	37.45	9.43	7.26	26.97	22.84
18-04-2025	25.1	1.67	108	0.18	12.72	0.99	37.57	12.08	9.3	27.05	21.27
19-04-2025	25.3	1.67	109	0.19	12.75	0.99	37.69	11.14	8.58	27.13	21.39
20-04-2025	26.2	1.67	110	0.19	12.77	0.99	37.8	11.57	8.91	27.21	23.8
21-04-2025	24.65	1.68	111	0.2	12.8	0.99	37.91	8.77	6.75	27.29	21.42
22-04-2025	26.2	1.68	112	0.21	12.82	0.99	38.02	8.79	6.77	27.37	23.63
23-04-2025	25.9	1.68	113	0.21	12.85	0.99	38.12	8.77	6.75	27.45	25.08
24-04-2025	25.55	1.68	114	0.22	12.87	0.99	38.23	8.79	6.77	27.52	25.05

Date	ez	Rnl	Rn	E _z ^o , Tmean	Δ	lamda	P	gamma	U2, km/hr	U2, m/s	ETo
20-12-2024	1.64	3.84	5.02	2.13	0.13	2.46	98.58	0.07	0.86	0.24	1.46
21-12-2024	1.64	3.66	4.97	2.08	0.13	2.46	98.58	0.07	0.79	0.22	1.41
22-12-2024	1.79	3.18	4.91	2.16	0.14	2.46	98.58	0.07	0.79	0.22	1.40
23-12-2024	1.68	3.76	5.08	2.12	0.13	2.46	98.58	0.07	0.74	0.21	1.45
24-12-2024	1.58	3.89	5.01	2.05	0.13	2.46	98.58	0.07	0.80	0.22	1.43
25-12-2024	1.52	3.94	4.95	1.99	0.13	2.46	98.58	0.07	0.77	0.22	1.40
26-12-2024	1.50	3.95	4.93	1.98	0.13	2.46	98.58	0.07	0.79	0.22	1.40
27-12-2024	1.62	3.85	5.05	2.08	0.13	2.46	98.58	0.07	0.78	0.22	1.44
28-12-2024	1.64	3.81	5.06	2.13	0.13	2.46	98.58	0.07	0.75	0.21	1.46
29-12-2024	1.62	1.26	3.73	2.10	0.13	2.46	98.58	0.07	0.71	0.20	1.09
30-12-2024	1.81	3.33	5.05	2.30	0.14	2.46	98.58	0.07	0.83	0.23	1.49
31-12-2024	1.73	3.61	5.17	2.15	0.13	2.46	98.58	0.07	0.78	0.22	1.48
01-01-2025	1.73	3.61	5.16	2.18	0.14	2.46	98.58	0.07	0.78	0.22	1.48
02-01-2025	1.66	3.77	5.16	2.13	0.13	2.46	98.58	0.07	0.81	0.23	1.48
03-01-2025	1.65	3.80	5.18	2.12	0.13	2.46	98.58	0.07	0.81	0.23	1.49
04-01-2025	1.62	3.87	5.17	2.14	0.13	2.46	98.58	0.07	0.84	0.23	1.50
05-01-2025	1.66	3.78	5.23	2.16	0.14	2.46	98.58	0.07	0.76	0.21	1.51
06-01-2025	1.75	3.19	5.03	2.23	0.14	2.46	98.58	0.07	0.80	0.22	1.46
07-01-2025	1.72	2.35	4.53	2.19	0.14	2.46	98.58	0.07	0.68	0.19	1.31
08-01-2025	2.22	0.65	3.70	2.44	0.15	2.45	98.58	0.07	0.95	0.27	1.08

09-01-2025	1.60	3.36	5.17	1.87	0.12	2.46	98.58	0.07	0.77	0.22	1.39
10-01-2025	1.72	3.49	5.41	2.03	0.13	2.46	98.58	0.07	0.81	0.22	1.49
11-01-2025	1.61	3.50	5.23	2.02	0.13	2.46	98.58	0.07	0.80	0.22	1.47
13-01-2025	1.65	2.33	4.61	2.10	0.13	2.46	98.58	0.07	0.77	0.21	1.32
14-01-2025	1.71	2.01	4.49	2.17	0.14	2.46	98.58	0.07	0.80	0.22	1.31
16-01-2025	1.60	3.76	5.49	2.10	0.13	2.46	98.58	0.07	0.81	0.22	1.58
17-01-2025	1.65	3.37	5.37	2.14	0.13	2.46	98.58	0.07	0.78	0.22	1.54
18-01-2025	1.61	3.53	5.45	2.11	0.13	2.46	98.58	0.07	0.78	0.22	1.56
20-01-2025	1.78	3.41	5.73	2.22	0.14	2.46	98.58	0.07	0.77	0.22	1.64
21-01-2025	1.76	3.24	5.60	2.25	0.14	2.46	98.58	0.07	0.78	0.22	1.62
22-01-2025	1.83	3.04	5.60	2.32	0.14	2.46	98.58	0.07	0.77	0.21	1.63
23-01-2025	2.10	1.84	5.04	2.38	0.15	2.46	98.58	0.07	0.67	0.19	1.44
24-01-2025	2.00	1.59	4.75	2.35	0.15	2.46	98.58	0.07	0.85	0.24	1.38
25-01-2025	2.04	1.64	4.86	2.37	0.15	2.46	98.58	0.07	0.75	0.21	1.41
26-01-2025	1.91	2.60	5.59	2.30	0.14	2.46	98.58	0.07	0.76	0.21	1.61
27-01-2025	1.78	2.85	5.65	2.23	0.14	2.46	98.58	0.07	0.75	0.21	1.62
28-01-2025	1.93	0.90	4.23	2.28	0.14	2.46	98.58	0.07	0.59	0.17	1.22
29-01-2025	1.92	0.73	4.11	2.23	0.14	2.46	98.58	0.07	0.55	0.15	1.17
30-01-2025	1.96	0.45	3.91	2.14	0.13	2.46	98.58	0.07	0.77	0.21	1.09
31-01-2025	1.81	0.78	4.19	2.03	0.13	2.46	98.58	0.07	1.02	0.28	1.16
01-02-2025	1.80	0.80	4.24	2.04	0.13	2.46	98.58	0.07	0.45	0.13	1.16
02-02-2025	1.78	1.31	4.69	2.09	0.13	2.46	98.58	0.07	0.69	0.19	1.31
03-02-2025	1.78	2.10	5.39	2.19	0.14	2.46	98.58	0.07	0.72	0.20	1.53
04-02-2025	1.89	2.21	5.64	2.30	0.14	2.46	98.58	0.07	0.74	0.21	1.63
05-02-2025	1.85	2.13	5.58	2.22	0.14	2.46	98.58	0.07	0.71	0.20	1.58
06-02-2025	2.02	2.23	5.96	2.27	0.14	2.46	98.58	0.07	0.58	0.16	1.67
07-02-2025	2.08	0.44	4.14	2.33	0.14	2.46	98.58	0.07	0.76	0.21	1.19
08-02-2025	1.85	1.90	5.53	2.20	0.14	2.46	98.58	0.07	0.72	0.20	1.56
09-02-2025	1.76	2.42	5.91	2.23	0.14	2.46	98.58	0.07	0.77	0.22	1.70
10-02-2025	1.83	2.04	5.67	2.30	0.14	2.46	98.58	0.07	0.75	0.21	1.65
11-02-2025	2.12	2.00	6.16	2.22	0.14	2.46	98.58	0.07	0.71	0.20	1.69
12-02-2025	1.56	0.73	4.45	1.82	0.12	2.47	98.58	0.07	0.94	0.26	1.20
13-02-2025	1.72	0.73	4.51	2.05	0.13	2.46	98.58	0.07	1.09	0.30	1.28
14-02-2025	1.58	1.79	5.47	1.97	0.13	2.46	98.58	0.07	0.94	0.26	1.53
15-02-2025	1.69	1.19	4.99	2.15	0.13	2.46	98.58	0.07	0.61	0.17	1.42
16-02-2025	1.74	0.86	4.72	2.27	0.14	2.46	98.58	0.07	0.74	0.21	1.39
17-02-2025	1.76	0.51	4.43	2.22	0.14	2.46	98.58	0.07	0.76	0.21	1.29
18-02-2025	1.64	1.18	5.08	2.08	0.13	2.46	98.58	0.07	0.86	0.24	1.45
19-02-2025	1.66	2.04	5.99	2.13	0.13	2.46	98.58	0.07	1.01	0.28	1.72
20-02-2025	1.62	1.91	5.83	2.20	0.14	2.46	98.58	0.07	0.79	0.22	1.70
21-02-2025	1.74	2.41	6.47	2.40	0.15	2.46	98.58	0.07	0.91	0.25	1.94
22-02-2025	2.03	0.58	4.76	2.33	0.14	2.46	98.58	0.07	0.82	0.23	1.37
23-02-2025	1.52	0.76	4.79	2.08	0.13	2.46	98.58	0.07	0.84	0.23	1.40
27-02-2025	1.73	2.45	6.86	2.37	0.15	2.46	98.58	0.07	0.83	0.23	2.03
28-02-2025	1.74	1.13	5.44	2.43	0.15	2.45	98.58	0.07	0.74	0.21	1.64
01-03-2025	2.44	0.83	5.59	2.74	0.17	2.45	98.58	0.07	0.60	0.17	1.65
02-03-2025	1.74	1.50	6.07	2.06	0.13	2.46	98.58	0.07	1.01	0.28	1.69
03-03-2025	2.12	0.68	5.26	2.47	0.15	2.45	98.58	0.07	1.09	0.30	1.55
04-03-2025	1.87	0.69	5.17	2.43	0.15	2.45	98.58	0.07	0.82	0.23	1.55

05-03-2025	2.04	0.45	4.95	2.55	0.16	2.45	98.58	0.07	0.82	0.23	1.50
06-03-2025	1.83	0.48	4.96	2.41	0.15	2.46	98.58	0.07	1.10	0.30	1.52
07-03-2025	1.77	0.49	4.98	2.41	0.15	2.46	98.58	0.07	0.93	0.26	1.53
08-03-2025	1.71	0.50	5.00	2.43	0.15	2.45	98.58	0.07	0.76	0.21	1.53
09-03-2025	1.87	0.48	5.06	2.62	0.16	2.45	98.58	0.07	0.89	0.25	1.59
10-03-2025	1.90	0.48	5.10	2.58	0.16	2.45	98.58	0.07	0.84	0.23	1.57
11-03-2025	1.96	0.47	5.14	2.60	0.16	2.45	98.58	0.07	0.76	0.21	1.57
12-03-2025	2.29	0.42	5.22	2.65	0.16	2.45	98.58	0.07	0.74	0.21	1.55
13-03-2025	2.12	0.43	5.24	2.22	0.14	2.46	98.58	0.07	0.72	0.20	1.44
14-03-2025	1.85	0.46	5.24	2.04	0.13	2.46	98.58	0.07	0.88	0.25	1.43
15-03-2025	1.94	0.45	5.28	2.16	0.14	2.46	98.58	0.07	0.78	0.22	1.47
16-03-2025	2.46	0.39	5.38	2.54	0.16	2.45	98.58	0.07	0.59	0.16	1.53
18-03-2025	1.88	2.09	7.58	2.78	0.17	2.45	98.58	0.07	1.75	0.49	2.48
19-03-2025	2.14	1.80	7.55	2.96	0.18	2.45	98.58	0.07	1.00	0.28	2.38
20-03-2025	2.17	0.53	5.62	2.63	0.16	2.45	98.58	0.07	0.57	0.16	1.67
21-03-2025	1.92	0.54	5.57	2.40	0.15	2.46	98.58	0.07	0.73	0.20	1.64
22-03-2025	2.08	0.59	5.75	2.53	0.16	2.45	98.58	0.07	0.62	0.17	1.69
23-03-2025	1.97	1.37	6.95	2.57	0.16	2.45	98.58	0.07	1.07	0.30	2.10
24-03-2025	2.13	1.35	7.09	2.81	0.17	2.45	98.58	0.07	0.77	0.22	2.17
25-03-2025	2.03	2.09	8.14	2.9	0.17	2.45	98.58	0.07	0.98	0.27	2.55
26-03-2025	2.04	2.04	8.15	2.84	0.17	2.45	98.58	0.07	0.99	0.27	2.53
27-03-2025	2.16	1.66	7.76	2.9	0.17	2.45	98.58	0.07	1.01	0.28	2.41
28-03-2025	2.29	0.52	5.86	3.05	0.18	2.45	98.58	0.07	0.84	0.23	1.86
29-03-2025	2.89	0.35	5.82	3.17	0.19	2.44	98.58	0.07	0.57	0.16	1.77
30-03-2025	2.22	0.57	6.03	2.72	0.17	2.45	98.58	0.07	0.81	0.23	1.82
31-03-2025	2.24	1.49	7.72	3.01	0.18	2.45	98.58	0.07	0.91	0.25	2.41
01-04-2025	2.27	0.65	6.22	2.97	0.18	2.45	98.58	0.07	0.75	0.21	1.94
03-04-2025	2.43	0.59	6.23	3.23	0.19	2.44	98.58	0.07	1.04	0.29	2.02
04-04-2025	2.42	0.78	6.64	3.25	0.19	2.44	98.58	0.07	0.94	0.26	2.14
05-04-2025	2.26	0.78	6.56	3.14	0.19	2.44	98.58	0.07	0.95	0.26	2.11
06-04-2025	2.43	1.15	7.52	3.08	0.18	2.45	98.58	0.07	1.07	0.3	2.35
07-04-2025	2.28	0.76	6.62	2.94	0.18	2.45	98.58	0.07	0.85	0.24	2.05
08-04-2025	2.31	1.34	7.8	3.12	0.19	2.45	98.58	0.07	0.92	0.26	2.46
09-04-2025	2.57	1.38	8.26	3.35	0.2	2.44	98.58	0.07	0.85	0.24	2.62
10-04-2025	2.6	0.73	6.84	3.39	0.2	2.44	98.58	0.07	0.84	0.23	2.2
11-04-2025	2.92	0.4	6.28	3.49	0.21	2.44	98.58	0.07	0.67	0.19	1.99
12-04-2025	2.35	0.86	7.02	3.08	0.18	2.45	98.58	0.07	0.96	0.27	2.22
13-04-2025	2.58	1.06	7.71	3.26	0.19	2.44	98.58	0.07	0.99	0.27	2.43
14-04-2025	2.49	1.27	8.07	3.35	0.2	2.44	98.58	0.07	0.86	0.24	2.57
15-04-2025	2.46	0.55	6.48	3.27	0.19	2.44	98.58	0.07	0.74	0.21	2.07
16-04-2025	2.56	0.78	7.08	3.46	0.2	2.44	98.58	0.07	1.31	0.36	2.35
17-04-2025	2.78	0.54	6.72	3.31	0.2	2.44	98.58	0.07	0.91	0.25	2.11
18-04-2025	2.53	1.19	8.11	3.19	0.19	2.44	98.58	0.07	0.81	0.23	2.52
19-04-2025	2.55	0.96	7.62	3.22	0.19	2.44	98.58	0.07	0.83	0.23	2.39
20-04-2025	2.95	0.91	8	3.4	0.2	2.44	98.58	0.07	1.07	0.3	2.5
21-04-2025	2.55	0.4	6.35	3.1	0.19	2.45	98.58	0.07	0.61	0.17	1.96
22-04-2025	2.92	0.36	6.41	3.4	0.2	2.44	98.58	0.07	0.92	0.26	2.02
23-04-2025	3.18	0.31	6.44	3.34	0.2	2.44	98.58	0.07	0.99	0.28	1.96
24-04-2025	3.18	0.31	6.46	3.27	0.19	2.44	98.58	0.07	0.99	0.27	1.94

4.5 Estimation of Water Balance Components

Understanding the water dynamics in the soil-plant system requires careful estimation of key water balance components. In this study, the components considered were irrigation water applied (Q), rainfall (R), and percolation losses (D). These water balance components were observed over the experimental period under various crop treatments and were subsequently used to calculate the crop evapotranspiration (ET_c), which represents the total water consumed by the crop through evaporation and transpiration. ET_c is instrumental in determining crop water demand and is the basis for planning efficient irrigation schemes.

Rainfall was obtained from an Automated Weather Station (AWS) located at the experimental site, which provided accurate, daily rainfall data. These rainfall values were incorporated into the water balance framework, particularly during periods when natural precipitation supplemented or replaced irrigation events. The irrigation water applied was determined using the discharge rate of surface emitters and the duration of irrigation for each pot. Percolation losses were quantified based on the volume of water drained from each pot and converted into depth using the surface area. Detailed explanations of latter two components are provided under their respective subheadings in the following sections.

4.5.1 Depth of Irrigation

The depth of irrigation is one of the fundamental components in understanding the water balance of crop production systems and in assessing crop water management and evaluating overall irrigation efficiency. It represents the depth of water applied to the soil surface during irrigation events and is closely linked to soil moisture replenishment and plant water requirements. In this study, irrigation was delivered to each pot through surface emitters, and the corresponding depth of water applied was estimated to analyse irrigation scheduling accuracy and uniformity across treatments.

Daily variations in irrigation depth were observed throughout the experimental duration based on the irrigation plan, growth stage of the crops, and behaviour of emitter discharge. Importantly, deeper irrigation depths were recorded following major watering events and during the mid-growth periods of the crops. Such variations are important for understanding water distribution patterns and their effects on percolation, evapotranspiration, as well as general crop water use efficiency.

The depth of irrigation for each pot was computed using the Equation 4.1.

$$\text{Depth of Irrigation (mm)} = \left(\frac{Q \times t}{\text{CSA}} \right) \times \text{No. of emitters} \quad 4.1$$

where Q is the discharge rate of the emitter, t is the irrigation duration, and CSA is the cross-sectional area of the pot. This standardized calculation enabled consistent comparison across different treatments and crop types.

To simplify representation and interpretation of the data, each pot was labeled using crop-specific identifiers. For instance, pots under beans were tagged as B1 to B5, carrots as C1 to C5, mustard as M1 to M5 and tomatoes as T1 to T5. These designations are maintained across all tables and graphical presentations to ensure consistency and allow easy identification of the pots of each crop.

1. Beans

Table 4.11 shows the daily irrigation depth values for beans, while Figure 4.14 illustrates the corresponding trends over time for pots B1 to B5.

Table 4.11. Daily irrigation depth (mm/day) for Beans

DATE	BEANS (mm/day)				
	B1	B2	B3	B4	B5
20-12-2024	1.15	1.01	0.98	1.1	1.08
21-12-2024	0.98	0.94	0.94	0.95	0.97
22-12-2024	0.95	0.95	0.95	0.95	0.95
23-12-2024	1.01	1.02	1.02	1.01	1.01
24-12-2024	1.01	1.01	1.01	1.01	1.01
25-12-2024	1.02	1.02	1.02	1.02	1.02
26-12-2024	8.58	8.49	3.99	1.04	3.33
27-12-2024	3.43	2.43	2.42	2.16	2.08
28-12-2024	1.41	2.77	2.44	2.44	2.77
29-12-2024	0.86	0.95	0.86	0.86	0.92
30-12-2024	1.2	1.2	1.2	1.2	1.2
31-12-2024	1.21	1.21	1.21	1.21	1.21
01-01-2025	1.24	1.24	1.24	1.24	1.24
02-01-2025	1.26	1.26	1.26	1.26	1.26
03-01-2025	1.29	1.29	1.29	1.34	1.32
04-01-2025	1.34	1.32	1.32	1.32	1.32
05-01-2025	1.35	1.35	1.35	1.35	1.35
06-01-2025	1.34	1.34	1.34	1.34	1.34
07-01-2025	1.22	1.22	1.22	1.22	1.22
08-01-2025	10.2	8.06	6.52	7.9	8.39
09-01-2025	1.52	1.58	1.44	1.4	1.35
10-01-2025	1.81	2.6	3.06	2.73	2.77
11-01-2025	1.45	1.45	1.47	1.45	1.45
13-01-2025	1.35	1.35	1.35	1.35	1.37
14-01-2025	1.36	1.36	1.36	1.36	1.36
16-01-2025	1.68	1.68	1.68	1.68	1.68
17-01-2025	1.67	1.67	1.67	1.67	1.67
18-01-2025	1.73	1.72	1.73	1.72	1.73

20-01-2025	1.81	1.81	1.81	1.81	1.81
21-01-2025	1.79	1.79	1.79	1.79	1.79
22-01-2025	1.8	1.8	1.8	1.8	1.8
23-01-2025	1.6	1.6	1.62	1.64	1.61
24-01-2025	1.53	1.53	1.53	1.53	1.53
25-01-2025	1.56	1.56	1.56	1.56	1.56
26-01-2025	1.78	1.78	1.78	1.78	1.78
27-01-2025	1.8	1.8	1.8	1.8	1.8
28-01-2025	1.35	1.35	1.35	1.35	1.35
29-01-2025	1.3	1.3	1.3	1.3	1.3
30-01-2025	1.21	1.21	1.21	1.21	1.21
31-01-2025	32.11	25.32	27.2	26.21	21.29
01-02-2025	1.29	1.29	1.29	1.29	1.29
02-02-2025	1.53	1.49	2.01	1.46	1.46
03-02-2025	1.71	1.71	1.71	1.71	1.71
04-02-2025	1.82	1.82	1.82	1.82	1.82
05-02-2025	1.8	1.79	1.8	1.79	1.79
06-02-2025	1.89	1.9	1.89	1.89	1.89
07-02-2025	1.33	1.33	1.33	1.33	1.33
08-02-2025	1.76	1.76	1.76	1.76	1.76
09-02-2025	1.91	1.91	1.91	1.91	1.91
10-02-2025	1.85	1.85	1.85	1.85	1.85
11-02-2025	20.59	14.36	17.64	12.39	11.74
12-02-2025	11.19	11.52	14.65	1.48	4.96
13-02-2025	1	0.99	0.99	0.96	1.85
14-02-2025	0	0	0	0	0
15-02-2025	20.95	9.48	4.76	10.79	2.48
16-02-2025	1.63	1.91	1.67	1.71	2.21
17-02-2025	1.48	1.48	42.45	1.48	11.63
18-02-2025	1.65	1.65	18.04	1.65	1.65
19-02-2025	1.95	1.95	1.95	1.95	1.95
20-02-2025	6.85	130.12	3.89	1.94	12.09
21-02-2025	2.23	2.24	2.22	2.22	2.23
22-02-2025	1.65	1.57	1.64	1.57	1.59

23-02-2025	1.77	1.65	1.71	1.61	1.64
27-02-2025	2.44	2.36	2.38	2.32	2.37
28-02-2025	1.88	1.88	1.88	1.88	1.91
01-03-2025	1.4	1.4	1.41	1.41	1.4
02-03-2025	1.94	1.94	1.94	1.94	1.94
03-03-2025	2.74	1.19	1.25	1.18	2.61
04-03-2025	1.82	1.79	1.78	1.79	1.78
05-03-2025	1.71	1.7	1.69	1.69	1.7
06-03-2025	1.71	1.7	1.69	1.7	1.7
07-03-2025	1.67	1.66	1.67	1.66	1.67
08-03-2025	1.64	1.63	1.63	1.63	1.64
09-03-2025	1.67	1.66	1.66	1.66	1.67
10-03-2025	1.61	1.61	1.61	1.61	1.61
11-03-2025	1.57	1.57	1.57	1.57	5.18
12-03-2025	1.53	1.51	1.51	1.51	1.54
13-03-2025	2.36	12.2	1.61	1.38	4
14-03-2025	96.41	75.76	66.91	60.35	90.18
15-03-2025	8.55	10.52	11.37	9.52	11.17
16-03-2025	5.58	4.47	6.44	4.14	8.08
18-03-2025	13.08	10.32	7.83	19.51	13.08
19-03-2025	1.51	1.49	1.51	1.51	1.49
20-03-2025	0.84	0.84	0.84	0.84	0.84
21-03-2025	0.78	0.78	0.79	0.78	0.78
22-03-2025	0.79	0.79	0.8	0.79	0.79
23-03-2025	1.05	1.05	1.05	1.06	1.05
24-03-2025	1.07	1.07	1.05	1.05	1.05
25-03-2025	0	0	0	0	0
26-03-2025	0	0	0	0	0
27-03-2025	0	0	0	0	0
28-03-2025	0	0	0	0	0
29-03-2025	0.08	0.08	0.08	0.08	0.08
30-03-2025	0.07	0.07	0.07	0.07	0.07
31-03-2025	0.86	0.86	0.86	0.86	0.86
01-04-2025	0.55	0.55	0.55	0.55	0.55

03-04-2025	1.01	1.01	1.01	1.01	1.01
04-04-2025	1.02	1.02	1.02	1.02	1.02
06-04-2025	0	0	0	0	0
07-04-2025	0	0	0	0	0
08-04-2025	0	0	0	0	0
09-04-2025	0	0	0	0	0
10-04-2025	0	0	0	0	0
11-04-2025	0	0	0	0	0
12-04-2025	0.18	0.18	0.18	0.18	0.18
13-04-2025	0.19	0.19	0.19	0.19	0.19
14-04-2025	63.88	20.01	5.59	0.67	86.57
15-04-2025	0.5	0.5	0.5	0.5	0.5
16-04-2025	36.21	12.14	2.37	1.06	131.16
17-04-2025	14.7	122.57	56.34	18.96	79.62
18-04-2025	0	26.64	0	0	0
19-04-2025	0	0	0	0	0
20-04-2025	3.75	31.22	17.85	129.98	2.76
21-04-2025	0	1.84	0.53	54.3	0
22-04-2025	0	0	0	0	0
23-04-2025	0	0	0	0	0
24-04-2025	47.79	56.31	52.05	49.43	42.02

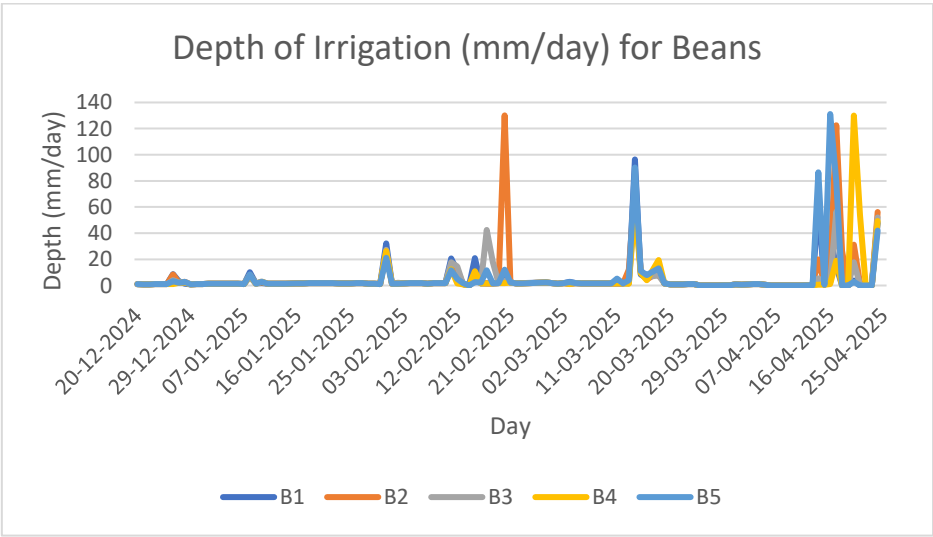


Figure 4.14 Daily irrigation depth (mm/day) for Beans.

2. Carrot

Table 4.12 provides the irrigation depth data for carrots for pots C1 to C5, with graphical trends displayed in Figure 4.15.

Table 4.12. Daily irrigation depth (mm/day) for Carrot

Date	CARROT (mm/day)				
	C1	C2	C3	C4	C5
20-12-2024	1.58	1.56	1.75	1.68	1.61
21-12-2024	1.51	1.48	1.48	1.5	1.5
22-12-2024	1.53	1.53	1.53	1.53	1.53
23-12-2024	1.58	1.58	1.58	1.58	1.58
24-12-2024	1.54	1.54	1.54	1.54	1.54
25-12-2024	1.51	1.51	1.51	1.51	1.51
26-12-2024	20.52	32.32	51.01	25.43	13.89
27-12-2024	3.86	3.18	3.18	3.07	3.83
28-12-2024	3.43	4.27	3.51	4.49	3.51
29-12-2024	1.15	1.15	1.17	1.15	1.15
30-12-2024	1.56	1.56	1.56	1.56	1.56
31-12-2024	1.54	1.54	1.54	1.54	1.54
01-01-2025	1.54	1.54	1.54	1.54	1.54
02-01-2025	1.53	1.53	1.53	1.53	1.53
03-01-2025	1.61	1.66	1.53	1.54	1.57
04-01-2025	1.53	1.53	1.53	1.53	1.53
05-01-2025	1.53	1.53	1.53	1.53	1.53
06-01-2025	1.48	1.48	1.48	1.48	1.48
07-01-2025	1.31	1.31	1.31	1.31	1.31
08-01-2025	14.71	14.49	14.59	14.49	14.49
09-01-2025	2.81	1.55	1.51	1.47	1.51
10-01-2025	3.3	4.02	3.33	4.1	1.6
11-01-2025	1.44	1.44	1.44	1.44	1.44
13-01-2025	1.28	1.28	1.28	1.28	1.28
14-01-2025	1.26	1.26	1.26	1.26	1.26
16-01-2025	1.5	1.5	1.5	1.5	1.5
17-01-2025	1.46	1.46	1.46	1.46	1.46

18-01-2025	1.48	1.47	1.47	1.47	1.47
20-01-2025	1.52	1.52	1.52	1.52	1.52
21-01-2025	1.5	1.5	1.5	1.5	1.5
22-01-2025	1.5	1.5	1.5	1.5	1.5
23-01-2025	1.32	1.36	1.33	1.36	1.32
24-01-2025	1.25	1.25	1.26	1.26	1.26
25-01-2025	1.27	1.27	1.27	1.27	1.28
26-01-2025	1.44	1.44	1.44	1.44	1.44
27-01-2025	1.44	1.44	1.44	1.44	1.44
28-01-2025	1.07	1.07	1.07	1.07	1.07
29-01-2025	1.05	1.02	1.02	1.04	1.02
30-01-2025	0.95	0.95	0.95	0.95	0.95
31-01-2025	22.64	15.2	25.6	23.3	27.23
01-02-2025	1	1	1.02	1.01	1.01
02-02-2025	1.13	1.13	1.13	1.12	1.12
03-02-2025	1.32	1.3	1.3	1.3	1.3
04-02-2025	1.37	1.37	1.37	1.37	1.38
05-02-2025	1.34	1.34	1.35	1.34	1.34
06-02-2025	1.41	1.41	1.42	1.41	1.41
07-02-2025	0.98	0.98	0.98	0.98	0.98
08-02-2025	1.28	1.28	1.28	1.28	1.28
09-02-2025	1.38	1.38	1.38	1.38	1.38
10-02-2025	1.33	1.33	1.33	1.33	1.33
11-02-2025	16.44	26.93	19.72	25.62	20.04
12-02-2025	12.43	15.49	11.45	14.07	14.07
13-02-2025	0.58	0.61	0.58	0.59	0.55
14-02-2025	-21.8	-21.8	-21.76	-21.8	-21.74
15-02-2025	1.11	11.27	58.36	6.36	32.85
16-02-2025	1.08	1.12	1.16	1.17	1.11
17-02-2025	1.01	0.99	0.99	0.99	0.99
18-02-2025	1.11	1.11	1.11	1.11	1.11
19-02-2025	1.37	1.32	1.5	1.32	1.32
20-02-2025	5.41	1.27	14.59	1.27	15.7
21-02-2025	1.46	1.46	1.46	1.46	1.55

22-02-2025	1.04	1.04	1.05	1.07	1.09
23-02-2025	1.09	1.05	1.09	1.17	1.12
27-02-2025	1.47	1.52	1.53	1.45	1.47
28-02-2025	1.16	1.16	1.16	1.16	1.16
01-03-2025	0.67	0.66	0.66	0.66	0.66
02-03-2025	1.17	1.17	1.17	1.17	1.17
03-03-2025	0.89	2.62	0.52	0.46	0.46
04-03-2025	1.16	1.11	1.07	1.06	1.06
05-03-2025	1.03	1.03	1.01	1.01	1.01
06-03-2025	1.02	1.04	1.04	1.02	1.04
07-03-2025	1.02	1.04	1.03	1.02	1.02
08-03-2025	1.01	1.01	1.01	1.01	1.01
09-03-2025	1.04	1.04	1.05	1.04	1.04
10-03-2025	1.02	1.02	1.02	1.02	1.02
11-03-2025	1.01	1.01	1.02	1.01	1.01
12-03-2025	0.99	0.99	0.99	0.99	0.99
13-03-2025	0.91	0.91	0.91	0.91	0.91
14-03-2025	55.48	65.48	60.24	59.91	50.07
15-03-2025	11.07	14.02	10.42	11.4	10.35
16-03-2025	0.44	6.15	5.69	6.74	0.44
18-03-2025	12.47	1.94	1.08	2.64	1.06
19-03-2025	1	0.93	0.93	0.93	0.96
20-03-2025	0.49	0.49	0.49	0.49	0.49
21-03-2025	0.46	0.46	0.46	0.46	0.46
22-03-2025	0.48	0.48	0.48	0.48	0.48
23-03-2025	0.72	0.71	0.71	0.71	0.71
24-03-2025	0.73	0.73	0.73	0.75	0.75
25-03-2025	0	0	0	0	0
26-03-2025	0	0	0	0	0
27-03-2025	0	0	0	0	0
28-03-2025	0	0	0	0	0
29-03-2025	0.01	0.01	0.01	0.01	0.01
30-03-2025	0.04	0.03	0.03	0.04	0.03
31-03-2025	0.87	0.87	0.87	0.87	0.87

01-04-2025	0.6	0.6	0.6	0.6	0.6
03-04-2025	19.84	12.63	10.07	17.54	10.46
04-04-2025	1.22	1.22	1.22	1.22	1.22
06-04-2025	22.17	20.53	20.92	10.04	21.71
07-04-2025	0	0	0	0	0
08-04-2025	0.56	0.4	0.45	0.43	0.44
09-04-2025	0	0	0	0	0
10-04-2025	0.26	0.25	0.25	0.25	0.25
11-04-2025	0.14	0.14	0.16	0.13	0.13
12-04-2025	0.76	0.76	0.76	0.76	0.76
13-04-2025	0.88	0.88	0.88	0.88	0.88
14-04-2025	132.61	75.89	1.47	1.46	120.48
15-04-2025	1.17	1.17	1.17	1.17	1.17
16-04-2025	1.06	1.25	0.93	4.45	131.99
17-04-2025	12.86	6.1	9.58	7.22	21.58
18-04-2025	0	0	0	0	0
19-04-2025	0.36	0.36	0.36	0.99	0.36
20-04-2025	17.95	18.61	73.03	38.48	20.58
21-04-2025	3.13	3	4.04	3.13	3.72
22-04-2025	0	0	0	0	0
23-04-2025	0	0	0	0	0
24-04-2025	64.87	60.02	131.76	60.61	59.63

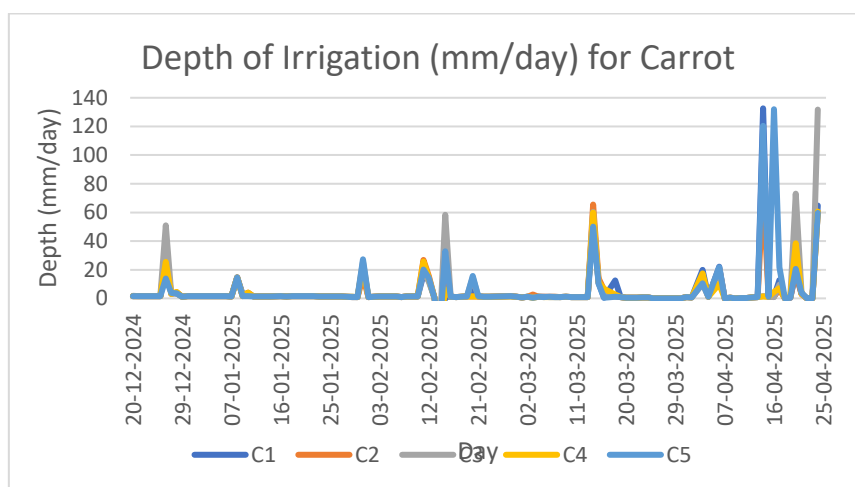


Figure 4.15. Daily irrigation depth (mm/day) for Carrot.

3. Mustard

Daily irrigation data for mustard are shown in Table 4.13 and the corresponding trend for each pot for mustard is also depicted in Figure 4.16, highlighting day-to-day variations.

Table 4.13. Daily irrigation depth (mm/day) for Mustard

Date	MUSTARD (mm/day)				
	M1	M2	M3	M4	M5
20-12-2024	1.63	1.61	1.79	1.62	1.61
21-12-2024	1.43	1.43	1.41	1.4	1.4
22-12-2024	1.51	1.35	1.35	1.35	1.35
23-12-2024	1.37	1.37	1.37	1.37	1.37
24-12-2024	1.33	1.34	1.33	1.33	1.33
25-12-2024	1.29	1.29	1.29	1.29	1.29
26-12-2024	5.52	59.62	1.26	1.26	5.52
27-12-2024	3.38	3.7	1.28	3.57	3.25
28-12-2024	2.58	4.54	1.27	3.18	5.53
29-12-2024	0.93	1.02	0.93	0.93	0.93
30-12-2024	1.25	1.25	1.25	1.25	1.25
31-12-2024	1.21	1.21	1.21	1.21	1.21
01-01-2025	1.2	1.2	1.2	1.2	1.2
02-01-2025	1.21	1.17	1.17	1.2	1.24
03-01-2025	1.15	1.17	1.15	1.15	1.15
04-01-2025	1.14	1.14	1.14	1.14	1.14
05-01-2025	1.12	1.12	1.12	1.12	1.12
06-01-2025	1.06	1.06	1.06	1.06	1.06
07-01-2025	0.93	0.93	0.93	0.93	0.93
08-01-2025	1.67	3.04	6.29	4.35	3.88
09-01-2025	1.06	1.07	0.97	1.01	1.02
10-01-2025	2.55	2.63	1	3.12	6.37
11-01-2025	0.95	0.96	0.95	0.95	0.96

13-01-2025	0.84	0.81	0.81	0.81	0.81
14-01-2025	0.78	0.78	0.78	0.78	0.78
16-01-2025	0.89	0.89	0.89	0.89	0.89
17-01-2025	0.85	0.85	0.85	0.85	0.85
18-01-2025	0.84	0.85	0.84	0.85	0.88
20-01-2025	0.83	0.83	0.83	0.83	0.83
21-01-2025	0.79	0.79	0.79	0.79	0.79
22-01-2025	0.77	0.78	0.77	0.78	0.77
23-01-2025	0.66	0.66	0.66	0.69	1.12
24-01-2025	0.61	0.61	1.24	0.62	3.23
25-01-2025	0.6	0.6	1.59	0.6	1.21
26-01-2025	0.68	0.68	0.69	0.68	0.7
27-01-2025	0.68	0.68	0.69	0.68	0.69
28-01-2025	0.5	0.5	0.5	0.5	0.52
29-01-2025	0.48	0.48	0.48	0.48	0.48
30-01-2025	0.44	0.44	0.44	0.44	0.44
31-01-2025	23.42	26.04	31.62	21.92	23.83
01-02-2025	0.46	0.46	0.46	0.46	0.46
02-02-2025	0.56	0.56	0.64	0.69	0.57
03-02-2025	0.6	0.6	0.6	0.6	0.6
04-02-2025	0.63	0.63	0.63	0.63	0.63
05-02-2025	0.62	0.61	0.61	0.61	0.61
06-02-2025	0.64	0.65	0.65	0.64	0.64
07-02-2025	0.45	0.45	0.46	0.46	0.46
08-02-2025	0.59	0.59	0.59	0.59	0.59
09-02-2025	0.63	0.63	0.63	0.63	0.63
10-02-2025	0.61	0.61	0.61	0.61	0.61
11-02-2025	13.4	19.97	31.43	28.81	104.22

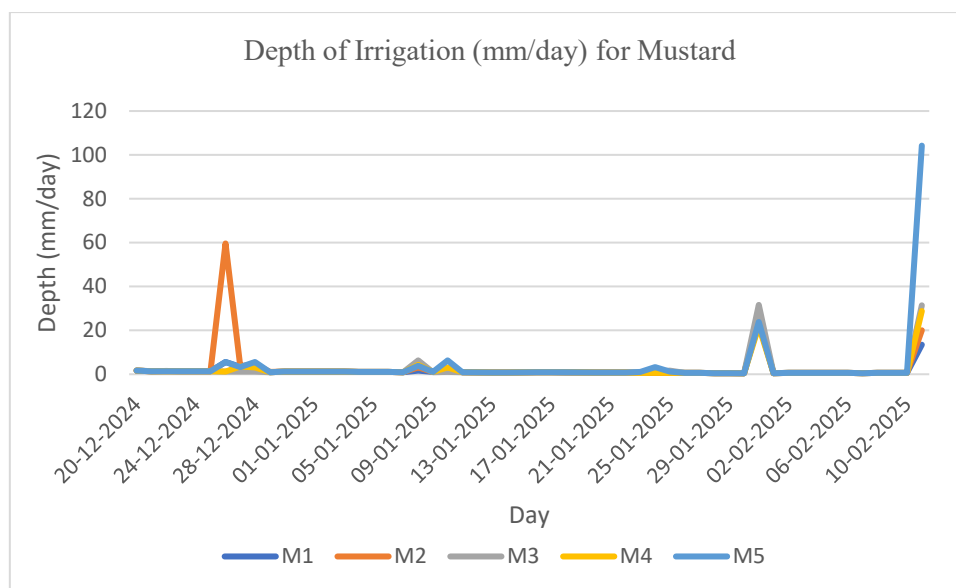


Figure 4.16. Daily irrigation depth (mm/day) for Mustard.

4. Tomato

Tomato irrigation data are presented in Table 4.14, and the irrigation trends for pots T1 to T5 are depicted in Figure 4.17.

Table 4.14. Daily irrigation depth (mm/day) for Tomato

Date	TOMATO, mm/day				
	T1	T2	T3	T4	T5
20-12-2024	1.77	1.69	1.98	1.74	1.69
21-12-2024	1.53	1.54	1.53	1.58	1.56
22-12-2024	1.54	1.54	1.54	1.55	1.55
23-12-2024	1.61	1.61	1.62	1.61	1.61
24-12-2024	1.6	1.6	1.6	1.6	1.62
25-12-2024	1.59	1.59	1.59	1.59	1.59
26-12-2024	1.61	1.61	2.04	1.61	1.61
27-12-2024	1.7	1.65	2.18	1.65	1.65
28-12-2024	1.64	1.64	3.61	1.64	1.64
29-12-2024	1.22	1.22	1.26	1.22	1.22
30-12-2024	1.65	1.65	1.65	1.65	1.65
31-12-2024	1.62	1.62	1.62	1.62	1.62
01-01-2025	1.62	1.62	1.62	1.62	1.62
02-01-2025	1.6	1.6	1.6	1.6	1.6

03-01-2025	1.59	1.59	1.59	1.59	1.61
04-01-2025	1.59	1.59	1.59	1.59	1.59
05-01-2025	1.58	1.58	1.58	1.58	1.58
06-01-2025	1.52	1.52	1.52	1.52	1.52
07-01-2025	1.35	1.35	1.35	1.35	1.35
08-01-2025	1.11	1.11	5.37	1.11	1.56
09-01-2025	1.42	1.42	1.52	1.48	1.47
10-01-2025	1.51	1.49	2.81	1.51	1.91
11-01-2025	1.47	1.45	1.48	1.48	1.47
13-01-2025	1.28	1.28	1.3	1.28	1.3
14-01-2025	1.25	1.25	1.25	1.25	1.25
16-01-2025	1.48	1.48	1.48	1.48	1.49
17-01-2025	1.43	1.43	1.43	1.43	1.43
18-01-2025	1.45	1.44	1.44	1.44	1.44
20-01-2025	1.48	1.48	1.48	1.48	1.48
21-01-2025	1.45	1.45	1.45	1.45	1.45
22-01-2025	1.44	1.44	1.44	1.44	1.44
23-01-2025	1.27	1.26	6.92	1.26	1.26
24-01-2025	1.19	1.19	1.25	1.19	1.19
25-01-2025	1.2	1.2	1.2	1.2	1.22
26-01-2025	1.36	1.35	1.35	1.35	1.35
27-01-2025	1.35	1.35	1.35	1.35	1.35
28-01-2025	1	1	2.64	1	1
29-01-2025	0.95	0.95	0.95	0.95	0.95
30-01-2025	0.87	0.87	0.99	0.87	0.87
31-01-2025	25.84	27.97	39.28	18.62	18.62
01-02-2025	0.91	0.91	0.91	0.91	0.91
02-02-2025	1.01	1.01	1.09	1.05	2.32
03-02-2025	1.17	1.17	1.17	1.17	1.17
04-02-2025	1.22	1.22	1.22	1.22	1.22
05-02-2025	1.17	1.17	1.17	1.17	1.17
06-02-2025	1.24	1.24	1.24	1.22	1.22
07-02-2025	0.87	0.85	0.85	0.85	0.88
08-02-2025	1.11	1.11	1.11	1.11	1.11

09-02-2025	1.19	1.19	1.19	1.19	1.19
10-02-2025	1.14	1.14	1.14	1.14	1.14
11-02-2025	17.54	12.3	29.67	13.28	19.84
12-02-2025	13.59	11.62	12.61	8.28	14.9
13-02-2025	0.45	0.45	0.38	0.46	0.52
14-02-2025	0	0	0	0	0
15-02-2025	82.22	29.44	7.07	6.81	42.35
16-02-2025	0.94	0.89	0.93	1.18	0.92
17-02-2025	14.23	18.64	5.06	7.1	24.74
18-02-2025	0.95	0.93	0.9	0.9	5.15
19-02-2025	1.03	1.03	1.03	1.03	1.03
20-02-2025	1	23.62	4.28	1	15.1
21-02-2025	1.14	1.22	1.24	1.14	2.76
22-02-2025	0.79	0.79	0.94	0.79	1.76
23-02-2025	0.8	0.81	0.95	0.84	2.42
27-02-2025	1.12	1.12	1.18	1.18	7.61
28-02-2025	0.84	0.84	0.84	0.84	0.93
01-03-2025	0.34	0.33	0.33	0.33	0.34
02-03-2025	0.83	0.83	0.83	0.83	0.83
03-03-2025	0.15	0.13	0.13	0.71	1
04-03-2025	0.74	0.74	0.73	0.73	2.43
05-03-2025	0.7	0.69	0.69	0.7	1.2
06-03-2025	0.69	0.69	0.7	0.7	0.74
07-03-2025	0.67	0.67	0.68	0.67	0.68
08-03-2025	0.66	0.66	0.66	0.66	0.67
09-03-2025	0.67	0.67	0.67	0.67	0.68
10-03-2025	0.65	0.65	0.65	0.65	0.65
11-03-2025	0.63	0.63	0.63	0.63	12.83
12-03-2025	0.6	0.6	0.6	0.6	2.31
13-03-2025	0.63	0.56	0.9	4.48	41.14
14-03-2025	80.53	77.84	76.59	71.68	38.5
15-03-2025	10.04	9.84	9.38	9.05	20.72
16-03-2025	5.61	5.28	6.99	4.63	5.94
18-03-2025	11.47	16.98	19.99	14.22	32.78

19-03-2025	0.46	0.38	0.36	0.27	0.92
20-03-2025	0.02	0.02	0.02	0.02	0.02
21-03-2025	0	0	0	0	0
22-03-2025	0	0	0	0	0
23-03-2025	0.09	0.09	0.09	0.09	0.11
24-03-2025	0.09	0.09	0.16	0.09	2.38
25-03-2025	0	0	0	0	0
26-03-2025	0	0	0	0	0
27-03-2025	0	0	0	0	0
28-03-2025	0	0	0	0	0
29-03-2025	0	0	0	0	0
30-03-2025	0	0	0	0	0
31-03-2025	0	0	0	0	0
01-04-2025	0	0	0	0	0
03-04-2025	1.98	4.6	2.96	4.34	13.19
04-04-2025	0.34	0.34	0.34	0.34	0.34
06-04-2025	19.45	12.83	18.21	18.14	5.88
07-04-2025	0	0	0	0	0
08-04-2025	0	0	0	0	0
09-04-2025	0	0	0	0	0
10-04-2025	0	0	0	0	0
11-04-2025	0	0	0	0	0
12-04-2025	0	0	0	0	0
13-04-2025	0	0	0	0	0
14-04-2025	8.02	84.74	12.61	2.45	4.02
15-04-2025	0.1	0.1	0.1	0.1	0.1
16-04-2025	1.89	1.69	2.87	22.87	25.85
17-04-2025	0	8.11	0	9.68	0
18-04-2025	0	0	0	0	0
19-04-2025	0	0	0	0	0
20-04-2025	31.88	0	0	0	0
21-04-2025	0.31	2.28	0	1.62	17.1
22-04-2025	0	0	0	0	0
23-04-2025	0	0	0	0	0

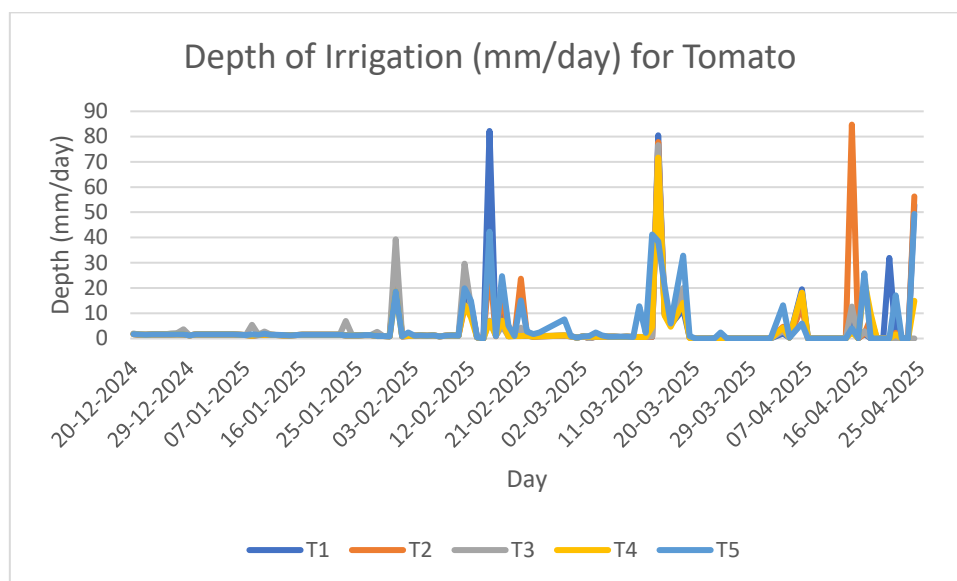


Figure 4.17. Daily irrigation depth (mm/day) for Tomato.

4.5.2 Percolation

In this study, daily percolation losses were recorded for each pot under different crop treatments (beans, carrot, mustard, and tomato) based on the drained water collected beneath each pot, as described in the methodology section. The percolation depth was estimated by collecting drained water from each pot and converting the volume to depth using the known surface area of the pots. Percolation data for each crop are presented separately to highlight crop-wise differences in water movement through the soil. These variations may result from differences in root structure, crop water uptake, and evapotranspiration rates.

Daily percolation data for each crop are presented both in tables and corresponding line graphs to facilitate a clearer interpretation of water movement patterns over the study period. For clarity in data representation, each pot was assigned a unique identifier based on the crop: B1, B2, etc., for beans; C1, C2, etc., for carrot; M1, M2, etc., for mustard; and T1, T2, etc., for tomato. These designations are used throughout the tables and graphs to differentiate between individual pots.

1. Beans

Table 4.15 shows daily percolation depths under beans, while Figure 4.18 graphically illustrates the trend over time for each pot.

Table 4.15. Daily Percolation Depths (mm) for Beans

DATE	BEANS (mm)				
	B1	B2	B3	B4	B5
20-12-2024	0.21	0.07	0.04	0.16	0.13

21-12-2024	0.04	0	0	0.01	0.03
22-12-2024	0	0	0	0	0
23-12-2024	0	0.01	0.01	0	0
24-12-2024	0	0	0	0	0
25-12-2024	0	0	0	0	0
26-12-2024	7.54	7.45	2.95	0	2.3
27-12-2024	2.33	1.34	1.32	1.06	0.98
28-12-2024	0.29	1.64	1.31	1.31	1.64
29-12-2024	0	0.09	0	0	0.07
30-12-2024	0	0	0	0	0
31-12-2024	0	0	0	0	0
01-01-2025	0	0	0	0	0
02-01-2025	0	0	0	0	0
03-01-2025	0	0	0	0.05	0.03
04-01-2025	0.01	0	0	0	0
05-01-2025	0	0	0	0	0
06-01-2025	0	0	0	0	0
07-01-2025	0	0	0	0	0
08-01-2025	9.18	7.04	5.51	6.89	7.37
09-01-2025	0.18	0.25	0.1	0.07	0.01
10-01-2025	0.35	1.14	1.6	1.27	1.31
11-01-2025	0	0	0.01	0	0
12-01-2025	0	0	0	0	0
13-01-2025	0	0	0	0	0.01
14-01-2025	0	0	0	0	0
15-01-2025	0.04	0.07	0	0.05	0.05
16-01-2025	0	0	0	0	0
17-01-2025	0	0	0	0	0
18-01-2025	0.01	0	0.01	0	0.01
19-01-2025	0	0	0	0	0
20-01-2025	0	0	0	0	0
21-01-2025	0	0	0	0	0
22-01-2025	0	0	0	0	0
23-01-2025	0	0	0.03	0.04	0.01
24-01-2025	0	0	0	0	0
25-01-2025	0	0	0	0	0
26-01-2025	0	0	0	0	0
27-01-2025	0	0	0	0	0
28-01-2025	0	0	0	0	0
29-01-2025	0.01	0.01	0	0	0
30-01-2025	0	0	0	0	0
31-01-2025	30.82	24.03	25.9	24.92	20
01-02-2025	0.04	0	0	0.01	0.03
02-02-2025	0.07	0.03	0.55	0	0
03-02-2025	0	0	0.01	0	0
04-02-2025	0.01	0.03	0	0.01	0.01
05-02-2025	0.03	0.01	0.03	0.01	0.01

06-02-2025	0.01	0.03	0.01	0.01	0.01
07-02-2025	0	0	0	0	0
08-02-2025	0	0	0	0	0
09-02-2025	0	0	0	0	0
10-02-2025	0	0	0	0	0
11-02-2025	18.69	12.46	15.74	10.49	9.84
12-02-2025	9.84	10.16	13.3	0.13	3.61
13-02-2025	0.05	0.04	0.04	0.01	0.9
14-02-2025	20.66	10.16	21.64	6.56	12.46
15-02-2025	19.34	7.87	3.15	9.18	0.87
16-02-2025	0.05	0.33	0.09	0.13	0.63
17-02-2025	0.01	0.01	40.98	0.01	10.16
18-02-2025	0	0	16.39	0	0
19-02-2025	0.07	0.98	0.33	0.03	0.04
20-02-2025	4.92	128.2	1.97	0.01	10.16
21-02-2025	0.03	0.04	0.01	0.01	0.03
22-02-2025	0.09	0.01	0.08	0.01	0.03
23-02-2025	0.17	0.05	0.12	0.01	0.04
24-02-2025	69.18	3.21	16.07	4.92	4.59
25-02-2025	61.7	2.95	15.87	4.26	4.33
26-02-2025	91.48	25.31	0.07	0.08	1.47
27-02-2025	0.12	0.04	0.07	0	0.05
28-02-2025	0	0	0	0	0.03
01-03-2025	0	0	0.01	0.01	0
02-03-2025	0	0	0	0	0
03-03-2025	1.57	0.03	0.08	0.01	1.44
04-03-2025	0.04	0.01	0	0.01	0
05-03-2025	0.03	0.01	0	0	0.01
06-03-2025	0.03	0.01	0	0.01	0.01
07-03-2025	0.01	0	0.01	0	0.01
08-03-2025	0.01	0	0	0	0.01
09-03-2025	0.01	0	0	0	0.01
10-03-2025	0	0	0	0	0
11-03-2025	0	0	0	0	3.61
12-03-2025	0.01	0	0	0	0.03
13-03-2025	0.98	10.82	0.24	0	2.62
14-03-2025	95.08	74.43	65.57	59.02	88.85
15-03-2025	7.21	9.18	10.03	8.18	9.84
16-03-2025	4.72	3.61	5.57	3.28	7.21
17-03-2025	0.66	0.79	29.57	23.28	19.34
18-03-2025	11.48	8.72	6.23	17.9	11.48
19-03-2025	0.05	0.03	0.05	0.05	0.03
20-03-2025	0	0	0	0	0
21-03-2025	0	0	0.01	0	0
22-03-2025	0	0	0.01	0	0
23-03-2025	0	0	0	0.01	0
24-03-2025	0.01	0.01	0	0	0

25-03-2025	0	0	0	0	0
26-03-2025	0	0	0	0	0
27-03-2025	0	0	0.01	0	0
28-03-2025	0	0	0	0	0
29-03-2025	0	0	0	0	0
30-03-2025	0	0	0	0	0
31-03-2025	0	0	0	0	0
01-04-2025	0	0	0	0	0
02-04-2025	0	0	0	0	0.01
03-04-2025	3.28	11.15	0	0	0
04-04-2025	0	0	0	0	0
05-04-2025	0	0	0	0	0
06-04-2025	0	0	0	0	0
07-04-2025	0	0	0	0	0
08-04-2025	0.01	0.04	0.07	0	0
09-04-2025	0	0	0	0	0
10-04-2025	0	0	0	0	0
11-04-2025	0.01	0	0	0	0
12-04-2025	0	0	0	0	0
13-04-2025	0	0	0	0	0
14-04-2025	81.31	131.15	33.44	31.87	0
15-04-2025	0	0	0	0	0
16-04-2025	61.97	1.64	1.57	1.31	0.2
17-04-2025	131.15	131.15	49.57	44.26	31.8
18-04-2025	4.59	20	13.11	0.01	0.08
19-04-2025	0	0	0	0	0
20-04-2025	29.84	28.46	18.69	21.97	21.64
21-04-2025	0.13	3.61	5.9	6.23	2.62
22-04-2025	0.01	0.01	31.48	0.03	0.03
23-04-2025	0	0	0	0	0
24-04-2025	131.15	131.15	131.15	131.15	131.15

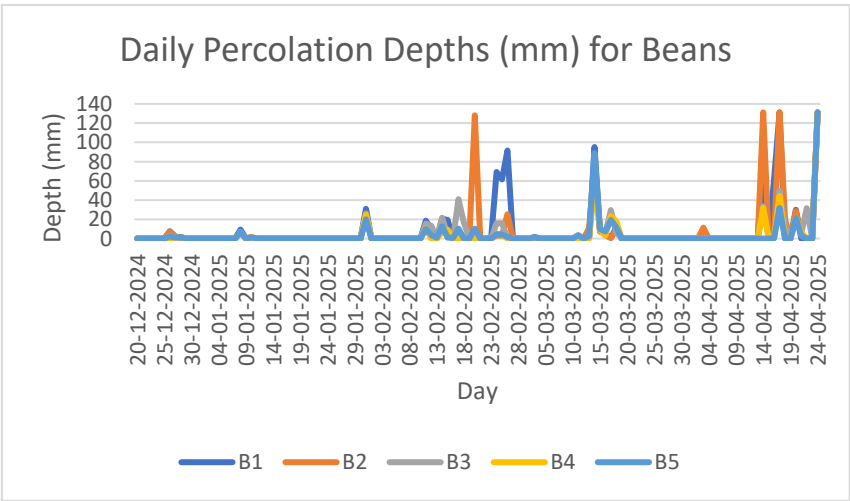


Figure 4.18. Daily percolation depth under bean cultivation for pots B1, B2, B3, B4 and B5.

2. Carrot

Daily percolation data for carrot are shown in Table 4.16 and the corresponding trend for each pot for carrot is also depicted in Figure 4.19, highlighting day-to-day variations.

Table 4.16. Daily Percolation Depths (mm) for Carrot

DATE	CARROT (mm)				
	C1	C2	C3	C4	C5
20-12-2024	0.07	0.05	0.24	0.17	0.1
21-12-2024	0.03	0	0	0.01	0.01
22-12-2024	0	0	0	0	0
23-12-2024	0	0	0	0	0
24-12-2024	0	0	0	0	0
25-12-2024	0	0	0	0	0
26-12-2024	19.02	30.82	49.51	23.93	12.39
27-12-2024	2.32	1.64	1.64	1.53	2.3
28-12-2024	1.89	2.73	1.97	2.95	1.97
29-12-2024	0	0	0.03	0	0
30-12-2024	0	0	0	0	0
31-12-2024	0	0	0	0	0
01-01-2025	0	0	0	0	0
02-01-2025	0	0	0	0	0
03-01-2025	0.08	0.13	0	0.01	0.04
04-01-2025	0	0	0	0	0
05-01-2025	0	0	0	0	0
06-01-2025	0	0	0	0	0
07-01-2025	0	0	0	0	0
08-01-2025	13.64	13.42	13.52	13.42	13.42
09-01-2025	1.43	0.17	0.13	0.09	0.13
10-01-2025	1.82	2.54	1.86	2.62	0.13
11-01-2025	0	0	0	0	0
12-01-2025	0	0	0	0	0
13-01-2025	0	0	0	0	0
14-01-2025	0	0	0	0	0
15-01-2025	0.08	0.09	0.03	0.08	0.07
16-01-2025	0	0	0	0	0
17-01-2025	0	0	0	0	0
18-01-2025	0.01	0	0	0	0
19-01-2025	0	0	0	0	0
20-01-2025	0	0	0	0	0
21-01-2025	0	0	0	0	0
22-01-2025	0	0	0	0	0
23-01-2025	0	0.04	0.01	0.04	0
24-01-2025	0	0	0.01	0.01	0.01

25-01-2025	0	0	0	0	0.01
26-01-2025	0	0	0	0	0
27-01-2025	0	0	0	0	0
28-01-2025	0	0	0	0	0
29-01-2025	0.03	0	0	0.01	0
30-01-2025	0	0	0	0	0
31-01-2025	21.64	14.19	24.59	22.3	26.23
01-02-2025	0	0	0.03	0.01	0.01
02-02-2025	0.01	0.01	0.01	0	0
03-02-2025	0.01	0	0	0	0
04-02-2025	0	0	0	0	0.01
05-02-2025	0.01	0.01	0.03	0.01	0.01
06-02-2025	0.01	0.01	0.03	0.01	0.01
07-02-2025	0	0	0	0	0
08-02-2025	0	0	0	0	0
09-02-2025	0	0	0	0	0
10-02-2025	0	0	0	0	0
11-02-2025	15.08	25.57	18.36	24.26	18.69
12-02-2025	11.48	14.53	10.49	13.11	13.11
13-02-2025	0.07	0.09	0.07	0.08	0.04
14-02-2025	0	0	0.04	0	0.07
15-02-2025	0	10.16	57.25	5.25	31.74
16-02-2025	0	0.04	0.08	0.09	0.03
17-02-2025	0.01	0	0	0	0
18-02-2025	0	0	0	0	0
19-02-2025	0.07	0.01	0.2	0.01	0.01
20-02-2025	4.13	0	13.31	0	14.43
21-02-2025	0.01	0.01	0.01	0.01	0.1
22-02-2025	0.03	0.03	0.04	0.05	0.08
23-02-2025	0.07	0.03	0.07	0.14	0.09
24-02-2025	84.92	66.56	102.49	11.48	93.44
25-02-2025	46.14	62.24	28.55	3.74	37.53
26-02-2025	3.67	3.61	0.07	0.17	0.2
27-02-2025	0.03	0.08	0.09	0.01	0.03
28-02-2025	0	0	0	0	0
01-03-2025	0.01	0	0	0	0
02-03-2025	0	0	0	0	0
03-03-2025	0.43	2.16	0.07	0	0
04-03-2025	0.1	0.05	0.01	0	0
05-03-2025	0.01	0.01	0	0	0
06-03-2025	0	0.01	0.01	0	0.01
07-03-2025	0	0.03	0.01	0	0
08-03-2025	0	0	0	0	0
09-03-2025	0	0	0.01	0	0
10-03-2025	0	0	0	0	0
11-03-2025	0	0	0.01	0	0
12-03-2025	0	0	0	0	0
13-03-2025	0	0	0	0	0

14-03-2025	54.58	64.59	59.34	59.02	49.18
15-03-2025	10.16	13.11	9.51	10.49	9.44
16-03-2025	0	5.7	5.25	6.3	0
17-03-2025	5.25	2.49	0.98	3.93	1.64
18-03-2025	11.48	0.94	0.08	1.64	0.07
19-03-2025	0.08	0.01	0.01	0.01	0.04
20-03-2025	0	0	0	0	0
21-03-2025	0	0	0	0	0
22-03-2025	0	0	0	0	0
23-03-2025	0.01	0	0	0	0
24-03-2025	0	0	0	0.01	0.01
25-03-2025	0	0	0.01	0	0
26-03-2025	0	0	0	0	0.01
27-03-2025	0.01	0	0	0	0.03
28-03-2025	0	0	0	0	0
29-03-2025	0	0	0	0	0
30-03-2025	0	0	0	0	0
31-03-2025	0	0	0	0	0
01-04-2025	0	0	0	0	0
02-04-2025	0	0	0	0.01	0
03-04-2025	0	0	0	0	0
04-04-2025	0	0	0	0	0
05-04-2025	0	0	0	0	0
06-04-2025	0	0	0	0	0
07-04-2025	0	0	0	0	0
08-04-2025	0.01	0.07	0	0	0
09-04-2025	0	0	0	0	0
10-04-2025	0	0	0	0	0
11-04-2025	0	0	0	0	0
12-04-2025	0	0	0	0	0
13-04-2025	0	0	0	0	0
14-04-2025	63.21	19.34	4.92	0	85.9
15-04-2025	0	0	0	0	0
16-04-2025	36.2	12.13	2.36	1.05	131.15
17-04-2025	23.28	131.15	64.92	27.54	88.2
18-04-2025	0.13	28.2	0.12	0.5	0.05
19-04-2025	0	0	0	0	0
20-04-2025	4.92	32.39	19.02	131.15	3.93
21-04-2025	0.38	2.62	1.31	55.08	0.04
22-04-2025	0.03	0.03	0.01	31.8	0.01
23-04-2025	0	0	0	0	0
24-04-2025	48.2	56.72	52.46	49.84	42.43

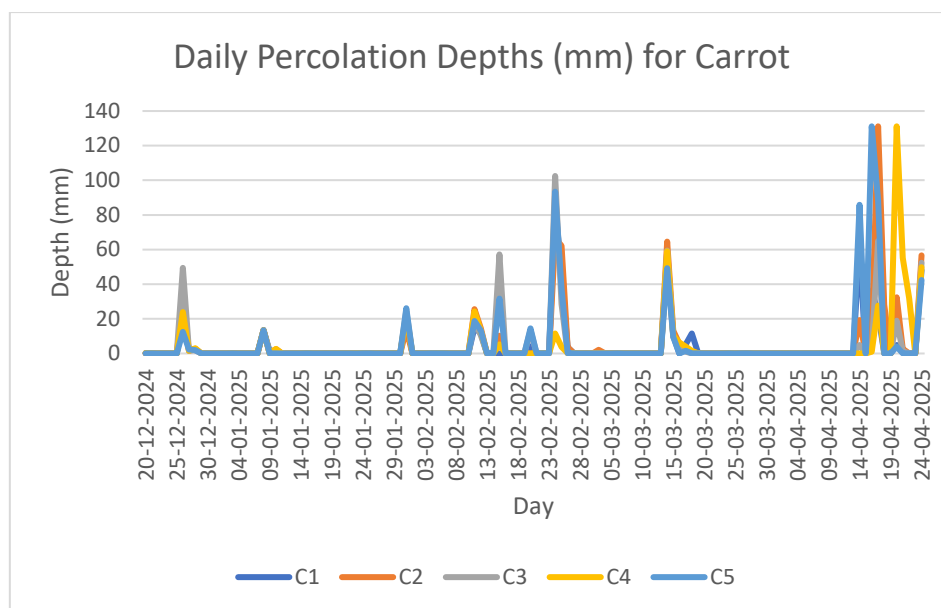


Figure 4.19. Daily percolation depth under carrot cultivation for pots C1, C2, C3, C4 and C5.

3. Mustard

As shown in Table 4.17, the percolation values for mustard were recorded for each pot. These are further visualized in Figure 4.20, showing the fluctuation in percolation over the observation period.

Table 4.17. Daily Percolation Depths (mm) for Mustard

DATE	MUSTARD (mm)				
	M1	M2	M3	M4	M5
20-12-2024	0.18	0.16	0.34	0.17	0.16
21-12-2024	0.04	0.04	0.03	0.01	0.01
22-12-2024	0.16	0	0	0	0
23-12-2024	0	0	0	0	0
24-12-2024	0	0.01	0	0	0
25-12-2024	0	0	0	0	0
26-12-2024	4.26	58.36	0	0	4.26
27-12-2024	2.1	2.43	0	2.3	1.97
28-12-2024	1.31	3.28	0	1.91	4.26
29-12-2024	0	0.09	0	0	0

30-12-2024	0	0	0	0	0
31-12-2024	0	0	0	0	0
01-01-2025	0	0	0	0	0
02-01-2025	0.04	0	0	0.03	0.07
03-01-2025	0	0.01	0	0	0
04-01-2025	0	0	0	0	0
05-01-2025	0	0	0	0	0
06-01-2025	0	0	0	0	0
07-01-2025	0	0	0	0	0
08-01-2025	0.92	2.3	5.55	3.61	3.13
09-01-2025	0.12	0.13	0.03	0.07	0.08
10-01-2025	1.56	1.64	0.01	2.12	5.38
11-01-2025	0	0.01	0	0	0.01
12-01-2025	0	0	0	0	0
13-01-2025	0.03	0	0	0	0
14-01-2025	0	0	0	0	0
15-01-2025	0.03	0.01	0	0.04	0.01
16-01-2025	0	0	0	0	0
17-01-2025	0	0	0	0	0
18-01-2025	0	0.01	0	0.01	0.04
19-01-2025	0	0	0	0	0
20-01-2025	0	0	0	0	0
21-01-2025	0	0	0	0	0
22-01-2025	0	0.01	0	0.01	0
23-01-2025	0	0	0	0.03	0.46
24-01-2025	0	0	0.63	0.01	2.62
25-01-2025	0	0	1	0	0.62
26-01-2025	0	0	0.01	0	0.03
27-01-2025	0	0	0.01	0	0.01

28-01-2025	0	0	0	0	0.01
29-01-2025	0	0	0	0.04	0
30-01-2025	0	0	0	0	0
31-01-2025	22.95	25.57	31.15	21.46	23.36
01-02-2025	0.01	0	0.01	0	0.07
02-02-2025	0.04	0.04	0.12	0.17	0.05
03-02-2025	0	0.01	0.01	0	0
04-02-2025	0.01	0	0.01	0	0
05-02-2025	0.01	0	0	0	0
06-02-2025	0	0.01	0.01	0	0
07-02-2025	0	0	0.01	0.01	0.01
08-02-2025	0	0	0	0	0
09-02-2025	0	0	0	0	0
10-02-2025	0	0	0	0	0
11-02-2025	12.79	19.36	30.82	28.2	103.61

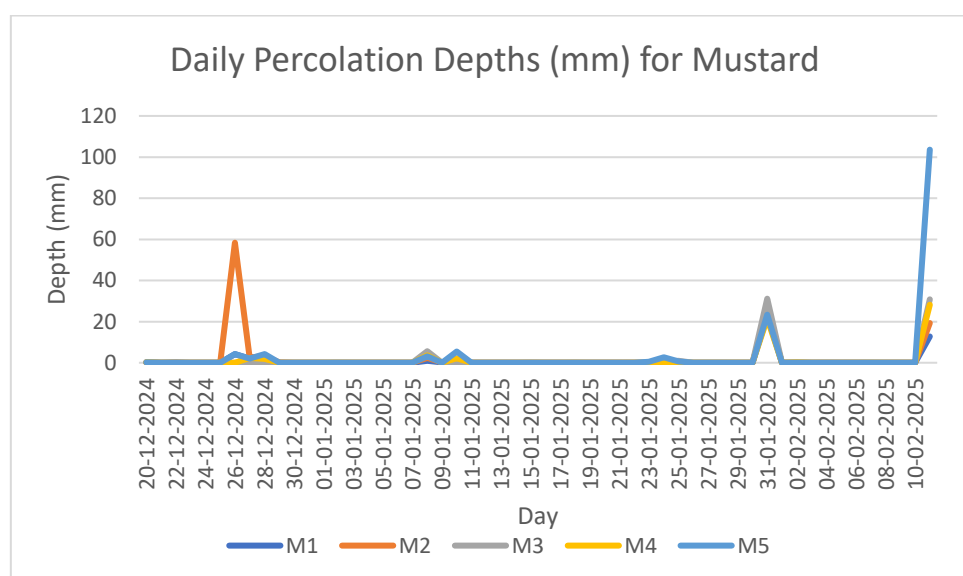


Figure 4.20. Daily percolation depth under mustard cultivation for pots M1, M2, M3, M4 and M5.

4. Tomato

Table 4.18 displays the percolation values observed under tomato cultivation. The variation among the pots is also represented in Figure 4.21 to facilitate a clearer comparison of percolation trends.

Table 4.18. Daily Percolation Depths (mm) for Tomato

DATE	TOMATO (mm)				
	T1	T2	T3	T4	T5
20-12-2024	0.21	0.13	0.42	0.18	0.13
21-12-2024	0	0.01	0	0.05	0.03
22-12-2024	0	0	0	0.01	0.01
23-12-2024	0	0	0.01	0	0
24-12-2024	0	0	0	0	0.01
25-12-2024	0	0	0	0	0
26-12-2024	0	0	0.43	0	0
27-12-2024	0.05	0	0.54	0	0
28-12-2024	0	0	1.97	0	0
29-12-2024	0	0	0.04	0	0
30-12-2024	0	0	0	0	0
31-12-2024	0	0	0	0	0
01-01-2025	0	0	0	0	0
02-01-2025	0	0	0	0	0
03-01-2025	0	0	0	0	0.01
04-01-2025	0	0	0	0	0
05-01-2025	0	0	0	0	0
06-01-2025	0	0	0	0	0
07-01-2025	0	0	0	0	0
08-01-2025	0.01	0.01	4.28	0.01	0.46
09-01-2025	0.01	0.01	0.12	0.08	0.07
10-01-2025	0.01	0	1.31	0.01	0.42
11-01-2025	0.01	0	0.03	0.03	0.01
12-01-2025	0	0	0	0	0
13-01-2025	0	0	0.01	0	0.01
14-01-2025	0	0	0	0	0
15-01-2025	0	0	0	0.01	0.01
16-01-2025	0	0	0	0	0.01
17-01-2025	0	0	0	0	0
18-01-2025	0.01	0	0	0	0
19-01-2025	0	0	0	0	0
20-01-2025	0	0	0	0	0
21-01-2025	0	0	0	0	0
22-01-2025	0	0	0	0	0
23-01-2025	0.01	0	5.67	0	0
24-01-2025	0	0	0.07	0	0
25-01-2025	0	0	0	0	0.03

26-01-2025	0.01	0	0	0	0
27-01-2025	0	0	0	0	0
28-01-2025	0	0	1.64	0	0
29-01-2025	0.01	0	0	0	0.03
30-01-2025	0	0	0.12	0	0
31-01-2025	24.92	27.06	38.36	17.7	17.7
01-02-2025	0.03	0	0	0.04	0.01
02-02-2025	0	0	0.08	0.04	1.31
03-02-2025	0	0.01	0.01	0.01	0
04-02-2025	0	0	0	0.01	0.01
05-02-2025	0	0	0	0	0
06-02-2025	0.01	0.01	0.01	0	0
07-02-2025	0.01	0	0	0	0.03
08-02-2025	0	0	0	0	0
09-02-2025	0	0	0	0	0
10-02-2025	0	0	0	0	0
11-02-2025	16.39	11.15	28.52	12.13	18.69
12-02-2025	12.79	10.82	11.8	7.48	14.1
13-02-2025	0.1	0.1	0.04	0.12	0.17
14-02-2025	9.18	9.84	11.48	10.49	8.85
15-02-2025	81.31	28.52	6.16	5.9	41.44
16-02-2025	0.07	0.01	0.05	0.3	0.04
17-02-2025	13.43	17.84	4.26	6.3	23.93
18-02-2025	0.07	0.04	0.01	0.01	4.26
19-02-2025	0	0.01	0.04	0.01	0.98
20-02-2025	0	22.62	3.28	0	14.1
21-02-2025	0.01	0.09	0.12	0.01	1.64
22-02-2025	0.01	0.01	0.16	0.01	0.98
23-02-2025	0.01	0.03	0.17	0.05	1.64
24-02-2025	0.98	2.82	70.16	83.87	97.25
25-02-2025	0.33	1.08	60.66	47.21	31.28
26-02-2025	17.7	22.62	71.15	73.77	44.92
27-02-2025	0.07	0.07	0.13	0.13	6.56
28-02-2025	0	0	0	0	0.09
01-03-2025	0.01	0	0	0	0.01
02-03-2025	0	0	0	0	0
03-03-2025	0.01	0	0	0.58	0.87
04-03-2025	0.01	0.01	0	0	1.7
05-03-2025	0.01	0	0	0.01	0.51
06-03-2025	0	0	0.01	0.01	0.05
07-03-2025	0	0	0.01	0	0.01
08-03-2025	0	0	0	0	0.01
09-03-2025	0	0	0	0	0.01
10-03-2025	0	0	0	0	0
11-03-2025	0	0	0	0	12.2

12-03-2025	0	0	0	0	1.7
13-03-2025	0.08	0.01	0.35	3.93	40.59
14-03-2025	80	77.31	76.07	71.15	37.97
15-03-2025	9.51	9.31	8.85	8.52	20.2
16-03-2025	5.57	5.25	6.95	4.59	5.9
17-03-2025	3.02	0.79	29.11	0.98	5.57
18-03-2025	11.15	16.66	19.67	13.9	32.46
19-03-2025	0.2	0.12	0.1	0.01	0.66
20-03-2025	0	0	0	0	0
21-03-2025	0	0	0	0.03	0.03
22-03-2025	0	0	0	0	0.01
23-03-2025	0	0	0	0	0.03
24-03-2025	0	0	0.08	0	2.3
25-03-2025	0	0	0	0.04	0.07
26-03-2025	0	0	0	0	0
27-03-2025	0.01	0	0	0	0
28-03-2025	0	0	0	0	0
29-03-2025	0	0	0	0	0
30-03-2025	0	0	0	0	0
31-03-2025	0	0	0	0	0
01-04-2025	0	0	0	0	0
02-04-2025	0	0	0	0	0
03-04-2025	1.64	4.26	2.62	4	12.85
04-04-2025	0	0	0	0	0
05-04-2025	0	0	0	0	0
06-04-2025	22.62	16	21.38	21.31	9.05
07-04-2025	0	0	0	0	0
08-04-2025	0.09	0.04	0.04	0.07	0.07
09-04-2025	0	0	0	0	0
10-04-2025	0	0	0	0	0
11-04-2025	0.01	0.01	0	0	0.03
12-04-2025	0	0	0	0	0
13-04-2025	0	0	0	0	0
14-04-2025	7.87	84.59	12.46	2.3	3.87
15-04-2025	0	0	0	0	0
16-04-2025	2.3	2.1	3.28	23.28	26.26
17-04-2025	0.79	17.05	0.59	18.62	5.9
18-04-2025	0.01	0.08	0.26	0.14	1.64
19-04-2025	0	0	0	0	0
20-04-2025	33.38	0	0.64	37.7	8.52
21-04-2025	1.31	3.28	0.13	2.62	18.1
22-04-2025	0.1	0.05	0.29	0.05	16
23-04-2025	0	0	0	0	0
24-04-2025	53.44	56.72	0.13	15.48	49.84

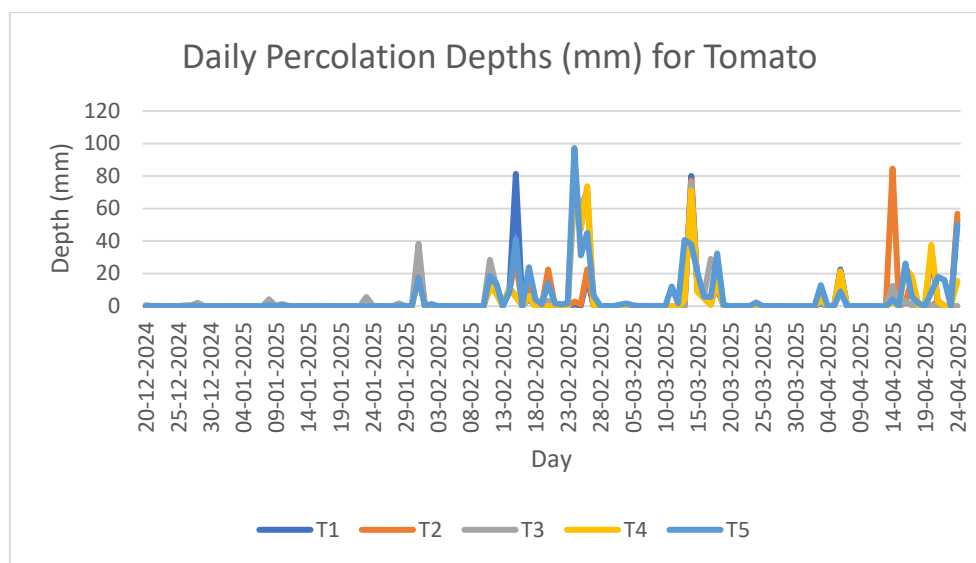


Figure 4.21. Daily percolation depth under tomato cultivation for pots T1, T2, T3, T4 and T5.

4.5 Determination of ETc of different crops

Irrigation for crop production constitutes the most part of agricultural water worldwide. As most of the water withdrawn by agriculture is lost through crop evapotranspiration (ETC), efficient water management requires accurate and reliable estimation of ETC. There are several methods for estimating ETC developed by hydrologists. Each has its concept and developed for a specific regional climate. The ETC was based on the calculations of the water balance of the system. This was estimated daily by using the plastic pots as micro-lysimeters. In this way, water provided by irrigation (Q), stored by the substrate (W), and drained (D) was calculated daily:

$$ETc = Q + R - D \quad 4.2$$

The daily and monthly variation of ETc is shown in the following Table 4.19.

Table 4.19. Daily variation of Etc values for Beans (mm)

Date	B1	B2	B3	B4	B5
20-12-2024	0.94	0.94	0.94	0.94	0.95
21-12-2024	0.94	0.94	0.94	0.94	0.94
22-12-2024	0.95	0.95	0.95	0.95	0.95
23-12-2024	1.01	1.01	1.01	1.01	1.01
24-12-2024	1.01	1.01	1.01	1.01	1.01
25-12-2024	1.02	1.02	1.02	1.02	1.02
26-12-2024	1.04	1.04	1.04	1.04	1.03

27-12-2024	1.1	1.09	1.1	1.1	1.1
28-12-2024	1.12	1.13	1.13	1.13	1.13
29-12-2024	0.86	0.86	0.86	0.86	0.85
30-12-2024	1.2	1.2	1.2	1.2	1.2
31-12-2024	1.21	1.21	1.21	1.21	1.21
01-01-2025	1.24	1.24	1.24	1.24	1.24
02-01-2025	1.26	1.26	1.26	1.26	1.26
03-01-2025	1.29	1.29	1.29	1.29	1.29
04-01-2025	1.33	1.32	1.32	1.32	1.32
05-01-2025	1.35	1.35	1.35	1.35	1.35
06-01-2025	1.34	1.34	1.34	1.34	1.34
07-01-2025	1.22	1.22	1.22	1.22	1.22
08-01-2025	1.02	1.02	1.01	1.01	1.02
09-01-2025	1.34	1.33	1.34	1.33	1.34
10-01-2025	1.46	1.46	1.46	1.46	1.46
11-01-2025	1.45	1.45	1.46	1.45	1.45
13-01-2025	1.35	1.35	1.35	1.35	1.36
14-01-2025	1.36	1.36	1.36	1.36	1.36
16-01-2025	1.68	1.68	1.68	1.68	1.68
17-01-2025	1.67	1.67	1.67	1.67	1.67
18-01-2025	1.72	1.72	1.72	1.72	1.72
20-01-2025	1.81	1.81	1.81	1.81	1.81
21-01-2025	1.79	1.79	1.79	1.79	1.79
22-01-2025	1.8	1.8	1.8	1.8	1.8
23-01-2025	1.6	1.6	1.59	1.6	1.6
24-01-2025	1.53	1.53	1.53	1.53	1.53
25-01-2025	1.56	1.56	1.56	1.56	1.56
26-01-2025	1.78	1.78	1.78	1.78	1.78
27-01-2025	1.8	1.8	1.8	1.8	1.8

28-01-2025	1.35	1.35	1.35	1.35	1.35
29-01-2025	1.3	1.3	1.3	1.3	1.3
30-01-2025	1.21	1.21	1.21	1.21	1.21
31-01-2025	1.29	1.29	1.3	1.29	1.29
01-02-2025	1.29	1.29	1.29	1.29	1.29
02-02-2025	1.46	1.46	1.46	1.46	1.46
03-02-2025	1.71	1.71	1.71	1.71	1.71
04-02-2025	1.82	1.82	1.82	1.82	1.82
05-02-2025	1.77	1.78	1.77	1.78	1.78
06-02-2025	1.88	1.87	1.88	1.88	1.88
07-02-2025	1.33	1.33	1.33	1.33	1.33
08-02-2025	1.76	1.76	1.76	1.76	1.76
09-02-2025	1.91	1.91	1.91	1.91	1.91
10-02-2025	1.85	1.85	1.85	1.85	1.85
11-02-2025	1.9	1.9	1.9	1.9	1.9
12-02-2025	1.35	1.36	1.35	1.35	1.35
13-02-2025	1.45	1.45	1.45	1.45	1.45
14-02-2025	1.72	1.73	1.72	1.72	1.72
15-02-2025	1.61	1.61	1.61	1.61	1.61
16-02-2025	1.58	1.58	1.58	1.58	1.58
17-02-2025	1.47	1.47	1.47	1.47	1.47
18-02-2025	1.65	1.65	1.65	1.65	1.65
19-02-2025	1.95	1.95	1.95	1.95	1.95
20-02-2025	1.93	1.92	1.92	1.93	1.93
21-02-2025	2.2	2.2	2.21	2.21	2.2
22-02-2025	1.56	1.56	1.56	1.56	1.56
23-02-2025	1.6	1.6	1.59	1.6	1.6
27-02-2025	2.32	2.32	2.31	2.32	2.32
28-02-2025	1.88	1.88	1.88	1.88	1.88

01-03-2025	1.9	1.9	1.9	1.9	1.9
02-03-2025	1.94	1.94	1.94	1.94	1.94
03-03-2025	1.78	1.77	1.78	1.78	1.78
04-03-2025	1.78	1.78	1.78	1.78	1.78
05-03-2025	1.68	1.69	1.69	1.69	1.69
06-03-2025	1.68	1.69	1.69	1.69	1.69
07-03-2025	1.66	1.66	1.66	1.66	1.66
08-03-2025	1.63	1.63	1.63	1.63	1.63
09-03-2025	1.66	1.66	1.66	1.66	1.66
10-03-2025	1.61	1.61	1.61	1.61	1.61
11-03-2025	1.57	1.57	1.57	1.57	1.57
12-03-2025	1.52	1.51	1.51	1.51	1.51
13-03-2025	1.38	1.38	1.37	1.38	1.38
14-03-2025	1.33	1.33	1.34	1.33	1.33
15-03-2025	1.34	1.34	1.34	1.34	1.33
16-03-2025	1.36	1.36	1.37	1.36	1.37
18-03-2025	2.1	2.1	2.1	2.11	2.1
19-03-2025	1.96	1.96	1.96	1.96	1.96
20-03-2025	1.34	1.34	1.34	1.34	1.34
21-03-2025	1.28	1.28	1.28	1.28	1.28
22-03-2025	1.29	1.29	1.29	1.29	1.29
23-03-2025	1.55	1.55	1.55	1.55	1.55
24-03-2025	1.56	1.56	1.55	1.55	1.55
25-03-2025	1.77	1.77	1.77	1.77	1.77
26-03-2025	1.7	1.7	1.7	1.7	1.7
27-03-2025	1.57	1.57	1.57	1.57	1.57
28-03-2025	1.17	1.17	1.17	1.17	1.17
29-03-2025	1.08	1.08	1.08	1.08	1.08
30-03-2025	1.07	1.07	1.07	1.07	1.07

31-03-2025	1.36	1.36	1.36	1.36	1.36
01-04-2025	1.05	1.05	1.05	1.05	1.05
03-04-2025	1.01	1.01	1.01	1.01	1.01
04-04-2025	1.02	1.02	1.02	1.02	1.02
05-04-2025	0.97	0.97	0.97	0.97	0.97
06-04-2025	1.02	1.02	1.02	1.02	1.02
07-04-2025	0.85	0.85	0.85	0.85	0.85
08-04-2025	0.96	0.96	0.96	0.96	0.96
09-04-2025	0.97	0.97	0.97	0.97	0.97
10-04-2025	0.76	0.76	0.76	0.76	0.76
11-04-2025	0.65	0.65	0.65	0.65	0.65
12-04-2025	0.68	0.68	0.68	0.68	0.68
13-04-2025	0.69	0.69	0.69	0.69	0.69
14-04-2025	0.67	0.67	0.67	0.67	0.67
15-04-2025	0.5	0.5	0.5	0.5	0.5
16-04-2025	0.51	0.51	0.51	0.51	0.51
17-04-2025	0.42	0.42	0.42	0.42	0.42
18-04-2025	0.44	0.44	0.44	0.44	0.44
19-04-2025	0.37	0.37	0.37	0.37	0.37
20-04-2025	0.33	0.33	0.33	0.33	0.33
21-04-2025	0.22	0.22	0.22	0.22	0.22
22-04-2025	0.18	0.18	0.18	0.18	0.18
23-04-2025	0.13	0.13	0.13	0.13	0.13
24-04-2025	0.09	0.09	0.09	0.09	0.09

Table 4.20. Monthly variations of Etc values for Beans, mm

Month	B1	B2	B3	B4	B5
December, 2024	1.03	1.03	1.03	1.03	1.03
January, 2025	1.46	1.46	1.46	1.46	1.46

February, 2025	1.72	1.72	1.72	1.72	1.72
March, 2025	1.55	1.55	1.55	1.55	1.55
April, 2025	0.63	0.63	0.63	0.63	0.63

Table 4.21. Daily variation of Etc values for Carrot (mm)

Date	C1	C2	C3	C4	C5
20-12-2024	1.51	1.51	1.51	1.51	1.51
21-12-2024	1.48	1.48	1.48	1.48	1.48
22-12-2024	1.53	1.53	1.53	1.53	1.53
23-12-2024	1.58	1.58	1.58	1.58	1.58
24-12-2024	1.54	1.54	1.54	1.54	1.54
25-12-2024	1.51	1.51	1.51	1.51	1.51
26-12-2024	1.5	1.5	1.5	1.5	1.5
27-12-2024	1.54	1.54	1.54	1.54	1.54
28-12-2024	1.54	1.54	1.54	1.54	1.54
29-12-2024	1.15	1.15	1.15	1.15	1.15
30-12-2024	1.56	1.56	1.56	1.56	1.56
31-12-2024	1.54	1.54	1.54	1.54	1.54
01-01-2025	1.54	1.54	1.54	1.54	1.54
02-01-2025	1.53	1.53	1.53	1.53	1.53
03-01-2025	1.53	1.53	1.53	1.53	1.53
04-01-2025	1.53	1.53	1.53	1.53	1.53
05-01-2025	1.53	1.53	1.53	1.53	1.53
06-01-2025	1.48	1.48	1.48	1.48	1.48
07-01-2025	1.31	1.31	1.31	1.31	1.31
08-01-2025	1.07	1.07	1.07	1.07	1.07
09-01-2025	1.38	1.38	1.38	1.38	1.38
10-01-2025	1.47	1.47	1.47	1.47	1.47
11-01-2025	1.44	1.44	1.44	1.44	1.44

13-01-2025	1.28	1.28	1.28	1.28	1.28
14-01-2025	1.26	1.26	1.26	1.26	1.26
16-01-2025	1.5	1.5	1.5	1.5	1.5
17-01-2025	1.46	1.46	1.46	1.46	1.46
18-01-2025	1.47	1.47	1.47	1.47	1.47
20-01-2025	1.52	1.52	1.52	1.52	1.52
21-01-2025	1.5	1.5	1.5	1.5	1.5
22-01-2025	1.5	1.5	1.5	1.5	1.5
23-01-2025	1.32	1.32	1.32	1.32	1.32
24-01-2025	1.25	1.25	1.25	1.25	1.25
25-01-2025	1.27	1.27	1.27	1.27	1.27
26-01-2025	1.44	1.44	1.44	1.44	1.44
27-01-2025	1.44	1.44	1.44	1.44	1.44
28-01-2025	1.07	1.07	1.07	1.07	1.07
29-01-2025	1.02	1.02	1.02	1.02	1.02
30-01-2025	0.95	0.95	0.95	0.95	0.95
31-01-2025	1	1	1	1	1
01-02-2025	1	1	1	1	1
02-02-2025	1.12	1.12	1.12	1.12	1.12
03-02-2025	1.3	1.3	1.3	1.3	1.3
04-02-2025	1.37	1.37	1.37	1.37	1.37
05-02-2025	1.33	1.33	1.33	1.33	1.33
06-02-2025	1.39	1.39	1.39	1.39	1.39
07-02-2025	0.98	0.98	0.98	0.98	0.98
08-02-2025	1.28	1.28	1.28	1.28	1.28
09-02-2025	1.38	1.38	1.38	1.38	1.38
10-02-2025	1.33	1.33	1.33	1.33	1.33
11-02-2025	1.36	1.36	1.36	1.36	1.36
12-02-2025	0.96	0.96	0.96	0.96	0.96

13-02-2025	1.01	1.01	1.01	1.01	1.01
14-02-2025	1.2	1.2	1.2	1.2	1.2
15-02-2025	1.11	1.11	1.11	1.11	1.11
16-02-2025	1.08	1.08	1.08	1.08	1.08
17-02-2025	0.99	0.99	0.99	0.99	0.99
18-02-2025	1.11	1.11	1.11	1.11	1.11
19-02-2025	1.3	1.3	1.3	1.3	1.3
20-02-2025	1.27	1.27	1.27	1.27	1.27
21-02-2025	1.45	1.45	1.45	1.45	1.45
22-02-2025	1.01	1.01	1.01	1.01	1.01
23-02-2025	1.03	1.03	1.03	1.03	1.03
27-02-2025	1.44	1.44	1.44	1.44	1.44
28-02-2025	1.16	1.16	1.16	1.16	1.16
01-03-2025	1.16	1.16	1.16	1.16	1.16
02-03-2025	1.17	1.17	1.17	1.17	1.17
03-03-2025	1.07	1.07	1.07	1.07	1.07
04-03-2025	1.06	1.06	1.06	1.06	1.06
05-03-2025	1.01	1.01	1.01	1.01	1.01
06-03-2025	1.02	1.02	1.02	1.02	1.02
07-03-2025	1.02	1.02	1.02	1.02	1.02
08-03-2025	1.01	1.01	1.01	1.01	1.01
09-03-2025	1.04	1.04	1.04	1.04	1.04
10-03-2025	1.02	1.02	1.02	1.02	1.02
11-03-2025	1.01	1.01	1.01	1.01	1.01
12-03-2025	0.99	0.99	0.99	0.99	0.99
13-03-2025	0.91	0.91	0.91	0.91	0.91
14-03-2025	0.89	0.89	0.89	0.89	0.89
15-03-2025	0.91	0.91	0.91	0.91	0.91
16-03-2025	0.94	0.94	0.94	0.94	0.94

18-03-2025	1.5	1.5	1.5	1.5	1.5
19-03-2025	1.42	1.42	1.42	1.42	1.42
20-03-2025	0.99	0.99	0.99	0.99	0.99
21-03-2025	0.96	0.96	0.96	0.96	0.96
22-03-2025	0.98	0.98	0.98	0.98	0.98
23-03-2025	1.21	1.21	1.21	1.21	1.21
24-03-2025	1.23	1.23	1.23	1.23	1.23
25-03-2025	1.43	1.43	1.43	1.43	1.43
26-03-2025	1.41	1.41	1.41	1.41	1.41
27-03-2025	1.33	1.33	1.33	1.33	1.33
28-03-2025	1.02	1.02	1.02	1.02	1.02
29-03-2025	0.96	0.96	0.96	0.96	0.96
30-03-2025	0.97	0.97	0.97	0.97	0.97
31-03-2025	1.27	1.27	1.27	1.27	1.27
01-04-2025	1.01	1.01	1.01	1.01	1.01
03-04-2025	1.03	1.03	1.03	1.03	1.03
04-04-2025	1.08	1.08	1.08	1.08	1.08
05-04-2025	1.06	1.06	1.06	1.06	1.06
06-04-2025	1.16	1.16	1.16	1.16	1.16
07-04-2025	1	1	1	1	1
08-04-2025	1.19	1.19	1.19	1.19	1.19
09-04-2025	1.25	1.25	1.25	1.25	1.25
10-04-2025	1.04	1.04	1.04	1.04	1.04
11-04-2025	0.93	0.93	0.93	0.93	0.93
12-04-2025	1.02	1.02	1.02	1.02	1.02
13-04-2025	1.11	1.11	1.11	1.11	1.11
14-04-2025	1.15	1.15	1.15	1.15	1.15
15-04-2025	0.91	0.91	0.91	0.91	0.91
16-04-2025	1.03	1.03	1.03	1.03	1.03

17-04-2025	0.91	0.91	0.91	0.91	0.91
18-04-2025	1.07	1.07	1.07	1.07	1.07
19-04-2025	1	1	1	1	1
20-04-2025	1.03	1.03	1.03	1.03	1.03
21-04-2025	0.8	0.8	0.8	0.8	0.8
22-04-2025	0.81	0.81	0.81	0.81	0.81
23-04-2025	0.78	0.78	0.78	0.78	0.78
24-04-2025	0.76	0.76	0.76	0.76	0.76

Table 4.22. Monthly variations of Etc values for Carrot, mm

Month	C1	C2	C3	C4	C5
December, 2024	1.5	1.5	1.5	1.5	1.5
January, 2025	1.36	1.36	1.36	1.36	1.36
February, 2025	1.2	1.2	1.2	1.2	1.2
March, 2025	1.1	1.1	1.1	1.1	1.1
April, 2025	1.01	1.01	1.01	1.01	1.01

Table 4.23. Daily variation of Etc values for Mustard (mm)

Date	M1	M2	M3	M4	M5
20-12-2024	1.45	1.45	1.45	1.45	1.45
21-12-2024	1.39	1.39	1.38	1.39	1.39
22-12-2024	1.35	1.35	1.35	1.35	1.35
23-12-2024	1.37	1.37	1.37	1.37	1.37
24-12-2024	1.33	1.33	1.33	1.33	1.33
25-12-2024	1.29	1.29	1.29	1.29	1.29
26-12-2024	1.26	1.26	1.26	1.26	1.26
27-12-2024	1.28	1.27	1.28	1.27	1.28
28-12-2024	1.27	1.26	1.27	1.27	1.27

29-12-2024	0.93	0.93	0.93	0.93	0.93
30-12-2024	1.25	1.25	1.25	1.25	1.25
31-12-2024	1.21	1.21	1.21	1.21	1.21
01-01-2025	1.2	1.2	1.2	1.2	1.2
02-01-2025	1.17	1.17	1.17	1.17	1.17
03-01-2025	1.15	1.16	1.15	1.15	1.15
04-01-2025	1.14	1.14	1.14	1.14	1.14
05-01-2025	1.12	1.12	1.12	1.12	1.12
06-01-2025	1.06	1.06	1.06	1.06	1.06
07-01-2025	0.93	0.93	0.93	0.93	0.93
08-01-2025	0.75	0.74	0.74	0.74	0.75
09-01-2025	0.94	0.94	0.94	0.94	0.94
10-01-2025	0.99	0.99	0.99	1	0.99
11-01-2025	0.95	0.95	0.95	0.95	0.95
13-01-2025	0.81	0.81	0.81	0.81	0.81
14-01-2025	0.78	0.78	0.78	0.78	0.78
16-01-2025	0.89	0.89	0.89	0.89	0.89
17-01-2025	0.85	0.85	0.85	0.85	0.85
18-01-2025	0.84	0.84	0.84	0.84	0.84
20-01-2025	0.83	0.83	0.83	0.83	0.83
21-01-2025	0.79	0.79	0.79	0.79	0.79
22-01-2025	0.77	0.77	0.77	0.77	0.77
23-01-2025	0.66	0.66	0.66	0.66	0.66
24-01-2025	0.61	0.61	0.61	0.61	0.61
25-01-2025	0.6	0.6	0.59	0.6	0.59
26-01-2025	0.68	0.68	0.68	0.68	0.67
27-01-2025	0.68	0.68	0.68	0.68	0.68
28-01-2025	0.5	0.5	0.5	0.5	0.51
29-01-2025	0.48	0.48	0.48	0.48	0.48

30-01-2025	0.44	0.44	0.44	0.44	0.44
31-01-2025	0.47	0.47	0.47	0.46	0.47
01-02-2025	0.46	0.46	0.46	0.46	0.46
02-02-2025	0.52	0.52	0.52	0.52	0.52
03-02-2025	0.6	0.6	0.6	0.6	0.6
04-02-2025	0.63	0.63	0.63	0.63	0.63
05-02-2025	0.61	0.61	0.61	0.61	0.61
06-02-2025	0.64	0.64	0.64	0.64	0.64
07-02-2025	0.45	0.45	0.45	0.45	0.45
08-02-2025	0.59	0.59	0.59	0.59	0.59
09-02-2025	0.63	0.63	0.63	0.63	0.63
10-02-2025	0.61	0.61	0.61	0.61	0.61
11-02-2025	0.61	0.61	0.61	0.61	0.61

Table 4.24. Monthly variations of Etc values for Mustard, mm

Month	M1	M2	M3	M4	M5
December, 2024	1.28	1.28	1.28	1.28	1.28
January, 2025	0.82	0.82	0.82	0.82	0.82
February, 2025	0.58	0.58	0.58	0.58	0.58

Table 4.25. Daily variation of Etc values for Tomato (mm)

Date	T1	T2	T3	T4	T5
20-12-2024	1.56	1.56	1.56	1.56	1.56
21-12-2024	1.53	1.53	1.53	1.53	1.53
22-12-2024	1.54	1.54	1.54	1.54	1.54
23-12-2024	1.61	1.61	1.61	1.61	1.61
24-12-2024	1.6	1.6	1.6	1.6	1.61
25-12-2024	1.59	1.59	1.59	1.59	1.59

26-12-2024	1.61	1.61	1.61	1.61	1.61
27-12-2024	1.65	1.65	1.64	1.65	1.65
28-12-2024	1.64	1.64	1.64	1.64	1.64
29-12-2024	1.22	1.22	1.22	1.22	1.22
30-12-2024	1.65	1.65	1.65	1.65	1.65
31-12-2024	1.62	1.62	1.62	1.62	1.62
01-01-2025	1.62	1.62	1.62	1.62	1.62
02-01-2025	1.6	1.6	1.6	1.6	1.6
03-01-2025	1.59	1.59	1.59	1.59	1.6
04-01-2025	1.59	1.59	1.59	1.59	1.59
05-01-2025	1.58	1.58	1.58	1.58	1.58
06-01-2025	1.52	1.52	1.52	1.52	1.52
07-01-2025	1.35	1.35	1.35	1.35	1.35
08-01-2025	1.1	1.1	1.09	1.1	1.1
09-01-2025	1.41	1.41	1.4	1.4	1.4
10-01-2025	1.5	1.49	1.5	1.5	1.49
11-01-2025	1.46	1.45	1.45	1.45	1.46
13-01-2025	1.28	1.28	1.29	1.28	1.29
14-01-2025	1.25	1.25	1.25	1.25	1.25
16-01-2025	1.48	1.48	1.48	1.48	1.48
17-01-2025	1.43	1.43	1.43	1.43	1.43
18-01-2025	1.44	1.44	1.44	1.44	1.44
20-01-2025	1.48	1.48	1.48	1.48	1.48
21-01-2025	1.45	1.45	1.45	1.45	1.45
22-01-2025	1.44	1.44	1.44	1.44	1.44
23-01-2025	1.26	1.26	1.25	1.26	1.26
24-01-2025	1.19	1.19	1.18	1.19	1.19
25-01-2025	1.2	1.2	1.2	1.2	1.19
26-01-2025	1.35	1.35	1.35	1.35	1.35

27-01-2025	1.35	1.35	1.35	1.35	1.35
28-01-2025	1	1	1	1	1
29-01-2025	0.95	0.95	0.95	0.95	0.95
30-01-2025	0.87	0.87	0.87	0.87	0.87
31-01-2025	0.92	0.91	0.92	0.92	0.92
01-02-2025	0.91	0.91	0.91	0.91	0.91
02-02-2025	1.01	1.01	1.01	1.01	1.01
03-02-2025	1.17	1.17	1.17	1.17	1.17
04-02-2025	1.22	1.22	1.22	1.22	1.22
05-02-2025	1.17	1.17	1.17	1.17	1.17
06-02-2025	1.23	1.23	1.23	1.22	1.22
07-02-2025	0.86	0.85	0.85	0.85	0.85
08-02-2025	1.11	1.11	1.11	1.11	1.11
09-02-2025	1.19	1.19	1.19	1.19	1.19
10-02-2025	1.14	1.14	1.14	1.14	1.14
11-02-2025	1.15	1.15	1.15	1.15	1.15
12-02-2025	0.8	0.8	0.81	0.8	0.8
13-02-2025	0.85	0.85	0.84	0.84	0.85
14-02-2025	0.99	0.99	0.99	0.99	0.99
15-02-2025	0.91	0.92	0.91	0.91	0.91
16-02-2025	0.87	0.88	0.88	0.88	0.88
17-02-2025	0.8	0.8	0.8	0.8	0.81
18-02-2025	0.88	0.89	0.89	0.89	0.89
19-02-2025	1.03	1.03	1.03	1.03	1.03
20-02-2025	1	1	1	1	1
21-02-2025	1.13	1.13	1.12	1.13	1.12
22-02-2025	0.78	0.78	0.78	0.78	0.78
23-02-2025	0.79	0.78	0.78	0.79	0.78
27-02-2025	1.05	1.05	1.05	1.05	1.05

28-02-2025	0.84	0.84	0.84	0.84	0.84
01-03-2025	0.83	0.83	0.83	0.83	0.83
02-03-2025	0.83	0.83	0.83	0.83	0.83
03-03-2025	0.75	0.74	0.74	0.74	0.74
04-03-2025	0.73	0.73	0.73	0.73	0.73
05-03-2025	0.69	0.69	0.69	0.69	0.69
06-03-2025	0.69	0.69	0.69	0.69	0.69
07-03-2025	0.67	0.67	0.67	0.67	0.67
08-03-2025	0.66	0.66	0.66	0.66	0.66
09-03-2025	0.67	0.67	0.67	0.67	0.67
10-03-2025	0.65	0.65	0.65	0.65	0.65
11-03-2025	0.63	0.63	0.63	0.63	0.63
12-03-2025	0.6	0.6	0.6	0.6	0.61
13-03-2025	0.55	0.55	0.55	0.55	0.55
14-03-2025	0.53	0.53	0.52	0.53	0.53
15-03-2025	0.53	0.53	0.53	0.53	0.52
16-03-2025	0.54	0.53	0.54	0.54	0.54
18-03-2025	0.82	0.82	0.82	0.82	0.82
19-03-2025	0.76	0.76	0.76	0.76	0.76
20-03-2025	0.52	0.52	0.52	0.52	0.52
21-03-2025	0.49	0.49	0.49	0.49	0.49
22-03-2025	0.49	0.49	0.49	0.49	0.49
23-03-2025	0.59	0.59	0.59	0.59	0.58
24-03-2025	0.59	0.59	0.58	0.59	0.58
25-03-2025	0.66	0.66	0.66	0.66	0.66
26-03-2025	0.63	0.63	0.63	0.63	0.63
27-03-2025	0.58	0.58	0.58	0.58	0.58
28-03-2025	0.43	0.43	0.43	0.43	0.43
29-03-2025	0.39	0.39	0.39	0.39	0.39

30-03-2025	0.38	0.38	0.38	0.38	0.38
31-03-2025	0.48	0.48	0.48	0.48	0.48
01-04-2025	0.37	0.37	0.37	0.37	0.37
03-04-2025	0.34	0.34	0.34	0.34	0.34
04-04-2025	0.34	0.34	0.34	0.34	0.34
05-04-2025	0.32	0.32	0.32	0.32	0.32
06-04-2025	0.33	0.33	0.33	0.33	0.33
07-04-2025	0.27	0.27	0.27	0.27	0.27
08-04-2025	0.3	0.3	0.3	0.3	0.3
09-04-2025	0.29	0.29	0.29	0.29	0.29
10-04-2025	0.22	0.22	0.22	0.22	0.22
11-04-2025	0.18	0.18	0.18	0.18	0.18
12-04-2025	0.18	0.18	0.18	0.18	0.18
13-04-2025	0.17	0.17	0.17	0.17	0.17
14-04-2025	0.15	0.15	0.15	0.15	0.15
15-04-2025	0.1	0.1	0.1	0.1	0.1
16-04-2025	0.09	0.09	0.09	0.09	0.09
17-04-2025	0.06	0.06	0.06	0.06	0.06
18-04-2025	0.05	0.05	0.05	0.05	0.05
19-04-2025	0.02	0.02	0.02	0.02	0.02
20-04-2025	0	0	0	0	0
21-04-2025	0	0	0	0	0
22-04-2025	0	0	0	0	0
23-04-2025	0	0	0	0	0
24-04-2025	0	0	0	0	0

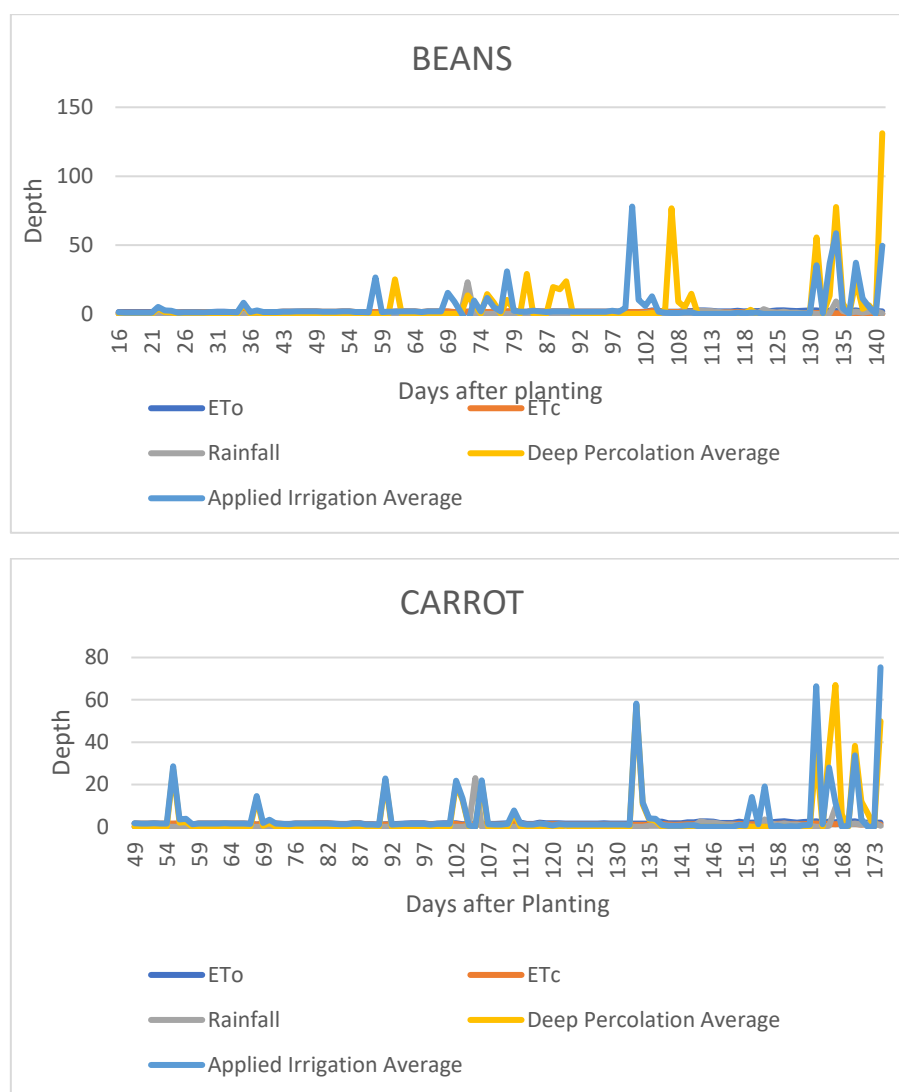
Table 4.26. Monthly variations of Etc values for Tomato, mm

Month	T1	T2	T3	T4	T5
December, 2024	1.57	1.57	1.57	1.57	1.57

January, 2025	1.35	1.34	1.34	1.34	1.34
February, 2025	1	1	0.99	0.99	0.99
March, 2025	0.61	0.61	0.61	0.61	0.61
April, 2025	0.21	0.21	0.21	0.21	0.21

4.5.1 Daily rainfall, ETo and ETc rates, deep percolation and applied irrigation of Vegetable Crops Tested

Figure 4.22 illustrates the daily variation in five key parameters - Daily Rainfall, Reference Crop Evapotranspiration (ET_o), Actual Crop Evapotranspiration (ET_c), Deep Percolation Average and Applied Irrigation Average for vegetable crops across different days after planting. The graph shows a similar trend of ET_o and ET_c in that an increase in ET_o recorded a corresponding increase in ET_c of the vegetable crops studied. During the early stages of growth, it was observed that the ET_c values of all vegetable crops were lower than the ET_o.



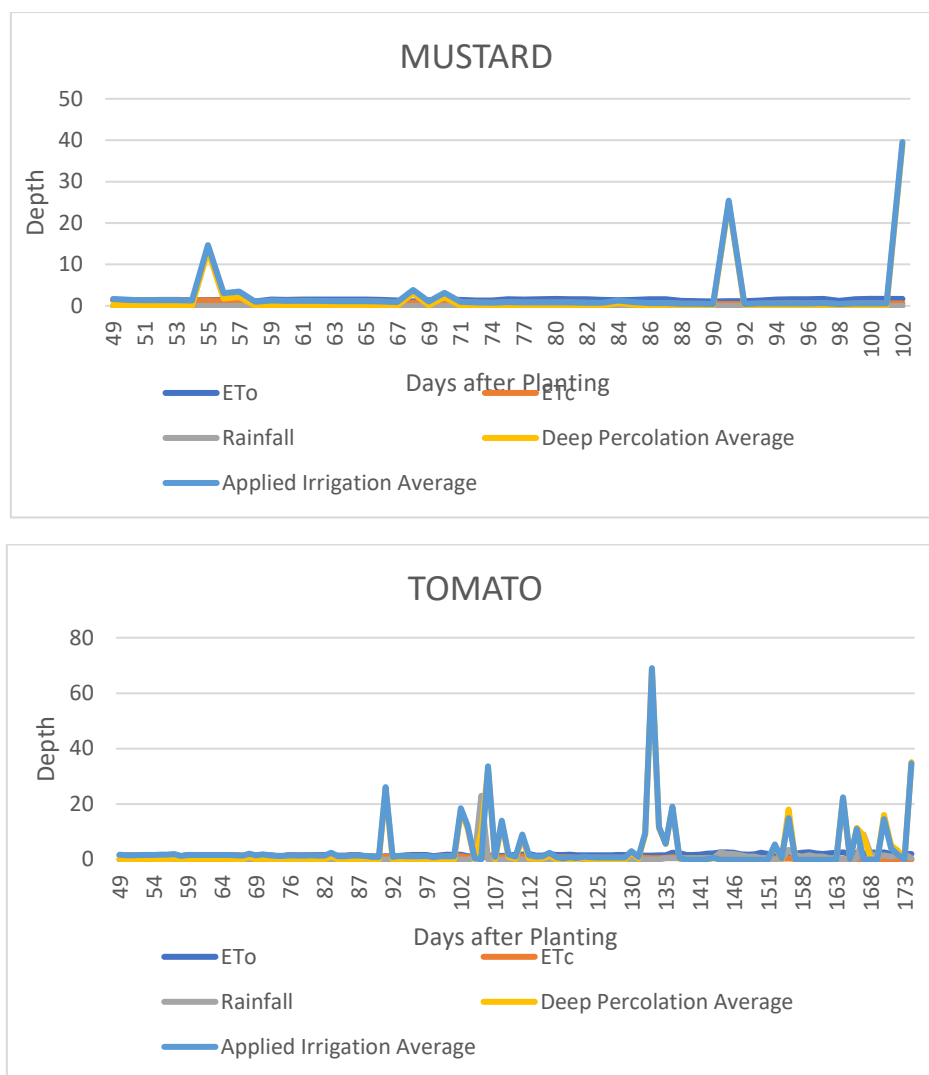


Figure 4.22. Daily Rainfall, Reference Crop Evapotranspiration (ETo, mm/day), Actual Crop Evapotranspiration (ETc, mm/day), Deep Percolation Average (mm/day) and Applied Irrigation Average (mm/day) of vegetable crops at different days after planting.

4.6 Determination of Kc

Crop coefficients (Kc) are the ratio of the evapotranspiration of the crop (ETC) to a reference crop (ETO). It is used to determine exact plant water usage and crop evapotranspiration (ETc). Irrigation scheduling can then be improved for private consultants and growers to avoid water over use and to more precisely meet the crop water demand to produce greater yields, crop quality, and enhanced water use efficiency.

In this study, Kc values were determined for different crops at various growth stages. The results revealed that Kc values vary significantly depending on crop type, growth stage, and climates.

4.6.1 Crop coefficient of Beans

Beans exhibited a relatively rapid rise in Kc values.

Initial Stage (0–20 Days): Lower Kc (~0.4) due to low leaf area.

Mid-Season (30–80 Days): Sharp increase in Kc as the crop rapidly develops.

Late Stage (after 90 Days): Decline in Kc reflects maturity and drying phases.

The Kc curves of Beans crop and its values are presented in the below Figure 4.23 and Table 4.27 respectively.

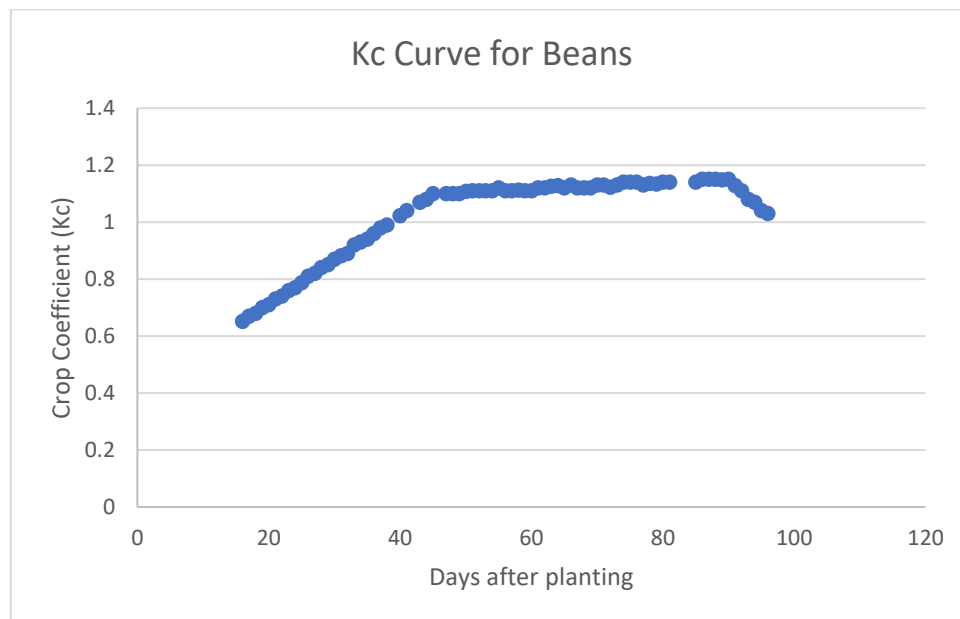


Figure 4.23. Crop coefficient of beans as a function of days after planting.

Table 4.27. KC values of Beans

Beans	Kc (From FAO)
Kc mid	1.10–1.15
Kc end	0.60–0.75

4.6.2 Crop Coefficient of Carrot

The Carrot Kc curve showed a moderate peak (~1.05) at around 90–100 days.

Initial Stage (0–30 Days): Lowest Kc (~0.35), typical for root vegetables with slow initial growth.

Mid-Season (40–100 Days): Gradual increase to peak as foliage develops, enhancing ET.

Late Stage (after 110 Days): Reduction in Kc as foliage declines, root maturity completes. The Kc curve represents the crop water use pattern throughout its growth stages.

The Kc curves of Beans crop and its values are presented in the below Figure 4.24 and Table 4.28 respectively.

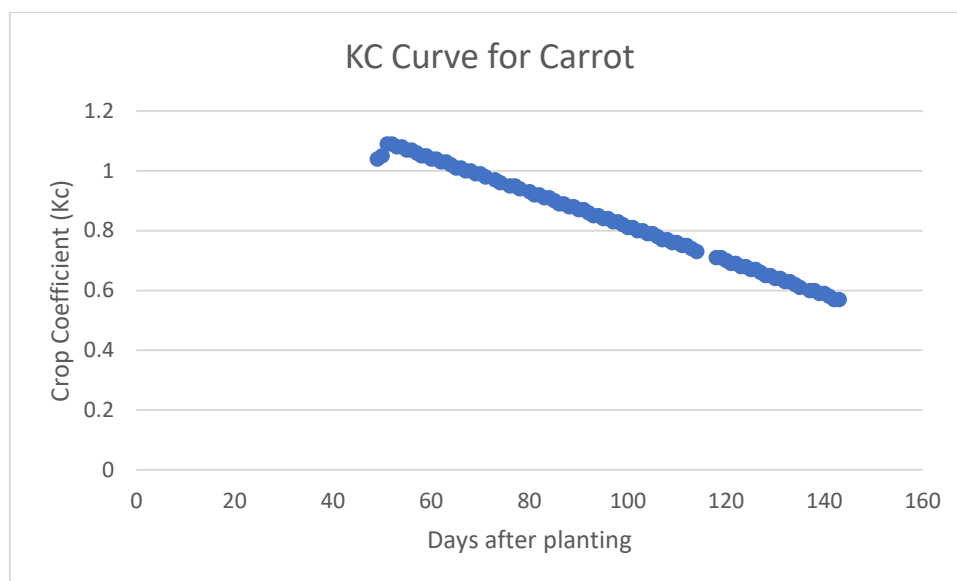


Figure 4.24. Crop coefficient of carrot as a function of days after planting.

Table 4.28. KC values of Carrot

Carrot	Kc (from FAO)
Kc mid	1.00–1.05
Kc end	0.55–0.70

4.6.3 Crop coefficient of Mustard

Initial Stage (0–25 Days): $K_c \approx 0.35$ This low K_c value reflects minimal canopy cover and low evapotranspiration (ET) during seedling emergence and early vegetative development.

Mid-Season Stage (50–80 Days): K_c peaks at approximately 1.06, showing the highest water requirement during full bloom and active vegetative expansion. The canopy is fully developed, and ET_c approaches E_{to} .

Late Season (80–110 Days) K_c declines to about 0.425 as the crop begins maturing and leaf senescence reduces transpiration. The K_c values of mustard crops is shown in Figure 4.25 and presented on Table 4.29 below respectively.

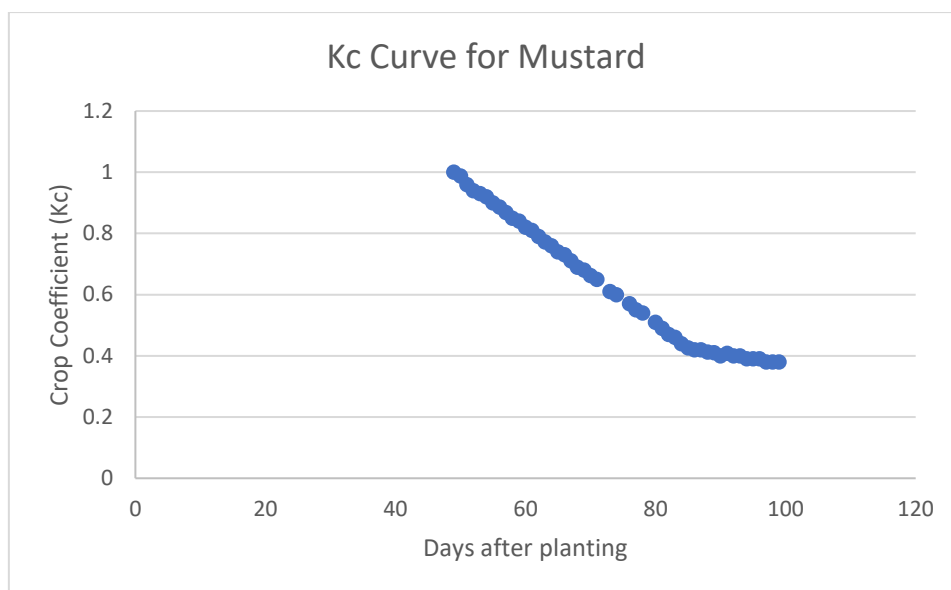


Figure 4.25. Crop coefficient of mustard as a function of days after planting.

Table 4.29. KC values of Mustard

Mustard	Kc (From FAO)
Kc mid	1.05–1.10
Kc end	0.40–0.50

4.6.4 Crop coefficient of Tomato

The Kc values for Tomato showed a progressive increase from the initial stage (~0.45) to the mid-season peak (~1.2) before declining during the late season.

Initial Stage (0–30 Days): Low Kc values indicate minimal evapotranspiration due to small canopy coverage.

Mid-Season (60–100 Days): The Kc peaked at ~1.2, coinciding with full crop cover and maximum water requirement.

Late Season (after 110 Days): A gradual decline in Kc reflects plant senescence and reduced evapotranspiration.

The crop coefficient (Kc) values for tomatoes have been analyzed and are presented in both graphical (Figure 4.26) and tabular formats (Table 4.30) to illustrate their variation over the growth period.

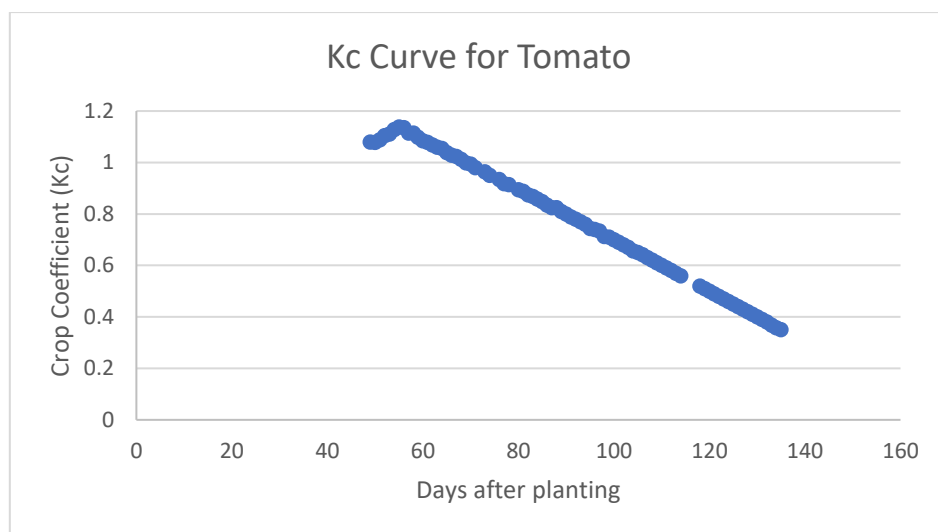


Figure 4.26. Crop coefficient of tomato as a function of days after planting.

Table 4.30. KC values of Tomato

Tomato	Kc (from FAO)
Kc mid	1.10–1.15
Kc end	0.60–0.70

The below Table 4.31 shows the Kc values of all the experimented crops after planting.

Table 4.31. KC values of each crop

Days after planting	Beans	Days after planting	Carrot	Days after planting	Mustard	Days after planting	Tomato
16	0.65	49	1.04	49	1	49	1.07
17	0.66	50	1.05	50	0.98	50	1.08
18	0.68	51	1.09	51	0.96	51	1.1
19	0.7	52	1.09	52	0.95	52	1.11
20	0.71	53	1.08	53	0.93	53	1.12
21	0.73	54	1.08	54	0.92	54	1.14
22	0.74	55	1.07	55	0.9	55	1.15
23	0.76	56	1.07	56	0.89	56	1.14
24	0.77	57	1.06	57	0.87	57	1.13
25	0.79	58	1.05	58	0.85	58	1.12
26	0.8	59	1.05	59	0.84	59	1.11
27	0.82	60	1.04	60	0.82	60	1.1
28	0.84	61	1.04	61	0.81	61	1.09
29	0.85	62	1.03	62	0.79	62	1.08
30	0.87	63	1.03	63	0.77	63	1.07

31	0.88	64	1.02	64	0.76	64	1.06
32	0.9	65	1.01	65	0.74	65	1.05
33	0.91	66	1.01	66	0.73	66	1.04
34	0.93	67	1	67	0.71	67	1.03
35	0.94	68	1	68	0.69	68	1.02
36	0.96	69	0.99	69	0.68	69	1.01
37	0.98	70	0.99	70	0.66	70	1
38	0.99	71	0.98	71	0.65	71	0.99
39	1.01	72	0.97	72	0.63	72	0.98
40	1.02	73	0.97	73	0.62	73	0.97
41	1.04	74	0.96	74	0.6	74	0.96
42	1.05	75	0.96	75	0.58	75	0.95
43	1.07	76	0.95	76	0.57	76	0.94
44	1.08	77	0.95	77	0.55	77	0.93
45	1.1	78	0.94	78	0.54	78	0.92
46	1.1	79	0.93	79	0.52	79	0.91
47	1.1	80	0.93	80	0.5	80	0.9
48	1.1	81	0.92	81	0.49	81	0.89
49	1.1	82	0.92	82	0.47	82	0.88
50	1.11	83	0.91	83	0.46	83	0.87
51	1.11	84	0.91	84	0.44	84	0.86
52	1.11	85	0.9	85	0.43	85	0.85
53	1.11	86	0.89	86	0.42	86	0.84
54	1.11	87	0.89	87	0.42	87	0.83
55	1.11	88	0.88	88	0.41	88	0.82
56	1.11	89	0.88	89	0.41	89	0.81
57	1.11	90	0.87	90	0.41	90	0.8
58	1.11	91	0.87	91	0.4	91	0.79
59	1.12	92	0.86	92	0.4	92	0.78
60	1.12	93	0.85	93	0.4	93	0.77
61	1.12	94	0.85	94	0.39	94	0.76
62	1.12	95	0.84	95	0.39	95	0.75
63	1.12	96	0.84	96	0.39	96	0.74
64	1.12	97	0.83	97	0.38	97	0.73
65	1.12	98	0.83	98	0.38	98	0.72
66	1.12	99	0.82	99	0.38	99	0.71
67	1.12	100	0.81	100	0.37	100	0.7
68	1.13	101	0.81	101	0.37	101	0.69
69	1.13	102	0.8	102	0.36	102	0.68
70	1.13	103	0.8	0	0	103	0.67
71	1.13	104	0.79	0	0	104	0.66
72	1.13	105	0.79	0	0	105	0.65
73	1.13	106	0.78	0	0	106	0.64
74	1.13	107	0.77	0	0	107	0.63
75	1.13	108	0.77	0	0	108	0.62

76	1.13	109	0.76	0	0	109	0.61
77	1.14	110	0.76	0	0	110	0.6
78	1.14	111	0.75	0	0	111	0.59
79	1.14	112	0.75	0	0	112	0.58
80	1.14	113	0.74	0	0	113	0.57
81	1.14	114	0.73	0	0	114	0.56
82	1.14	115	0.73	0	0	115	0.55
83	1.14	116	0.72	0	0	116	0.54
84	1.14	117	0.72	0	0	117	0.53
85	1.14	118	0.71	0	0	118	0.52
86	1.15	119	0.71	0	0	119	0.51
87	1.15	120	0.7	0	0	120	0.5
88	1.15	121	0.69	0	0	121	0.49
89	1.15	122	0.69	0	0	122	0.48
90	1.15	123	0.68	0	0	123	0.47
91	1.13	124	0.68	0	0	124	0.46
92	1.11	125	0.67	0	0	125	0.45
93	1.09	126	0.67	0	0	126	0.44
94	1.06	127	0.66	0	0	127	0.43
95	1.04	128	0.65	0	0	128	0.42
96	1.02	129	0.65	0	0	129	0.41
97	1	130	0.64	0	0	130	0.4
98	0.98	131	0.64	0	0	131	0.39
99	0.96	132	0.63	0	0	132	0.38
100	0.93	133	0.63	0	0	133	0.37
101	0.91	134	0.62	0	0	134	0.36
102	0.89	135	0.61	0	0	135	0.35
103	0.87	136	0.61	0	0	136	0.34
104	0.85	137	0.6	0	0	137	0.33
105	0.83	138	0.6	0	0	138	0.32
106	0.8	139	0.59	0	0	139	0.31
107	0.78	140	0.59	0	0	140	0.3
108	0.76	141	0.58	0	0	141	0.29
109	0.74	142	0.57	0	0	142	0.28
110	0.72	143	0.57	0	0	143	0.27
111	0.7	144	0.56	0	0	144	0.26
112	0.67	145	0.56	0	0	145	0.25
113	0.65	146	0.55	0	0	146	0.24
114	0.63	147	0.55	0	0	147	0.23
115	0.61	148	0.54	0	0	148	0.22
116	0.59	149	0.53	0	0	149	0.21
117	0.57	150	0.53	0	0	150	0.2
118	0.54	151	0.52	0	0	151	0.19
119	0.52	152	0.52	0	0	152	0.18
120	0.5	153	0.51	0	0	153	0.17

121	0.48	154	0.51	0	0	154	0.16
122	0.46	155	0.5	0	0	155	0.15
123	0.44	156	0.49	0	0	156	0.14
124	0.41	157	0.49	0	0	157	0.13
125	0.39	158	0.48	0	0	158	0.12
126	0.37	159	0.48	0	0	159	0.11
127	0.35	160	0.47	0	0	160	0.1
128	0.33	161	0.47	0	0	161	0.09
129	0.31	162	0.46	0	0	162	0.08
130	0.28	163	0.45	0	0	163	0.07
131	0.26	164	0.45	0	0	164	0.06
132	0.24	165	0.44	0	0	165	0.05
133	0.22	166	0.44	0	0	166	0.04
134	0.2	167	0.43	0	0	167	0.03
135	0.18	168	0.43	0	0	168	0.02
136	0.15	169	0.42	0	0	169	0.01
137	0.13	170	0.41	0	0	170	0
138	0.11	171	0.41	0	0	171	0
139	0.09	172	0.4	0	0	172	0
140	0.07	173	0.4	0	0	173	0
141	0.05	174	0.39	0	0	174	0

4.6.5 Discussion

For all studied crops, the Kc values followed the typical pattern of low values during the initial stage, increasing through the development stage, peaking during the mid-season stage when crop cover and water demand are highest, and then decreasing during the late season stage as the crops reach maturity. The variations in Kc values are consistent with crop growing stages and water requirements. The Kc peaks correspond to the crop development and peak evapotranspiration.

Determination of crop coefficient (Kc) values for different crops is essential for accurate irrigation scheduling and water resource management.

This study confirmed that:

- Kc values are crop-specific and vary by growth stage.
- The highest water requirement occurs during the mid-season stage for most crops.

Hence use of micro pot lysimeters proved effective for direct ET measurement in controlled conditions. The Kc values obtained can aid in precise irrigation scheduling for the region, especially under limited water conditions.

CHAPTER 5

CONCLUSION

- i. The soil sample analyzed was predominantly loamy sand in texture, as determined by particle size distribution using the hydrometer method.
- ii. Based on gravimetric analysis, the average field capacity of the soil used in this study was found to be 25.97%.
- iii. The simulation period spanned from December 2024 to April 2025. During this period, monthly reference evapotranspiration (ET_o) ranged from 1.41 to 2.21 mm/day, with the highest values recorded in April and the lowest in December and January.
- iv. Beans exhibited the highest percolation losses (5.01 mm/day), while mustard showed the lowest (1.63 mm/day) as observed from this study.
- v. Beans required the highest irrigation depth (4.98 mm/day), whereas mustard required the lowest (2.60 mm/day) as observed from this study.
- vi. Crop evapotranspiration (ET_c) determined using percolation micro-lysimeters indicated that beans had the highest water requirement, followed by carrot, tomato, and mustard.
- vii. Beans exhibited a steady increase in ET_c from 1.03 mm/day in December to 1.54 mm/day in March, corresponding with rising temperatures and light availability.
- viii. Tomato ET_c peaked in December (1.56–1.57 mm/day) and declined to 0.62 mm/day in March, reflecting reduced water demand during the later growth stages.
- ix. Mustard ET_c peaked in December (1.28–1.29 mm/day), dropped sharply in January, showed anomalous negative values, and recovered to 1.08 mm/day in March.
- x. Carrot ET_c peaked in December (1.54–1.57 mm/day), with the lowest values observed in March (0.89–0.90 mm/day).
- xi. Crop coefficient estimation of tomatoes, mustard, carrots, and beans using the FAO-56 Penman-Monteith equation and measured ET_c showed distinct variations among the crops, despite identical climatic conditions.
- xii. The tomato crop recorded the highest mid-season K_c value (1.20), followed by beans (1.12), mustard (1.06), and carrot (1.05), indicating the highest water use by tomato during peak growth.
- xiii. The K_c curves of the experimental crops closely followed the generalized FAO K_c curve, particularly during the mid and late growth stages, validating the reliability of the pot micro-lysimeter method in approximating standardized crop coefficients under local conditions.

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